

MUST-SCORE FACTS

Geography

Consolidated Must-Score Facts and must-score drill cards for every chapter.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

Chapters

1. Daily Read — High-Yield Revision Facts
2. Topic 1 — Indian Physical Geography: Mountains & Hills
3. Topic 2 — Climate of India
4. Topic 3 — Drainage System
5. Topic 4 — Lakes, Waterfalls & Water Resources
6. Topic 5 — Soils
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Daily Read — High-Yield Revision Facts

1. Indian Mountain Passes (State-Wise Cheat-Sheet)

State / UT	Mountain Pass	Connects / Geographic Significance
Ladakh	Karakoram Pass	Ancient caravan route between India and Yarkand (Tarim Basin, China)
	Zoji La	Connects Srinagar with Kargil and Leh ; Great Himalayas; Zojila Tunnel
	Khardung La	Gateway to the Shyok and Nubra Valleys
	Chang La	Connects Leh to Pangong Tso lake
	Fotu La	Highest point on the Srinagar-Leh highway (Zaskar Range)
	Umling La	World's highest motorable pass/road (19,024 ft) built by BRO
Jammu & Kashmir	Banihal Pass	Connects Jammu with Srinagar ; Jawahar Tunnel under Pir Panjal Range
	Pir Panjal Pass	Traditional route from Jammu to Srinagar valley (<i>Peer Ki Gali</i>)
	Burzil Pass	Connects Srinagar with Gilgit; ancient gateway to Central Asia
Himachal Pradesh	Rohtang Pass	Connects Kullu Valley with Lahaul and Spiti Valleys; Atal Tunnel (9.02 km) under Rohtang
	Shipki La	Sutlej River enters India from Tibet through this pass; Zaskar Range
	Bara-lacha La	Connects Mandi/Manali to Leh (Zaskar Range); origin of Chandra & Bhaga rivers
	Debsa Pass	Connects Spiti and Parvati Valleys
Uttarakhand (UKPCS Top Priority!)	Lipulekh Pass	Tri-junction of India, Nepal, and Tibet ; primary route for Kailash Mansarovar Yatra (Pithoragarh)
	Mana Pass (Dungri La)	Connects Uttarakhand with Tibet; north of Badrinath (Chamoli)
	Niti Pass	Connects Chamoli (Uttarakhand) with Tibet
	Muling La	High mountain pass in Uttarkashi north of Gangotri connecting with Tibet
	Trail's Pass	Connects Pithoragarh with Bageshwar (Pindari Glacier to Milam Valley)
	Sin La	Connects Darma and Byans valleys in Pithoragarh
Sikkim	Nathu La	Strategic border post on India-China border; Old Silk Route branch into Chumbi Valley; reopened for trade in 2006
	Jelep La	Connects Sikkim with Lhasa (Tibet) passing through the Chumbi Valley
Arunachal Pradesh	Bom Di La	Connects Arunachal Pradesh (Kameng) with Lhasa (Tibet)
	Diphu Pass	Strategic pass at the tri-junction of India, China, and Myanmar
	Bum La	High mountain pass near Tawang on India-China McMahon line
	Dihang Pass	Connects Arunachal Pradesh with Myanmar
	Pangsau Pass	Historic Ledo Road (Stilwell Road) connecting Assam to Myanmar

Western Ghats Gaps (Strict North-to-South Chronology)

- Thalghat (Kasara Ghat):** Connects **Mumbai to Nashik / Nagpur / North India** (National Highway 160 & railway line).
- Bhorghat:** Connects **Mumbai to Pune / Bengaluru / South India** (Mumbai-Pune Expressway).
- Palghat (Palakkad Gap):** Located between **Nilgiri Hills (North) and Anaimalai Hills (South)**; connects **Coimbatore (Tamil Nadu) to Palakkad / Kochi (Kerala)**.

4. **Shenkottai Pass:** Located between **Cardamom Hills and Agasthyamalai Hills**; connects **Kollam (Kerala) to Madurai / Tenkasi (Tamil Nadu)**.

2. Highest Peaks of Major Ranges in India

Mountain Range / Region	Highest Peak	Elevation (m)	Location / Crucial Takeaway
Trans-Himalayas (Karakoram)	K2 (Godwin-Austen)	8,611 m	Second highest peak in the world ; highest point in Indian territory (POK)
Great Himalayas (India)	Kanchenjunga	8,586 m	Highest peak in undisputed Indian territory ; located on Sikkim-Nepal border
Garhwal Himalayas (Uttarakhand)	Nanda Devi	7,816 m	Highest peak located ENTIRELY within India ; Chamoli, UNESCO WHS
Garhwal (Second Highest)	Kamet	7,756 m	Chamoli district, Uttarakhand
Aravalli Range	Guru Shikhar	1,722 m	Mount Abu (Sirohi district, Rajasthan); oldest fold mountains in India
Satpura Range	Dhupgarh	1,350 m	Near Pachmarhi (Hoshangabad/Narmadapuram, Madhya Pradesh)
Vindhya Range	Sadbhavna Shikhar (Kalumar)	752 m	Damoh district, Madhya Pradesh
Western Ghats (Sahyadri)	Anamudi	2,695 m	Anaimalai Hills (Idukki, Kerala) — HIGHEST PEAK OF PENINSULAR / SOUTH INDIA
Nilgiri Hills	Doda Betta	2,637 m	Udhagamandalam / Ooty (Tamil Nadu); meeting point of Western & Eastern Ghats
Eastern Ghats	Jindhagada Peak / Arma Konda	1,690 m	Araku Valley (Visakhapatnam, Andhra Pradesh); Mahendragiri (1,501 m, Odisha) often given in older PCS options
Chhotanagpur Plateau	Parasnath Peak (Sammed Shikhar)	1,366 m	Giridih (Jharkhand); sacred Jain pilgrimage site (20 Tirthankaras attained Moksha)
Andaman & Nicobar Islands	Saddle Peak	732 m	North Andaman Island (Saddle Peak National Park)
Great Nicobar Island	Mount Thullier	642 m	Southernmost peak in India
Naga Hills (Purvanchal)	Saramati Peak	3,826 m	Indo-Myanmar border, Nagaland
Mizo Hills (Lushai Hills)	Phawngpui (Blue Mountain)	2,157 m	Mizoram

3. Panch Prayag & Indian River Drainage System

A. The Panch Prayag of Uttarakhand (Order along Alaknanda River)

Memorize downstream order from source towards plains: 1. **Vishnuprayag: Alaknanda + Dhauliganga** (Chamoli) 2. **Nandaprayag: Alaknanda + Nandakini** (Chamoli) 3. **Karnaprayag: Alaknanda + Pindar** (Chamoli) 4. **Rudraprayag: Alaknanda + Mandakini** (Rudraprayag) 5. **Devprayag: Alaknanda + Bhagirathi** → Official **GANGA** begins here! (Tehri Garhwal)

B. Ganga River Tributaries (Left vs Right Bank)

- Left Bank Tributaries (West to East):** 1. **Ramganga** (originates in Dudhatoli range, Pauri Garhwal; flows through Jim Corbett NP; meets Ganga at Kannauj) 2. **Gomati** (originates in plains at **Gomat Taal / Fulhar Lake, Pilibhit,**

UP — only Himalayan/Ganga tributary originating in the plains!; Lucknow & Jaunpur on its banks; joins Ganga near Saidpur, Ghazipur) 3. **Ghaghra / Karnali** (originates from Mapchachungo glacier near Mansarovar; joined by Sarda/Kali; **largest tributary by water volume**; joins Ganga at Chhapra, Bihar) 4. **Gandak** (originates in Nepal Himalayas; joins Ganga at Sonpur / Hajipur, Bihar) 5. **Kosi** (*Sorrow of Bihar*; 7 streams / Saptakoshi; notorious for course-shifting; joins at Kursela, Katihar) 6. **Mahananda** (originates in Darjeeling hills; **last left-bank tributary of Ganga in India**).

- **Right Bank Tributaries:** 1. **Yamuna** (longest tributary, 1376 km; Yamunotri glacier, Bandarpunch peak; joins at Prayagraj Triveni Sangam; its tributaries: Chambal, Sind, Betwa, Ken, Tons, Hindon) 2. **Son** (originates from **Amarkantak Plateau, MP**; flows north-eastwards through Sonbhadra UP; joins Ganga near Danapur/Patna, Bihar) 3. **Punpun** (originates in Palamu, Jharkhand; joins Ganga at Fatuha near Patna).

C. Peninsular Drainage: West-Flowing vs East-Flowing

- **West-Flowing (into Arabian Sea — form Estuaries, NO deltas):**
- **Narmada** (1312 km): Amarkantak plateau (MP); flows through **Rift Valley** between Vindhyas (North) and Satpuras (South); forms **Dhuandhar Falls** at Marble Rocks (Bhedaghat, Jabalpur); Sardar Sarovar Dam.
- **Tapi / Tapti** (724 km): Multai (Betul district, MP); flows in rift valley south of Satpuras; Ukai Dam & Kakrapar Project (Gujarat).
- **Sabarmati:** Dhebar Lake (Aravalli Range, Udaipur); flows through Ahmedabad and Gandhinagar into Gulf of Khambhat.
- **Mahi:** Vindhyan hills (Dhar, MP); **CUTS THE TROPIC OF CANCER TWICE!**
- **Luni:** Originates near Ajmer (Aravallis) as Sagarmati; **ends in the marshy Rann of Kutch (Inland drainage, never reaches the sea)**; sweet water up to Balotra, then saline.
- **Sharavati:** Shimoga (Karnataka); forms famous **Jog Falls (Gersoppa / Mahatma Gandhi Falls)**.
- **Periyar & Bharathappuzha (Ponnani):** Kerala.
- **East-Flowing (into Bay of Bengal — form large Deltas):**
- **Mahanadi** (851 km): Sihawa hills (Dhamtari, Chhattisgarh); flows through Odisha; Hirakud Dam.
- **Godavari** (1465 km — **Dakshin Ganga / Vridhha Ganga**): Trimbakeshwar (Nashik, Maharashtra); **longest river in Peninsular India**; Tributaries: Pravara, Manjra, Penganga, Wainganga, Wardha, **Pranhita** (largest), Indravati, Sabari.
- **Krishna** (1401 km): Mahabaleshwar (Maharashtra); Tributaries: Koyna, Ghataprabha, Malaprabha, Bhima, **Tungabhadra** (largest), Musi (Hyderabad on its banks), Munneru.
- **Kaveri** (800 km — **Ganga of the South**): Talakaveri (Brahmagiri Range, Coorg, Karnataka); flows year-round (perennial due to SW & NE monsoons); creates river islands: **Srirangapatna, Shivanasamudra, and Srirangam**; Tributaries: Hemavati, Harangi, Shimsha, Arkavathi (left); Lakshmantirtha, Kabini, Bhavani, Noyyal, Amaravati (right).

4. Multipurpose River Valley Projects, Dams & Waterfalls

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Tehri Dam	Bhagirathi + Bhilangana	Uttarakhand	Highest dam in India (260.5 m); rock and earth-fill embankment
Bhakra Nangal Dam	Sutlej	Himachal Pradesh / Punjab	Highest straight gravity dam in India; reservoir is Gobind Sagar
Hirakud Dam	Mahanadi	Odisha (Sambalpur)	Longest earthen dam in India / world (main dam 4.8 km, total 25.8 km)
Sardar Sarovar Dam	Narmada	Gujarat (Navagam)	Beneficiary states: Gujarat, MP, Maharashtra, Rajasthan; Statue of Unity nearby

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Nagarjuna Sagar Dam	Krishna	Andhra Pradesh / Telangana border	World's largest masonry dam; creates Nagarjunasagar reservoir
Rihand Project (Govind Ballabh Pant Sagar)	Rihand (tributary of Son)	Sonbhadra (Uttar Pradesh)	Largest man-made artificial reservoir / lake in India by volume
Mata Tila & Rajghat Dam	Betwa	Lalitpur (Uttar Pradesh / MP)	Joint venture between UP and MP
Kunchikal Falls	Varahi	Karnataka (Shimoga)	Highest tiered waterfall in India (455 m)
Jog Falls (Gersoppa)	Sharavati	Karnataka (Shimoga)	Highest untiered plunge waterfall (253 m)
Dhuandhar Falls	Narmada	Bhedaghat, Jabalpur (MP)	"Smoke cascade" cutting through marble gorge
Chitrakote Falls	Indravati	Bastar (Chhattisgarh)	Known as the " Niagara of India " (widest horseshoe waterfall in India)
Hundru Falls	Subarnarekha	Ranchi (Jharkhand)	98 m drop on Chotanagpur plateau
Nohkalikai Falls	Rain-fed stream	Cherrapunji (Meghalaya)	Tallest plunge waterfall in India (340 m)

5. Major Agricultural Crops & Top Producing States

Crop Category	Specific Crop	Top Producing State (1st Rank)	Other Key States / Prelims Note
Foodgrains	Total Foodgrains	Uttar Pradesh	Followed by Madhya Pradesh, Punjab
	Wheat	Uttar Pradesh	Followed by MP, Punjab (Punjab highest per hectare yield)
	Rice	West Bengal	Followed by Uttar Pradesh, Punjab
	Maize	Karnataka	Followed by Madhya Pradesh, Maharashtra
	Coarse Cereals (Nutri-cereals)	Rajasthan	Followed by Karnataka, Maharashtra
	Total Pulses	Madhya Pradesh	Followed by Maharashtra, Rajasthan
	Gram (Chana)	Maharashtra / MP	Accounts for ~40% of total pulses
Oilseeds	Total Oilseeds	Rajasthan	Followed by Maharashtra, Madhya Pradesh
	Rapeseed & Mustard	Rajasthan	(>45% of national production)
	Soyabean	Maharashtra / MP	MP traditionally called <i>Soyabean State</i>
	Groundnut	Gujarat	Followed by Rajasthan, Tamil Nadu
Cash / Commercial	Sugarcane	Uttar Pradesh	Followed by Maharashtra, Karnataka (Maha highest sugar recovery)
	Cotton	Gujarat	Followed by Maharashtra, Telangana
	Jute & Mesta	West Bengal	(>75% of national production - Golden Fibre)
	Tea	Assam	Followed by West Bengal (Darjeeling - 1st GI tag 2004), Tamil Nadu
	Coffee	Karnataka	(>70% of national production - Arabica & Robusta in Kodagu/Chikkamagaluru)
	Natural Rubber	Kerala	(>75% of national production)
	Tobacco	Gujarat / Andhra Pradesh	CTRI located at Rajahmundry (AP)

6. Major Tribes of India & State-Wise Matrix (Direct Match-Ups)

State / UT	Prominent Indigenous Tribes	Key Ethnographic / Exam Fact
Uttarakhand (UKPCS Top Priority!)	Tharu, Jaunsari, Bhotia, Buksa, Raji (Van Rawat)	Tharu: Largest tribe in UK & UP; historically celebrate Diwali as mourning (<i>Shok Parva</i>); live in Terai (US Nagar). Jaunsari: Second largest; polyandry tradition; Dehradun (Chakrata). Bhotia: Transhumance (<i>Ritu Pravas</i>); seasonal migration to bugyals; Indo-Tibetan trade. Buksa: Claim Rajput descent (Pawar); Terai region. Raji: Smallest tribe (<i>Van Rawat</i> / kings of forest); cave dwellings.
Uttar Pradesh	Tharu, Gond, Kharwar, Sahariya, Parahiya, Baiga, Agariya, Chero	Sonbhadra has highest ST population & diversity in UP; Sahariya in Lalitpur; Agariya (iron smelters) in Sonbhadra.
Madhya Pradesh	Bhil, Gond, Baiga, Sahariya, Kol, Korku, Bharia	Highest ST population in India; Bhils largest tribe of India; <i>Bhagoria</i> festival; <i>Pithora</i> painting.
Rajasthan	Bhil, Meena, Garasia, Sahariya, Damor	Meenas are the most populous tribe in Rajasthan; Sahariya in Baran.
Odisha	Santhal, Bonda, Dongria Kondh, Juang, Saora	Largest number of Particularly Vulnerable Tribal Groups (PVTGs: 13 out of 75 in India) ; Dongria Kondh worship Niyamgiri hills.
Tamil Nadu	Toda, Badaga, Kota, Irula, Kurumba	Toda: Inhabit Nilgiri Hills; pastoral tribe centered on water buffaloes; practice fraternal polyandry; barrel-shaped huts.
Andhra Pradesh / Telangana	Chenchu, Yanadi, Koya, Lambada	Chenchu: Primitive hunter-gatherers in Nallamala forest / Srisailem.
Sikkim	Lepcha, Bhutia, Limbu	Lepchas are indigenous inhabitants; practice sacred groves.
Himachal Pradesh	Gaddi, Gujjar, Kinnaura, Pangwala	Gaddis: Pastoral transhumance shepherds of Dhauladhar range.
Andaman & Nicobar Islands	Great Andamanese, Onge, Jarawa, Sentinelese, Shompen, Nicobarese	Negrito stock (Jarawa, Sentinelese, Onge - North Sentinel isolated); Shompen are Mongoloid stock.

7. Major Sea Ports & National Waterways of India

A. 13 Major Ports of India

- **Western Coast:** 1. **Deendayal Port (Kandla)** (Gulf of Kutch, Gujarat): Tidal port; free trade zone (SEZ); handles crude oil & grains. 2. **Mumbai Port** (Maharashtra): Largest natural port and harbour in India. 3. **Jawaharlal Nehru Port (JNPT / Nhava Sheva)** (Navi Mumbai): **Largest container port in India**; fully mechanized. 4. **Mormugao Port** (Goa - Zuari estuary): Premier port for iron ore exports from Goa. 5. **New Mangalore Port** (Karnataka): Handles iron ore exports from Kudremukh. 6. **Cochin Port** (Kerala - Willingdon Island in Vembanad Lake): Natural harbour; International Container Transshipment Terminal (Vallarpadam).
- **Eastern Coast:** 7. **V.O. Chidambaranar Port (Tuticorin)** (Tamil Nadu - Gulf of Mannar): Artificial deep-sea port; pearl fishery. 8. **Chennai Port** (Tamil Nadu): **Oldest artificial harbour on the east coast**; second busiest. 9. **Kamarajar Port (Ennore)** (Tamil Nadu): **First corporatized port in India** (public company). 10. **Visakhapatnam Port** (Andhra Pradesh): **Deepest landlocked and protected port in India** (Dolphin's Nose hill protects from storms). 11. **Paradip Port** (Odisha - Mahanadi delta): Deep-water port exporting iron ore to Japan. 12. **Syama Prasad Mookerjee Port (Kolkata-Haldia)** (West Bengal): **Only riverine major port in India** (on Hooghly River); twin dock system (Kolkata & Haldia).
- **Island:** 13. **Port Blair Port** (Andaman & Nicobar Islands).

B. Core National Waterways (NW) of India

- **NW-1: Ganga–Bhagirathi–Hooghly River System** from **Prayagraj to Haldia** (1,620 km) — **Longest National Waterway in India**; passes through UP, Bihar, Jharkhand, WB.
- **NW-2: Brahmaputra River** from **Sadiya to Dhubri** (891 km) in Assam.
- **NW-3: West Coast Canal** from **Kottapuram to Kollam** (205 km) in Kerala (includes Champakara & Udyogmandal canals).
- **NW-4: Krishna–Godavari Rivers & Canals** from Kakinada to Puducherry (1,078 km).
- **NW-5: Brahmani River, Deltaic channels of Mahanadi & East Coast Canal** (588 km) in Odisha & WB.
- **NW-16: Barak River** from Lakhimpur to Bhanga (121 km) in Assam.

8. Census 2011 Essential Fact Card (India vs UP vs UK)

Demographic Indicator	All-India	Uttar Pradesh	Uttarakhand
Total Population	121.08 Crore (17.5% of world)	19.98 Crore (16.51% of India - Rank 1)	1.01 Crore (0.83% of India - Rank 20)
Decadal Growth Rate (2001–11)	17.70% (Highest: Meghalaya 27.95%; Lowest/Negative: Nagaland -0.58%)	20.22% (Highest: Gautam Buddha Nagar 49.1%; Lowest: Kanpur Nagar 9.9%)	18.81% (Highest: US Nagar 33.4%; Negative: Almora -1.28%, Pauri Garhwal -1.41%)
Population Density	382 per sq km (Highest: Bihar 1106; Lowest: Arunachal 17)	829 per sq km (Rank 4 among states) • Highest: Ghaziabad (3,971) • Lowest: Lalitpur (242)	189 per sq km (Rank 25) • Highest: Haridwar (801) • Lowest: Uttarkashi (41)
Overall Sex Ratio	943 females / 1000 males (Highest: Kerala 1084; Lowest: Haryana 879)	912 females / 1000 males • Highest: Jaunpur (1,024), Azamgarh (1,019), Deoria (1,017) • Lowest: Gautam Buddha Nagar (851)	963 females / 1000 males • Highest: Almora (1,139), Rudraprayag (1,114) • Lowest: Haridwar (880), Dehradun (902)
Child Sex Ratio (0–6 yrs)	919 (Highest: Mizoram 970; Lowest: Haryana 834)	902 • Highest: Balrampur (950) • Lowest: Baghpat (841)	890 • Highest: Pithoragarh (916) • Lowest: Pithoragarh / Champawat
Literacy Rate (Overall)	73.0% (Male: 80.9%, Female: 64.6%; Gap: 16.3%)	67.7% (Male: 77.3%, Female: 57.2%; Gap: 20.1%) • Highest: Gautam Buddha Nagar (80.1%) • Lowest: Shravasti (46.7%)	78.82% (Male: 87.4%, Female: 70.0%; Gap: 17.4%) • Highest: Dehradun (84.25%) • Lowest: Haridwar (73.43%)
SC Population %	16.6% (Highest %: Punjab 31.9%; Highest No.: UP)	20.7% of state pop (Highest No.: Sitapur; Highest %: Kaushambi 34.7%; Lowest: Baghpat)	18.8% of state pop (Highest %: Bageshwar; Lowest %: Dehradun)
ST Population %	8.6% (Highest %: Lakshadweep/Mizoram; Highest No.: MP; Nil in Punjab, Haryana)	0.6% of state pop (Highest No. & %: Sonbhadra ; Lowest: Baghpat; Tharu is largest tribe)	2.9% of state pop (Highest %: Udham Singh Nagar ; Tharu & Jaunsari are largest)

9. Minerals of Uttar Pradesh & Uttarakhand (state-site map)

National mineral belts stay in the Minerals chapter. Daily sheet needs the **UP / UK places** that otherwise vanish (same gap as Hulas on the Indus table).

Mineral / resource	Uttar Pradesh	Uttarakhand	Trap
Limestone / cement	Sonbhadra (Churk, Dalla, Ghurma); also Banda–Mirzapur belt	Dehradun , Tehri, Pithoragarh, Pauri belts	Do not give Rajasthan's Makrana marble as a UP fact
Coal	Singrauli coalfield (Sonbhadra; shared with MP)	Negligible	Jharia / Raniganj are Jharkhand–WB , not UP
Silica sand / glass	Shankargarh (Prayagraj) , Banda; feeds Firozabad glass	—	Firozabad = glass craft; sand is Prayagraj belt
Bauxite	Banda, Vindhyan pockets (small)	—	Big bauxite = Odisha / Gujarat, not UP
Andalusite / sillimanite	Sonbhadra / Mirzapur (refractory)	—	
Magnesite	—	Jhironi / Bageshwar (Kumaon) — India's important magnesite belt	Magnesite ≠ magnesium metal plant trap
Gypsum	—	Dehradun / Nainital foothill pockets	Rajasthan remains India's gypsum giant
Steatite / soapstone	—	Almora / Pithoragarh (also used in local craft)	
Copper (old)	—	Askot (Pithoragarh) — historic Kumaon copper	Khetri copper is Rajasthan
Rock phosphate	—	Mussoorie / Durmala–Maldeota (Dehradun)	
Building stone / granite	Mirzapur–Sonbhadra; Lalitpur	Granite–quartzite of Lesser Himalaya	
Energy water	Rihand / Govind Ballabh Pant Sagar (Sonbhadra) — largest man-made lake by volume	Tehri (Bhagirathi–Bhilangana) — highest dam	Hydel ≠ mineral ore

Logic: Sonbhadra is UP's mineral hub (coal + limestone + refractories). **Bageshwar magnesite** and **Askot copper** are the UK pair. Do not park Rajasthan or Jharkhand fields in these two states.

10. Major Minerals of India & Famous Mines Master Matrix

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Iron Ore (Hematite & Magnetite)	Odisha (1st in production & reserves), Chhattisgarh, Karnataka, Jharkhand	• Odisha: Mayurbhanj (Gorumahisani, Badampahar), Keonjhar (Joda), Sundargarh • Chhattisgarh: Bailadila (Bastar/Dantewada — high-grade hematite exported to Japan via Visakhapatnam Port), Dalli-Rajhara (Durg — feeds Bhilai Steel Plant) • Karnataka: Kudremukh (Chikkamagaluru — horse-face shaped; slurry pipeline to Mangalore), Bababudan Hills, Bellary-Hospet (Sandur) • Jharkhand: Noamundi, Gua, Kiriburu (West Singhbhum) • Goa: Marmagao belt	• India is rich in Hematite (~70%) and Magnetite (~30%). • Dharwar and Cuddapah rock systems are primary sources. • <i>Trap:</i> Bailadila is in Chhattisgarh, NOT MP or Odisha!

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Bauxite (Aluminium Ore)	Odisha (>50% of national production), Gujarat, Jharkhand	• Odisha: Panchpatmali (Koraput — India's largest deposit; feeds NALCO refinery at Damanjodi), Kalahandi, Rayagada, Bolangir • Jharkhand: Lohardaga, Gumla, Latehar • Gujarat: Jamnagar, Junagadh, Kutch • Madhya Pradesh: Amarkantak plateau (Shahdol), Balaghat • Maharashtra: Kolhapur, Ratnagiri	• Associated with laterite rocks of tertiary age formed via intense leaching. • NALCO (Damanjodi & Angul) and HINDALCO (Renukoot, UP) are major processors.
Copper	Madhya Pradesh (1st in production), Rajasthan (1st in reserves), Jharkhand	• Madhya Pradesh: Malanjkhand (Balaghat district — largest open-cast copper mine in Asia; accounts for ~50% of Indian production) • Rajasthan: Khetri, Chandmari, Kolihaan (Jhunjhunu), Kho-Dariba (Alwar), Delwara (Udaipur) • Jharkhand: Mosabani, Ghatshila, Rakha (Singhbhum Copper Belt) • Uttarakhand: Askot (Pithoragarh — old workings)	• Hindustan Copper Limited (HCL, est. 1967) holds public monopoly. • India is heavily import-dependent for copper ore (~80% imported).
Manganese	Madhya Pradesh (1st in production), Maharashtra, Odisha	• Madhya Pradesh: Balaghat (Bharveli underground mine), Chhindwara • Maharashtra: Nagpur, Bhandara, Ratnagiri • Odisha: Sundargarh, Keonjhar, Koraput • Karnataka: Shimoga, Bellary	• Vital ferro-alloy: ~10 kg manganese is needed to manufacture 1 tonne of steel. • MOIL (Manganese Ore India Limited) headquartered in Nagpur.
Coal	Jharkhand (1st in reserves), Odisha / Chhattisgarh (1st in production)	• Jharkhand: Jharia (Dhanbad — richest and only storehouse of prime coking coal in India), Bokaro, Giridih, North/South Karanpura, Ramgarh • West Bengal: Raniganj (Asansol/Bardhaman — oldest commercial coalfield in India, started 1774 by Summer & Heatly) • Chhattisgarh: Korba (Hasdeo valley), Tatapani-Ramkola, Chirimiri • Odisha: Talcher (Brahmani valley — feeds NTPC super thermal plant), Ib valley • Madhya Pradesh: Singrauli (Jhingurdah seam — second thickest coal seam in the world; shared with UP Sonbhadra), Pench-Kanhan • Telangana: Singareni (Godavari valley — SCCL) • Tamil Nadu: Neyveli (Cuddalore — premier deposit of Lignite / Brown Coal in India) • Assam: Makum, Nazira, Ledo (Tertiary coal; high sulphur content)	• Gondwana Coal (~98% of India's reserves; 250 million yrs old; low moisture, low sulphur, high ash, non-coking/coking). • Tertiary Coal (~2% reserves; 15–60 million yrs old; NE states & Jammu; high sulphur & volatile matter).
Petroleum & Crude Oil	Offshore (Bombay High) (1st in production), Rajasthan, Gujarat, Assam	• Offshore: Bombay High (1974 discovery by ONGC vessel <i>Sagar Samrat</i>; 160 km off Mumbai), Bassein, Ravva (Krishna-Godavari) • Rajasthan: Barmer Basin (Mangala, Bhagyam, Aishwarya fields — discovered by Cairn/Vedanta; largest onshore producer) • Gujarat: Ankleshwar (<i>Fountain of Prosperity</i> named by Nehru), Kalol, Mehsana, Aliabet Island (Gulf of Khambhat) • Assam: Digboi (oldest producing oilfield & refinery in India and Asia, 1901), Naharkatiya, Moran-Hugrijan, Surma Valley	• Refineries: Jamnagar (Gujarat) is the world's largest grass-root refining complex (private - Reliance). • Indian Oil's Digboi is the oldest operational refinery.

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Gold	Karnataka (>90% of national production), Andhra Pradesh	• Karnataka: Kolar Gold Fields (KGF) (shut down in 2001 due to depth & costs), Hutti Gold Mines (Raichur — only active primary gold producing mine in India today) • Andhra Pradesh: Ramagiri (Anantapur) • Jharkhand: Alluvial placer gold in Subarnarekha river sands	• Hutti produces ~99% of India's indigenous mined gold.
Diamond	Madhya Pradesh (holds >90% of diamond resources)	• Madhya Pradesh: Panna (Majhgawan pipe mine) operated by NMDC — the ONLY mechanized active diamond mine in Asia/India • Historic: Kollur Mine (Guntur/Golconda, Andhra Pradesh — origin of the Koh-i-Noor, Hope Diamond, and Daria-i-Noor)	• Kimberly pipe formations in Vindhyan conglomerates.
Uranium	Jharkhand, Andhra Pradesh	• Jharkhand: Jaduguda (East Singhbhum — India's first uranium mine, 1967), Bhatin, Narwapahar, Turamdih, Banduhurang (all run by UCIL) • Andhra Pradesh: Tummalapalle (Kadapa/YSR district — massive global-scale reserves discovered in 2011) • Meghalaya: Domiasiat / Mawthab (Kasi Hills - sandstone deposit) • Rajasthan: Rohil (Sikar)	• Operated under Department of Atomic Energy (DAE) via Uranium Corporation of India Ltd (UCIL, Jaduguda).
Thorium	Kerala, Odisha, Tamil Nadu	• Kerala: Monazite beach sands of Chavara, Kollam, and Alappuzha • Odisha: Ganjam coast (Chhatrapur beach sands) • Tamil Nadu: Manavalakurichi (Kanyakumari)	• Derived from Monazite sand (contains thorium phosphate, cerium, and rare earths). • Indian Rare Earths Limited (IREL) processes beach sand minerals. India holds the world's largest thorium reserves.
Lead & Zinc / Silver	Rajasthan (near 100% monopoly)	• Rajasthan: Zawar Mines (Udaipur — ancient & modern lead-zinc-silver), Rampura-Agucha (Bhilwara — world-class open-pit zinc deposit), Rajpura-Dariba (Rajsamand), Sindesar Khurd	• Hindustan Zinc Limited (HZL, Vedanta group). • Galena (lead ore) and Sphalerite (zinc ore). Silver is recovered as a by-product.
Mica (Ruby Mica / Muscovite)	Andhra Pradesh (1st in production), Rajasthan, Jharkhand	• Andhra Pradesh: Nellore Mica Belt (Gudur) • Jharkhand: Koderma (historically called the "Mica Capital of the World"), Giridih, Hazaribagh • Rajasthan: Bhilwara, Ajmer, Jaipur	• India was historic world leader in sheet mica (Muscovite / Ruby mica). • Essential electrical insulator resistant to high voltages.

11. Ocean Currents & Planetary Local Winds of the World

A. Major Ocean Currents (Warm vs Cold Master Matrix)

Rule of Thumb: Currents moving from Equator to Poles = Warm; Currents moving from Poles to Equator = Cold.

Ocean Basin	Warm Ocean Currents	Cold Ocean Currents	High-Yield Prelims Exam Takeaways
Atlantic Ocean	• Gulf Stream (powerful warm current) • North Atlantic Drift (keeps NW European ports ice-free year-round) • Florida Current • Antilles Current • Brazil Current (South Atlantic) • Guinea Current (West Africa)	• Labrador Current (flows from Arctic past Greenland) • Canary Current (North Africa coast) • Benguela Current (South-West Africa coast; creates Namib & Kalahari Deserts) • Falkland / Malvinas Current (Patagonia coast) • East Greenland Current	• Grand Banks of Newfoundland: Confluence of warm Gulf Stream and cold Labrador Current produces dense fog and creates world's richest cod-fishing ground. • Sargasso Sea: Clockwise North Atlantic gyre creates a calm, landless sea surrounded by currents, filled with <i>Sargassum</i> seaweed.
Pacific Ocean	• Kuroshio (Japan Current) (warm counterpart of Gulf Stream) • North Pacific Drift • East Australian Current (flows south along eastern Australia) • Alaska Current (keeps Alaskan coast ice-free) • Tsushima Current (Sea of Japan)	• Oyashio (Kurul Current) (cold current from Bering Strait) • California Current (West coast of USA) • Peru / Humboldt Current (flows along Chile & Peru; high upwelling, rich in anchovies; suppressed during El Niño) • Okhotsk Current	• Meeting of warm Kuroshio and cold Oyashio creates dense fog and prime fishing grounds off Japan. • Atacama Desert (driest non-polar desert) created by the desiccating effect of the cold Humboldt/Peru Current.
Indian Ocean	• Mozambique Current (East Africa) • Agulhas Current (South Africa) • South-West Monsoon Drift (summer)	• West Australian Current (flows along West Australia; creates Great Sandy/Victoria Deserts) • Somali Current (reverses direction seasonally; cold upwelling in summer)	• Indian Ocean currents completely reverse their direction of flow twice a year due to the seasonal reversal of the Monsoon winds.
Circumpolar	—	• West Wind Drift (Antarctic Circumpolar Current)	• Largest ocean current by water volume; circles Antarctica unbroken from West to East under roaring westerlies.

B. Famous Local Winds of the World (Hot vs Cold & Nicknames)

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Chinook	Warm & Dry (Katabatic)	Rocky Mountains (USA & Canada)	"Snow Eater" — descends eastern slopes of Rockies, melting winter snow rapidly and opening pastures for cattle.
Foehn (Föhn)	Warm & Dry	Northern slopes of the Alps (Switzerland)	Accelerates melting of alpine snow, aids grape ripening in Swiss valleys (<i>the climber's wind</i>).
Harmattan	Warm, Dry & Dusty	Sahara Desert blowing south-west to the Gulf of Guinea (West Africa)	"The Doctor" — provides welcome relief from the sultry, humid coastal tropical heat and kills pathogens.
Sirocco	Hot, Dry & Dust-laden	Sahara Desert blowing north across Mediterranean into Southern Europe (Italy/Sicily)	Causes "Blood Rain" (red Sahara dust particles falling with condensation rain in Italy and Greece).
Khamsin	Hot & Dusty	Egypt	Blows from the southern desert for approximately 50 days (<i>Khamsin</i> = fifty in Arabic).
Gibli	Hot & Dry	Libya	Sahara desert wind blowing north towards Libyan coast.
Brickfielder	Hot & Dusty	Southern Australia	Blows from the central Australian desert towards coastal cities (Melbourne/Sydney).

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Black Roller	Hot & Dust-storm	Great Plains of North America / USA	Intense summer dust squall darkening the sky.
Zonda	Hot & Dry (Foehn-type)	Eastern slopes of the Andes (Argentina)	Blows down Argentine pampas from the Andes.
Santa Ana	Hot, Dry & Gusty	Southern California (USA)	Blows from Mojave desert down coastal mountain canyons; notoriously triggers catastrophic wildfires.
Loo	Scorching Hot & Dry	Northern and North-Western Plains of India (UP, Bihar, Punjab, Rajasthan)	Blows during May–June afternoons; causes severe heatstrokes.
Nor'wester / Kalbaishakhi	Hot-season thunderstorm	West Bengal, Odisha, Assam (Bardoisila)	Violent afternoon squall in Baisakh (April–May); beneficial for early tea, jute, and aus paddy crops.
Mistral	Extremely Cold & Dry	Blows from Alps down the Rhone Valley into Gulf of Lion (France & Spain)	Winter/spring gale-force cold wind; trains often disrupted; houses in Provence face south without north windows.
Bora	Extremely Cold, Gusty & Dry	Blows from Dinaric Alps down towards the Adriatic Sea (Italy, Croatia)	Fierce northeastern catabatic katabatic wind reaching hurricane speeds in winter.
Blizzard	Freezing Cold with Blinding Snow	Canada, Northern USA, Siberia, Polar regions	Gale-force winds with sub-zero temperatures and zero visibility.
Buran / Purga	Bitterly Cold & Fierce Blizzard	Siberia & Central Asian steppes (Russia)	Buran brings icy blizzard conditions; Purga is tundra version.
Pampero	Cold & Violent Wind	Pampas of Argentina and Uruguay	Cold polar squall blowing from southwest across the pampas.

12. Structure of Atmosphere & Key Meteorological Phenomena

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
1. Troposphere	0 to 18 km (Equator: ~18 km; Poles: ~8 km)	Decreases with altitude (Lapse Rate: 6.5°C per km or 1°C per 165 m)	• Thickest at Equator due to strong vertical convective currents. • Contains ~75% to 80% of total atmospheric mass and almost ALL atmospheric water vapour and dust particles. • ALL weather phenomena (clouds, rainfall, lightning, thunder, cyclones, fog) occur EXCLUSIVELY in this layer. • Upper boundary is the Tropopause (temperature drops to -80°C over Equator and -45°C over Poles).
2. Stratosphere	From Tropopause up to 50 km	Increases with altitude (from -60°C up to 0°C at Stratopause)	• Contains the Ozone Layer (Ozonosphere) between 15 km and 35 km (absorbs harmful solar ultraviolet UV-B and UV-C radiation). • Completely free from clouds and turbulent weather; ideal condition for flying commercial supersonic jet aircraft. • High-altitude Subtropical Jet Streams blow near the tropopause. • Rare iridescent Nacreous Clouds (Mother-of-Pearl clouds) form in polar winter stratosphere.

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
3. Mesosphere	50 km to 80 km	Decreases rapidly with altitude (reaches -100°C at Mesopause)	• COLDEST LAYER OF THE ATMOSPHERE (-100°C to -130°C). • Meteors burn up in this layer upon entering from space due to friction with thin gases. • Rare Noctilucent Clouds (night-shining polar clouds composed of ice crystals on meteor dust) appear here.
4. Thermosphere / Ionosphere	80 km to 400+ km	Increases rapidly with altitude (exceeds 1,500°C due to intense solar radiation absorption)	• Contains electrically charged ionized gas particles (Ions). • Reflects terrestrial radio waves back to Earth, enabling wireless global radio communication: - D-layer (60–90 km): Reflects low-frequency radio waves, absorbs high-frequency; disappears at night. - E-layer / Kennelly-Heaviside Layer (90–130 km): Reflects medium-frequency radio waves. - F-layer / Appleton-Barnett Layer (140–400 km): Reflects high-frequency short radio waves. • Auroras (polar light displays caused by solar wind hitting charged atmospheric gases): - Aurora Borealis (Northern Lights, Arctic) - Aurora Australis (Southern Lights, Antarctic) • International Space Station (ISS) orbits in this layer (~400 km).
5. Exosphere	>400 km to outer space (~10,000 km)	Extremely high temperature, but imperceptible due to ultra-low density	• Outermost rarefied fringe of Earth's atmosphere merging into interplanetary space. • Dominated by light gases: Hydrogen and Helium. • Artificial communication and weather satellites orbit in higher exosphere.

Topic 1 — Indian Physical Geography: Mountains & Hills

Consolidated — 32 Must-Score Facts

1. India has **six** relief divisions for map questions: Northern Mountains, Northern Plains, Peninsular Plateau, Indian Desert (Thar (थार)), Coastal Plains, and Islands. The Thar is a **Pleistocene and recent** sand sheet, not an older Tertiary desert.
2. From north to south the Himalayan belts are **Trans-Himalaya → Himadri (Greater) → Himachal (Lesser) → Shiwalik**. Himadri (हिमाद्रि) is crystalline and **fossil-less**. Himachal (हिमाचल) carries **marine fossils**. Shiwalik (शिवालिक) is the outermost belt and holds **human remains**.
3. Among the usual age options, **Himadri is the youngest** Himalayan belt and **Aravalli is the oldest** fold mountain system of India. Guru (गुरु) Shikhar (**1722 m**) on Mount Abu is Rajasthan's highest peak.
4. The Kashmir Valley sits between **Pir Panjal on the south** and **Himadri on the north**, with Zaskar toward the north-west. **Karewas** are old lake-bed terraces of this valley and grow saffron. A **dun** (दून) is a different landform — a longitudinal valley between Himachal and Shiwalik.
5. Central India hills from west to east run **Satpura → Mahadeo → Maikal → Chhotanagpur**. **Vindhya** (विन्ध्य) lies **north** of the Narmada (नर्मदा) and **Satpura** (सतपुड़ा) lies **south** of the Narmada. Do not reverse them.
6. The classic state–peak match set is **Tamil Nadu (नाडु)–Doddabetta, Rajasthan–Guru Shikhar, Nagaland–Saramati**, and **Madhya Pradesh–Dhupgarh**. Also fact **Kerala–Anaimudi (अनाइमुदी)** and **Uttarakhand (उत्तराखंड)–Nanda Devi (नंदा देवी)**.
7. **Lipulekh (लिपुलेख), Niti (नीति), and Mana (माना)** are all in **Uttarakhand (उत्तराखंड)**. Lipulekh is **not** in Ladakh. Mana is **not** in Himachal. **Nathu La (नाथू ला)** is Sikkim. **Shipki La (शिपकी ला)** is Himachal.
8. Tirupati's Venkateswara temple stands on the **Tirumala / Mallamalla Hills** of the **Eastern Ghats** (पूर्वी घाट) in Andhra Pradesh. It is **not** on the Shevaroy Hills of Tamil Nadu.
9. **Intertrappean beds** between Deccan lava flows hold **land and freshwater** fossils. They do **not** hold sea plants and animals. Keep the depth figures: upper about **450 m**, middle about **1200 m**, lower about **150 m**.
10. India is the **seventh**-largest country by area (about **3.28 million km²**, roughly **2.4%** of world land). The Tropic of Cancer passes through the **middle** of the country, so India is **not** wholly tropical.
11. The **Atal Tunnel** runs under **Rohtang (रोहतांग)** in the **Pir Panjal (पीर पंजाल)** of Himachal Pradesh. The safe line is longest highway tunnel **above 10,000 ft** — not an unqualified "world's longest."
12. The Tropic of Cancer (**23°30' N**) crosses **eight states only**, west to east: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram. It does **not** cross **Uttar Pradesh (उत्तर प्रदेश)** or **Ladakh (लद्दाख)**.
13. India's Standard Meridian is **82°30' E** near **Mirzapur (मिर्ज़ापुर), Uttar Pradesh**. It passes through **five states**: Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha, and Andhra Pradesh. **IST is GMT + 5 hours 30 minutes** for the whole country.
14. **Namcha Barwa** lies in **Tibet** and is **not** an Indian peak. **K2** stands in the **Karakoram**, not on Himadri. **Kanchenjunga (कंचनजंगा)** is the highest peak **fully in India**.
15. The **Konkan** coast is a coast of **submergence**. The **Malabar** and **Coromandel** coasts are coasts of **emergence**.
16. The **Marwar (मारवाड़) Plateau (मारवाड़ पठार)** lies **east** of the Aravalli. The **Marwar Plain (मैदान) / Thar** lies **west** of the Aravalli. Do not swap the two.
17. **Palghat (Palakkad) Gap** is a **rift** and the widest break in the Western Ghats, linking Kerala with Tamil Nadu. **Anaimudi (2695 m)** is the hub of Anamalai, Palani, and Cardamom and is the highest peak of South India.
18. The snowline is **lower in the western Himalaya** than in the east. **Siachen (सियाचिन)** is the long Karakoram glacier of the Nubra belt. **Zemu** in Sikkim feeds the **Teesta**.
19. The Western Ghats are a **continuous** wall from the Tapi (तापी) gap toward Kanyakumari (कन्याकुमारी). The Eastern Ghats are **discontinuous** and lower. Nilgiri is the junction of both Ghats with the southern hills.

20. **Purvanchal Hills** are the north-eastern fold hills beyond the Dihang (दिहांग) (Patkai (पटकाई), Naga (नागा), Mizo (मिज़ो), Mishmi (मिशमी)). They are **not** the eastern Uttar Pradesh region also called Purvanchal.
21. **Garo (गारो), Khasi (खासी), and Jaintia (जयंतिया)** hills form the **Meghalaya Plateau**. They are geologically **peninsular**, not Himalayan fold ranges. Mawsynram and Cherrapunji sit on the **Khasi Hills**.
22. India's northern extreme is **Indira Col** (इंदिरा कोल) (Siachen / Ladakh). The southernmost **territory** is **Indira Point** (इंदिरा पॉइंट) on Great Nicobar. The southernmost **mainland** point is **Kanyakumari** (कन्याकुमारी). East is **Kibithu** (Arunachal); west is **Guhar Moti** (Gujarat).
23. **Gujarat** has the longest **state** coastline. Mainland plus islands is about **7516 km**. Telangana is **not** a coastal state.
24. The **10° Channel** (दस डिग्री) separates Andaman from Nicobar. The **9° Channel** (नौ डिग्री) separates Minicoy from the rest of Lakshadweep. The **8° Channel** lies between Minicoy and the Maldives. Andaman–Nicobar (अंडमान) are largely **volcanic**; Lakshadweep is **coral**.
25. The Himalayan arc is about **2400 km** from the Indus (सिंधु) gorge to the Dihang gorge. The western syntaxial bend is around **Nanga Parbat** (नंगा पर्वत). The eastern bend is around **Namcha Barwa** in Tibet.
26. A **dun** is a longitudinal valley between Himachal and Shiwalik (Dehra Dun is the classic example). **Duars** are the West Bengal–Assam foothills that open toward Bhutan. They are not the same as the UP Terai (तराई) belt.
27. **Uttar Pradesh's** highest point is **Amsot** (about **941 m**) in the Kaimur / Sonbhadra (सोनभद्र) belt. It is a Vindhyan fringe peak, **not** a Himalayan summit.
28. In Uttar Pradesh, Bhabar (भाबर)–Terai foothills appear in Pilibhit, Lakhimpur Kheri, Bahraich, and Shravasti. The core plain is the **Ganga (गंगा)–Yamuna (यमुना) Doab (दोआब)**. Bundelkhand in south-west UP is granite–gneiss and drought-prone.
29. The Standard Meridian place fact for Uttar Pradesh is **Mirzapur** (मिर्ज़ापुर). The Tropic of Cancer does **not** enter Uttar Pradesh. Eastern UP "Purvanchal" is a **plain region**, not the NE Purvanchal Hills.
30. **Annamalai** and **Sirumalai** are **Tamil Nadu peninsular** hills. They are **not** Himalayan peaks and must not sit in a Himalayan match list.
31. The Deccan Plateau is the **southern** tableland of the old peninsular block. The Central Highlands (Malwa, Bundelkhand, Baghelkhand) are the **northern** part of the same block. The plateau slopes **high in the west and low in the east**.
32. The Aravalli runs about **800 km** from Palanpur (Gujarat) toward Delhi. It is a **relict** Archaean fold belt and helps cast the rain shadow (छाया) that feeds the Thar.

Must-score facts — belts, peaks, passes

Himalayan belts (N → S)

Trans-Himalaya → Himadri (Greater) → Himachal (Lesser) → Shiwalik

Item	Lock
Youngest belt (usual options)	Himadri (हिमाद्रि)
Oldest fold system (usual options)	Aravalli (अरावली)
Kashmir Valley	Pir Panjal south ; Himadri north

Central India hills (W → E)

Satpura → Mahadeo → Maikal → Chhotanagpur · Vindhya lies **north** of Narmada

State ↔ peak

State	Peak
Tamil Nadu	Doddabetta

State	Peak
Rajasthan	Guru Shikhar (Aravalli)
Nagaland	Saramati

Passes / temple

Item	Lock
Lipulekh / Niti / Mana	All in Uttarakhand (Lipulekh ≠ Ladakh)
Venkateswara / Tirupati	Tirumala / Mallamalla — Eastern Ghats (AP)
Intertrappean beds	Land / freshwater fossils (not marine)
India area rank	7th (~3.28 million km ² ; ~2.4% world land)

Topic 2 — Climate of India

Consolidated — 30 Must-Score Facts

1. India's climate is **tropical monsoon**, not wholly tropical, because the country stretches into the subtropical north. The Himalaya (हिमालय) acts as a winter **wall** against Central Asian cold and forces **orographic** rain on southern slopes in the monsoon.
2. The word monsoon comes from Arabic **mausim** (season). About **75–90%** of India's rain falls with the south-west monsoon from June to September. Onset usually begins first in **Kerala around 1 June**.
3. IMD seasons are Cold Weather (**December–February**, Western Disturbances in the north-west), Hot Weather (**March–May**, Loo, Nor'westers, mango and blossom showers), South-West Monsoon (**June–September**), and Retreating Monsoon (**October–November**, October heat and Bay of Bengal / Andaman (अंडमान) cyclones).
4. South-west monsoon starts when the **ITCZ** shifts north and a thermal low forms over north-west India. South-east trades cross the equator near about **40–60° E**, then the **Mascarene High** and **Somali Jet** feed the two monsoon branches.
5. The **Arabian Sea branch** soaks the Western Ghats (पश्चिमी घाट) windward slope and leaves the leeward Deccan (दक्कन) dry. The Chhattisgarh basin is comparatively weak on this branch. The **Bay of Bengal branch** waters the north-east and the Ganga (गंगा) plain.
6. The Tamil Nadu south-east coast stays **dry in the south-west monsoon** because it lies parallel to the Bay branch and in the rain shadow (छाया) of the Arabian Sea branch. Its main rain comes with the **north-east monsoon** later.
7. Typical onset dates are Kerala about **1 June**, Mumbai and Kolkata about **10 June**, Delhi about **29 June**, and Rajasthan last around **mid-July**.
8. An **active** monsoon keeps the trough on the **Ganga plain** so the plains stay wet. A **break** monsoon shifts the trough onto the **Himalaya** (हिमालय), so central India goes dry.
9. On the northern plains rainfall generally **declines east to west**. The rainy season is **shorter from south to north**. Rajasthan stays dry partly because the Aravalli (अरावली) runs **parallel** to the Arabian Sea branch. A rainy day is counted from about **2.5 mm** of rain.
10. The **north-east monsoon (October–December)** is mainly for Tamil Nadu, south Andhra Pradesh, the south-east Karnataka–Kerala fringe, and Puducherry. It is **not** all-India winter rain.
11. **Western Disturbances** are Mediterranean **extra-tropical** systems that bring winter rain to the north-west. Their rain **decreases west to east**. They are **not** the retreating monsoon. Core belt: Punjab, Haryana, Himachal (हिमाचल), Jammu & Kashmir, Uttarakhand (उत्तराखंड), western Uttar Pradesh (उत्तर प्रदेश), and northern Rajasthan.
12. Mid-latitude jet streams are **westerly**. In winter the **Subtropical Westerly Jet** south of the Himalaya steers Western Disturbances after Tibet splits the jet. The northern branch does **not** drive India's winter rain.
13. The **Tropical Easterly Jet** is a summer **easterly** around about **14° N** and supports the south-west monsoon. The **Somali Jet** is the cross-equatorial feed into that monsoon.
14. **El Niño** warms the eastern Pacific off Peru, usually weakens the Indian monsoon, and **reduces** plankton because warm water suppresses upwelling. **La Niña** cools that ocean and usually strengthens the monsoon.
15. The **Indian Ocean Dipole** compares the western and eastern **Indian Ocean**, not the Pacific. A **positive IOD** helps the Indian monsoon; a **negative IOD** hurts it.
16. Köppen map for India: **Am** = Kerala / Konkan / north-east; **Aw** = Deccan; **As** = Tamil Nadu dry-summer Coromandel; **BWh** = Thar; **Cwg** = Ganga / Great Plains / north Bihar (and most of the Uttar Pradesh plain).
17. The January **18°C** isotherm is the usual tropical–subtropical divide. **Diurnal range** is highest in the Rajasthan desert. **Leh** (लेह) is among the driest class stations; Mawsynram / Cherrapunji are the wettest class.
18. The troposphere (क्षोभमंडल) is heated mainly by **long-wave** radiation from the ground. Condensation releases **latent heat**. Normal lapse rate is about **6.5 °C per 1000 m**. Relative (सापेक्ष) humidity (सापेक्ष आर्द्रता) **falls as temperature rises**. Ozone peaks in the **stratosphere** (समतापमंडल).

19. Circulation cells are **Hadley (0–30°)**, **Ferrel (30–60°)**, and **Polar (60–90°)**. **Doldrums** (डोलड्रम) mark the rainy ITCZ calm belt. **Horse latitudes** (हॉर्स अक्षांश) are the dry subtropical highs near **30°**.
20. Rain types to fact are **orographic**, **convictional**, and **cyclonic**. An **isohyet** is a line of equal rainfall.
21. **Loo** is the hot dry wind of the Indo-Gangetic plains in May–June. **Kal Baisakhi / Nor’westers** are violent pre-monsoon thunderstorms of eastern and north-eastern India in April–May.
22. **Mango showers** are pre-monsoon rains of Kerala and Karnataka. **Blossom showers** help coffee in Tamil Nadu and Kerala. **Sea breeze** blows by day toward land; **land breeze** blows by night toward sea.
23. **Anabatic** winds climb slopes by day. **Katabatic** winds drain downslope by night. **Foehn / Chinook** (चिनूक) are warm dry lee winds — not the same as cold katabatic drainage.
24. Tropical cyclones form more often over the **Bay of Bengal** than the Arabian Sea. Sea surface temperature needs about **26–27°C**. The **eye** is calm; the **eyewall** is fiercest. They do **not** form on the equator.
25. Local cyclone names: **Hurricane** (USA / Atlantic), **Typhoon / Taifu** (NW Pacific / Japan), **Baguio** (Philippines), **Willy-willies** (Australia), and generic **Cyclone** in the North Indian Ocean. Willy-willy (विली-विली) is **not** the Australian hot wind **Brickfielder** (ब्रिकफील्डर).
26. Thornthwaite’s famous line is that **vegetation is the true index of climate**. That fact is **not** Köppen’s letter-code system.
27. In Uttar Pradesh, May–June **Loo** heat waves hit the plains. Winter rain and fog track **Western Disturbances**, and **western UP is wetter than eastern UP**.
28. Uttar Pradesh’s south-west monsoon rain often arrives with **Bay of Bengal depressions**. An active spell needs the trough on the plain. Pre-monsoon **lightning** and winter **inversion smog** (west UP / NCR fringe) are key hazards.
29. Absolute (निरपेक्ष) humidity (निरपेक्ष) is the **mass of vapour** in air. Relative humidity is the **percentage of saturation**. Relative humidity falls when temperature rises even if vapour mass stays the same.
30. The north-east monsoon on the Coromandel / Chennai coast is **NE-monsoon dependent**, not a south-west monsoon belt. Do not confuse retreating-monsoon October heat with Western Disturbance (पश्चिमी विक्षोभ) winter rain in the north-west.

Must-score facts — seasons, monsoon, onset

Frame

Item	Lock
Climate type	Tropical monsoon (not wholly tropical)
Etymology	Arabic mausim (season)
SW monsoon share	~ 75–90% of annual rain

IMD seasons

Season	Months	Tag
Cold	Dec–Feb	Western Disturbances (NW)
Hot	Mar–May	Pre-monsoon / loo
SW monsoon	Jun–Sep	Main rains
Retreating / NE	Oct–Dec	TN coast rain

Onset / branch tags

Place	Typical onset
Kerala	~1 June
Mumbai / Kolkata	~10 June
Delhi	~29 June
Branch	Lock
-----	-----
Arabian Sea	Windward Western Ghats wet; leeward Deccan dry
Tamil Nadu SE coast	Dry in SW monsoon ; rain in NE monsoon (उत्तर-पूर्व मानसून)
Active vs break	Trough on Ganga plain vs shift to Himalaya foothills
Plains rain	Declines east → west

Topic 3 — Drainage System

Consolidated — 32 Must-Score Facts

1. About **77%** of India's drainage **area** faces the **Bay of Bengal**, about **23%** faces the **Arabian Sea**, and inland basins cover about **8%** of area. Over **90% of river water** still goes to the Bay of Bengal.
2. Water-share fact: **Brahmaputra about 40%, Ganga about 25%, Godavari about 6%**. Brahmaputra leads water volume; Ganga leads Indian basin area.
3. The Western Ghats (पश्चिमी घाट) are the main peninsular divide. From **Amarkantak**, drainage is radial — **Narmada** (नर्मदा) west and **Son** toward the Ganga.
4. **Ganga** is the largest basin **inside India**. **Godavari** (गोदावरी) is the largest **peninsular** basin. Among Tapi (ताप्ती) / Narmada / Mahanadi / Cauvery alone, **Mahanadi** (महानदी) is the largest basin.
5. Himalayan rivers are mostly **perennial** (melt plus rain) and include **antecedent** (पूर्ववर्ती) giants. Peninsular rivers are mostly **rain-fed** and graded.
6. Drainage types: **consequent** (अनुवर्ती) follows original slope; **subsequent** (परवर्ती) follows weak belts later; **antecedent** (Indus, Sutlej (सतलुज), Brahmaputra) cuts rising Himalaya (हिमालय); **superimposed** (अध्यारोपित) (Chambal (चंबल)) inherits an older course.
7. **Trans-Himalayan** rivers rise north of the Great Himalaya in Tibet and cut through — fact **Sutlej** (सतलुज), and also Indus and Brahmaputra (Tsangpo). Among Jhelum (झेलम) / Sutlej / Ganga / Ravi (रावी), the trans-Himalayan pick is **Sutlej**.
8. The Indus rises in a Tibet gorge and reaches the Arabian Sea in Pakistan. The five Punjab rivers east to west are **Sutlej-Beas (ब्यास)-Ravi-Chenab (चिनाब)-Jhelum**. Under the Indus Treaty, India gets Ravi, Beas, and Sutlej; Pakistan gets Indus, Jhelum, and Chenab.
9. Dam (दाम -)-river pairs: **Baglihar = Chenab**, **Pandoh = Beas** (not Ravi), **Srisailem = Krishna** (not Tungabhadra (तुंगभद्रा)).
10. The Ganga is named only at **Devprayag** (देवप्रयाग) where Alaknanda (अलकनंदा) meets Bhagirathi (भागीरथी). It is about **2525 km** long. In Bangladesh it becomes the **Padma** (पद्मा), meets the Brahmaputra as **Jamuna** (जमुना), then joins the Meghna system to the Bay of Bengal.
11. Ganga joins: Yamuna (यमुना) at **Prayagraj** (प्रयागराज), Ramganga (रामगंगा) at **Kannauj** (कन्नौज), Ghaghara (घाघरा) at **Chhapra**, Gandak (गंडक) at **Sonpur**, Son at **Patna** (पटना), Kosi (कोसी) at **Kursela**. Left-bank Himalayan tributaries include Ramganga, Ghaghara, Gandak, Kosi, and Mahananda (महानंदा); **Gomti** is a plains-origin left-bank river. Right-bank facts are Yamuna, Son, and Damodar (दामोदर).
12. Yamuna right-bank tributaries west to east are **Chambal-Sind (सिंध)-Betwa-Ken**. Chambal joins at Etawah (इटवा); Betwa at **Hamirpur**; Ken at **Banda**. Betwa does **not** meet at Prayagraj.
13. **Jonk** (जोंक) joins the **Mahanadi**, not the Ganga. **Gandak** (गंडक) is not an Uttar Pradesh river in the usual state list (राज्य सूची). **Punpun** (पुनपुन) meets near Fatuha. Chambal ravines (चंबल) start near **Mhow**.
14. Panch Prayag upstream to downstream: **Vishnu** (Dhuli (धौली)) → **Nanda** (Nandakini) → **Karn** (Pindar (पिंडर)) → **Rudra** (Mandakini) → **Dev** (Bhagirathi) = Ganga. Badrinath (बद्रीनाथ) is on the **Alaknanda** (अलकनंदा); Kedarnath (केदारनाथ) is on the **Mandakini** (मंदाकिनी).
15. The eastern **Dhauliganga** (धौलीगंगा) joins the **Kali** (काली), not the Alaknanda. Do not confuse it with the Dhuli of Vishnuprayag (विष्णुप्रयाग).
16. The Brahmaputra rises as Angsi / Tsangpo, makes a **U-turn at the eastern syntaxial bend near Namcha Barwa**, and enters India as Siang / Dihang (दिहांग). **Majuli** is the classic island. Right-bank north tributaries include Subansiri, Manas, Sankosh, and Teesta; left-bank south tributaries include Burhi Dihing and Dhansiri.
17. **Teesta** was captured from the Ganga system into the Brahmaputra. Barak (बराक) becomes Meghna and reaches the Bay of Bengal — it is **not** inland drainage.
18. Peninsular length fact: **Godavari (~1465 km) > Krishna (~1400) > Narmada (~1312) > Mahanadi (~851) > Tapi (~724)**. Godavari is also called Dakshin / Vriddha Ganga and rises near **Trimbak**.

19. Origins: Yamuna–Yamunotri (यमुनोत्री), Krishna–Mahabaleshwar, Godavari–Nashik, Mahanadi–Sihawa, Narmada–Amarkantak, Tapi–**Multai (Satpura (सतपुड़ा))** not Western Ghats, Kaveri–Brahmagiri.
20. **Pranhita** (प्रणहिता) joins **Godavari**. **Hemavati** joins Kaveri. **Malaprabha** joins Krishna. Hyderabad (हैदराबाद) stands on the **Musi**, not the Krishna stem. Ludhiana is on the **Sutlej**, not the Ravi.
21. West-flowing set: Narmada, Tapi, Mahi (माही), Sabarmati (साबरमती), **Periyar** (पेरियार). East-flowing examples include Pennar (पेन्नार) and Palar. Narmada and Tapi form **estuaries** in rift valleys; most east-flowing peninsular rivers build **deltas**.
22. **Luni** (लूनी) is inland: upper course fresh, lower course **saline** (लवणीय). The classic fault-valley tag is **Damodar** (दामोदर); Narmada and Tapi are also rift valleys.
23. Delta (डेल्टा) types: **Arcuate** = Sundarbans (सुंदरबन); **Bird's-foot** (पक्षी-पाद) = Mississippi (not Ganga). **Moribund** as a **Bengal subdivision** ≠ **Cauvery** as a **moribund lobe** match — read the stem. **Trellis** (जालनुमा) marks folded structure; **radial** marks Amarkantak; **dendritic** (वृक्षाकार) is the common homogeneous pattern.
24. National Waterways: **NW-1** Ganga–Hooghly (हुगली), **NW-2** Brahmaputra, **NW-3** Kerala, **NW-4** Krishna–Godavari, **NW-5** Brahmani–Mahanadi.
25. **Ken–Betwa** is the only National Perspective Plan link under implementation, linking Madhya Pradesh and Uttar Pradesh in Bundelkhand.
26. Uttar Pradesh points: Sangam (संगम) at **Prayagraj**; **Gomti** is Lucknow (लखनऊ)'s river and the pollution tag; **Ghaghara** (घाघरा) is Ayodhya (अयोध्या)'s river; the Ganga–Yamuna **Doab** (दोआब) is the core plain.
27. Chambal sanctuary spans **Uttar Pradesh, Madhya Pradesh, and Rajasthan** — not Haryana.
28. Kaveri island sequence includes **Srirangapatna (श्रीरंगपट्टनम)–Shivasamudram–Srirangam**. Bhubaneswar is **not** a Mahanadi-bank city fact in the usual trap set.
29. East-flowing peninsular rivers north to south often run **Subarnarekha → Mahanadi → Godavari → Krishna → Pennar → Cauvery → Vaigai**.
30. Indus is the longest of the four (चातुर्थम) Himalayan systems overall, but most of its course lies in Pakistan. Ganga is the longest river **in India**.
31. Concordant (संगत) drainage follows structure; discordant (असंगत) drainage cuts across it — Himalayan antecedents are the classic discordant case.
32. **Kosi** (कोसी) is the Sorrow of Bihar; **Damodar** is the Sorrow of Bengal. Do not swap those titles.

Must-score facts — basins, types, dam–river

Area / water share

Item	Lock
Bay of Bengal drainage area	~77%
Arabian Sea drainage area	~23%
Water volume	Brahmaputra ~40% · Ganga ~25% · Godavari ~6%
Largest basin in India	Ganga
Largest peninsular basin	Godavari
Main peninsular divide	Western Ghats (पश्चिमी घाट)
Amarkantak radial	Narmada west · Son east

Drainage types

Type	Lock
Consequent	Follows original slope

Type	Lock
Subsequent	Follows weak belts later
Antecedent	Cuts rising land (Indus, Sutlej, Brahmaputra)
Himalayan vs peninsular	Mostly perennial vs mostly rain-fed / seasonal

Dam ↔ river

Dam	River
Baglihar	Chenab (चिनाब)
Pandoh	Beas (ब्यास) (≠ Ravi)
Srisaillam	Krishna (कृष्णा) (≠ Tungabhadra)

Master Table — Panch Prayag of Uttarakhand (पंच प्रयाग)

(Tested heavily in match-the-pair: UPPCS 2023 Q50, UKPCS)

Prayag (प्रयाग)	Headstream 1	Headstream 2 (Tributary)	District / Location	Key Shrine / Geographical Landmark
1. Vishnuprayag (विष्णुप्रयाग)	Alaknanda (अलकनंदा)	Dhauliganga (धौलीगंगा - Western)	Chamoli, Uttarakhand	Nearest prayag to Badrinath & Joshimath; origin from Satopanth glacier
2. Nandaprayag (नंदप्रयाग)	Alaknanda	Nandakini (नंदाकिनी)	Chamoli, Uttarakhand	Confluence named after King Nanda; Nandakini rises from Nanda Ghunti
3. Karnaprayag (कर्णप्रयाग)	Alaknanda	Pindar (पिंडर)	Chamoli, Uttarakhand	Pindar rises from Pindari glacier (Kumaon/Garhwal border)
4. Rudraprayag (रुद्रप्रयाग)	Alaknanda	Mandakini (मंदाकिनी / Kali Ganga)	Rudraprayag, Uttarakhand	Mandakini flows down from Kedarnath (Chorabari glacier)
5. Devprayag (देवप्रयाग)	Alaknanda	Bhagirathi (भागीरथी)	Tehri Garhwal, Uttarakhand	The river is officially named GANGA (गंगा) from this confluence onward ; Bhagirathi rises from Gaumukh (Gangotri glacier)

Master Table — Tributaries of Major Indian River Systems (बाएँ एवं दाएँ तट की सहायक नदियाँ)

(High-yield for list-matching and tributary identification questions)

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Ganga (2,525 km)	Devprayag (Bhagirathi + Alaknanda)	• Ramganga (meets at Kannauj) • Gomti (origin Gomat Taal/Phulhar Jheel, Pilibhit; meets near Saidpur) • Ghaghara / Saryu (meets at Chhapra) • Gandak (meets at Sonpur) • Burhi Gandak • Kosi ("Sorrow of Bihar"; meets at Kursela) • Mahananda (easternmost left tributary)	• Yamuna (largest tributary; meets at Prayagraj Sangam) • Tamas / Tons (meets near Sirsa) • Son (radial from Amarkantak; meets at Patna/Danapur) • Punpun (meets at Fatuha)	Enters Bangladesh as Padma ; combines with Jamuna (Brahmaputra) and Meghna to form Sundarbans delta

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Yamuna (1,376 km)	Yamunotri Glacier (Bandarpunch)	• Tons (largest Himalayan tributary) • Hindon (meets near Noida/Greater Noida) • Rind, Sasur Khaderi	• Chambal (meets near Muradganj/Etawah) • Sindh (meets in Jalaun) • Betwa (meets at Hamirpur) • Ken (meets at Chilla, Banda)	Parallel to Ganga; forms fertile Ganga-Yamuna Doab
Indus (Sindhu, 2,880 km)	Bokhar Chu (near Lake Mansarovar, Tibet)	• Zaskar • Suru, Soan • Panjinad (combined flow of Jhelum, Chenab, Ravi, Beas, Sutlej)	• Shyok (with Nubra) • Gilgit • Hunza • Kabul (with Swat) • Kurram, Gomai	Trans-Himalayan; enters Arabian Sea south of Karachi
Brahmaputra (2,900 km)	Angsi / Chemayungdung Glacier (Tibet as <i>Yarlung Tsangpo</i>)	• Lohit • Dibang (Sikang) • Burhi Dihing • Dhansiri (South) • Kopili	• Subansiri (gold-bearing; largest tributary) • Kameng (Jia Bhoreli) • Manas • Sankosh • Teesta (captured from Ganga system in 1787 flood)	Syntaxial U-turn at Namcha Barwa; world's largest river island Majuli (Assam)
Godavari (1,465 km) <i>"Dakshin Ganga"</i> / <i>"Vridhha Ganga"</i>	Trimbakeshwar (Nashik, Maharashtra)	• Dharna • Penganga • Wainganga • Wardha (combined: Pranhita) • Indravati (Chitrakote Falls) • Sabari & Sileru	• Pravara • Mula • Manjra / Manjira (longest right-bank tributary, flows through Nizamabad) • Maner	Largest Peninsular river basin; forms lobate delta in Andhra Pradesh
Krishna (1,400 km)	Near Mahabaleshwar (Western Ghats, Maharashtra)	• Bhima (longest tributary, 861 km) • Dindi • Peddavagu • Musi (Hyderabad sits on its banks) • Paleru, Munneru	• Venna • Koyna (Shivsagar reservoir) • Panchganga • Dudhganga • Ghataprabha (Gokak falls) • Malaprabha • Tungabhadra (Tunga + Bhadra; largest basin tributary)	Second largest east-flowing peninsular river; major dams: Almatti, Srisailem, Nagarjuna Sagar
Cauvery / Kaveri (800 km) <i>"Ganga of the South"</i>	Talakaveri, Brahmagiri Range (Kodagu, Karnataka)	• Harangi • Hemavati • Shimsha • Arkavathi	• Lakshmantirtha • Kabini • Suvarnavathi • Bhavani • Noyyal • Amaravati	Perennial due to dual monsoons (SW monsoon in upper basin + NE monsoon in lower basin); islands: Srirangapatna, Shivanasamudra, Srirangam
Mahanadi (851 km)	Sihawa highlands (Dhantari, Chhattisgarh)	• Seonath (longest tributary, 383 km) • Hasdeo • Mand • lb	• Jonk (UPPCS repeated trap: Jonk joins Mahanadi, NOT Ganga!) • Ong • Tel	Hirakud Dam (longest earthen dam); drains Chhattisgarh and Odisha into Bay of Bengal

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Narmada (1,312 km)	Amarkantak Plateau (Anuppur, MP)	• Sher • Shakkar • Dudhi • Tawa (longest tributary) • Ganjal, Kundi	• Hiran • Barna • Choral • Man, Uri, Orsang	Flows through rift valley between Vindhyas (north) & Satpuras (south); Dhuandhar falls; Sardar Sarovar Dam; forms estuary (no delta)
Tapi / Tapti (724 km)	Multai (Betul district, Satpura Range, MP)	• Purna (main tributary; drains Vidarbha) • Girna • Panjhra • Bori, Waghur	• Suki • Gomai • Arunavati	Parallel rift valley south of Satpuras; Ukai & Kakrapar dams; forms estuary at Surat

Master Table — Major Mountain Passes of India (भारत के प्रमुख पर्वतीय दर्रे)

(High-frequency match-the-pair: Pass ↔ State / Mountain Range)

Pass Name (दर्रा)	State / Union Territory	Mountain Range	Strategic / Historical Connection
Zoji La (ज़ोजिला)	Ladakh (UT)	Great Himalayas	Connects Srinagar to Leh ; Zojila Tunnel under construction
Karakoram Pass	Ladakh (UT)	Karakoram Range	Highest pass in India (~5,540 m); ancient silk route to Tarim Basin / Yarkand
Bara-lacha La (बारालाचा)	Himachal Pradesh	Zaskar Range	Connects Mandi/Manali to Leh ; source of Bhaga river (Chenab)
Shipki La (शिपकी ला)	Himachal Pradesh	Zaskar Range	Sutlej river enters India from Tibet through this gorge; India-China border trade post
Rohtang Pass (रोहतांग)	Himachal Pradesh	Pir Panjal Range	Connects Kullu Valley with Lahaul and Spiti; bypassed by Atal Tunnel (9.02 km)
Mana Pass (माना दर्रा) / <i>Dungri-La</i>	Uttarakhand	Greater Himalayas	Located in Nanda Devi Biosphere; connects Uttarakhand with Tibet; near Badrinath
Niti Pass (नीति दर्रा)	Uttarakhand	Greater Himalayas	Ancient trade route between Uttarakhand (Chamoli) and Tibet
Lipulekh Pass (लिपुलेख)	Uttarakhand (Pithoragarh)	Kumaon Himalayas	Tri-junction of India, Nepal, and Tibet ; pilgrimage route for Kailash-Mansarovar Yatra
Nathu La (नाथुला)	Sikkim	Dongkya Range	Ancient Silk Route to Lhasa (Chumbi Valley, Tibet); reopened for border trade in 2006
Jelep La (जेलेप ला)	Sikkim	Greater Himalayas	Passes through Chumbi Valley connecting Sikkim/Kalimpong with Lhasa (Tibet)
Bomdi La (बोमडिला)	Arunachal Pradesh	Greater Himalayas	Connects Arunachal Pradesh (Tawang) with Lhasa (Tibet)
Diphu Pass (दिफू दर्रा)	Arunachal Pradesh	Mishmi Hills	Tri-junction border of India, China, and Myanmar
Thal Ghat (थाल घाट) / <i>Kasara Ghat</i>	Maharashtra	Western Ghats (Sahyadri)	Links Mumbai to Nashik & Kolkata (NH-3 & Central Railway)
Bhor Ghat (भोर घाट)	Maharashtra	Western Ghats (Sahyadri)	Links Mumbai to Pune & Chennai (Mumbai-Pune Expressway & rail line)
Palghat Gap (पालघाट)	Kerala & Tamil Nadu	Between Nilgiris & Anaimalai	Major gap connecting Palakkad (Kerala) with Coimbatore (Tamil Nadu)
Shencottah Gap (शेनकोट्टा)	Kerala & Tamil Nadu	Between Cardamom & Agasthyamalai	Connects Kollam (Kerala) with Madurai (Tamil Nadu)

Topic 4 — Lakes, Waterfalls & Water Resources

Consolidated — 40 Must-Score Facts

1. First classify a lake as lagoon, inland saline, freshwater, crater, glacial, oxbow, or artificial. Origin pairs: **Wular** tectonic, **Chilika** lagoon, **Lonar** crater, **Sambhar** wind / playa, **Kabartal** oxbow.
2. **Chilika** is the largest **east-coast lagoon** (Odisha / Northern Circars). If a question says "largest saline" and omits "inland," Chilika is often the intended answer.
3. **Sambhar** is the largest **inland** saline lake (Rajasthan). Sister Rajasthan saline names include Didwana, Kuchaman, Sargol, and Khatu.
4. **Wular** is the largest-freshwater volume fact (Jammu & Kashmir, on the Jhelum (झेलम), tectonic). **Dal** is the tourism lake. **Roopkund** is Uttarakhand (उत्तराखंड)'s Mystery / Skeleton Lake.
5. **Loktak** (Manipur) holds **phumdis**, **Keibul Lamjao**, and the Sangai (सांगाई). **Kabartal** is Bihar's oxbow Ramsar lake.
6. **Pulicat** is the second major brackish lagoon (Andhra Pradesh–Tamil Nadu (नाडु), Sriharikota bar). **Kolleru** is mainly **freshwater**, not a lagoon king.
7. **Vembanad** is Kerala's largest lake and India's **longest**; it is a **west-coast** kayal / Ramsar, not India's overall largest lake. **Kayal** means a Kerala coastal lagoon / backwater.
8. **Phulhar** (Pilibhit) is linked to the Gomti source. **Kodaikanal** lake is artificial. Periyar (पेरियार) Lake is **not** a coastal lagoon.
9. World lake traps: **Onega** and **Ladoga** are in **Russia** (not Canada). Michigan is USA-only; Superior is USA and Canada.
10. Rudrasagar is **Tripura**; Sultanpur is **Haryana**; Surinsar–Mansar is **Jammu & Kashmir** — none of these is an Uttar Pradesh (उत्तर प्रदेश) Ramsar.
11. India has more than **17%** of world people, about **4%** of water, and about **2.45%** of land. Precipitation is about **4000 BCM**; available water about **1869 BCM**; utilisable about **1122 BCM** (surface ~690 + groundwater ~433).
12. The Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)–Barak (बराक) belt is about **one-third** of area but about **60%** of surface water. Irrigation takes about **89%** of surface water use and about **92%** of groundwater use.
13. Groundwater now covers more than **60%** of irrigated area. Canals dominate the Indo-Gangetic plains; tanks matter on the peninsula. Tamil Nadu leads south-Indian groundwater utilisation.
14. Quality points: **arsenic** in the Ganga belt; **fluoride** in Rajasthan and parts of Maharashtra; **usar** from over-irrigation in the north-west; Yamuna (यमुना)'s Delhi–Etawah (इटवा) stretch is the classic dirty stretch.
15. National Water Policy 2012 puts **drinking first**. PMKSY means har khet plus more crop per drop. **Atal Jal** covers seven states including **Uttar Pradesh** (उत्तर प्रदेश). Ken–Betwa (बेतवा) is the live interlinking project.
16. **Kunchikal** on the **Varahi** is the usual UPPCS **highest** waterfall key. **Nohkalikai** is the tallest **plunge**. **Jog** on the **Sharavati** is famous for **width**, not height.
17. **Shivanasamudra** on the **Cauvery** is the volume king. **Kapildhara** is on the **Narmada** (नर्मदा), not the Godavari (गोदावरी). Sharavati is a **west-flowing** Karnataka river, not a Cauvery tributary.
18. Match pairs: **Dudhsagar–Goa (गोवा) (Mandovi)**; Barkana–Karnataka; Khandadhar–Odisha; Palani–Himachal (हिमाचल) Pradesh.
19. **Hundru** is on the Subarnarekha. **Lodh / Budha Ghagh** is on the **Burha** (Jharkhand's highest), not the Kanchi (कांची). **Dhuandhar** is Narmada at Bhedaghat. **Chitrakote** is on the Indravati.
20. **Chachai–Bihad** is a wrong pair. **Bhalkund** is on the **Bina**, not the Narmada. Other falls: Amritdhara–Hasdeo, Chulia–Chambal (चंबल), Duduma–Machhkund, Gokak–Ghataprabha.
21. **Bhakra–Nangal** is on the **Sutlej** (सतलुज) with **Gobind Sagar** (joint **PB–HR–RJ**). **Tehri** is the tallest rock-fill dam on the **Bhagirathi** (भागीरथी) (+ Bhilangana confluence). **Hirakud** on the **Mahanadi** (महानदी) in Odisha is the longest-earthen narrative (1948–57).
22. **Indira Sagar** (Narmada Sagar) on the Narmada in **Madhya Pradesh** is the volume king (**Harsud** submerged). **Sardar Sarovar** (सरदार सरोवर) is Narmada in **Gujarat** (~163 m, ~1450 MW; beneficiaries GJ–RJ–MH–MP). **Ban**

- Sagar** is on the **Son**, not Narmada. **Idukki** is an arch dam on the **Periyar** (पेरियार). Koyna is linked to reservoir-induced seismicity in **1967**.
23. **Nagarjuna Sagar** (नागार्जुन सागर) and **Srisailem** are both on the **Krishna** (कृष्णा). **Tulbul** is on the Jhelum / Wular. **Baglihar** / **Dulhasti** are on the Chenab (चिनाब). **Pandoh** / **Pong** are on the Beas (ब्यास).
 24. Ramganga project points: length about **825.8 m**, power about **198 MW**, Garhwal (गढ़वाल) location; irrigation is **not** a neat **6 lakh ha**.
 25. **Indira Gandhi (गांधी) Canal** (इंदिरा गांधी नहर) is India's **longest** canal (Harike Barrage; Sutlej–Beas–Ravi (रावी) waters; western Rajasthan). **Gang Canal** (1927, Maharaja (महाराज) Ganga Singh) is among the oldest developed systems. Upper Ganga Canal offtake is at **Bhimgoda / Haridwar (हरिद्वार)**; Lower Ganga Canal at **Narora**.
 26. Himalayan hydropower potential is huge. Eastern Uttar Pradesh and Bihar cannot tap the ranges to their north because those ranges lie in **Nepal**.
 27. India–Bhutan hydro names to fact: **Chukha**, Tala (ताल), Kurichhu, and **Mangdechhu / Mungdechhu**.
 28. **Hokera** is in Jammu & Kashmir, not Punjab. **Sasthamkotta** is Kerala. **Renuka** is Himachal. **Yaya Tso** is a Ladakh (लद्दाख) biodiversity heritage site.
 29. A **dam stores** water; a **barrage diverts** it (**Farakka** 1975 diverts about **40,000 cusec** toward the Hooghly (हुगली) — not a storage dam).
 30. Oxbow lakes are cut-off meanders on floodplains; lagoons are barred coastal water bodies — Chilika is a lagoon, not an oxbow.
 31. Lonar is a **meteorite crater** in Maharashtra basalt, not a volcanic caldera and not a tectonic basin like Wular.
 32. India now has **100 Ramsar** sites; Uttar Pradesh has **13**, with **Surha Tal / JP Narayan Bird Sanctuary, Ballia** as the 100th site.
 33. Irrigation project size by **Culturable Command Area (CCA)**: **minor** up to **2,000 ha** (~62% of irrigation potential); **medium** **2,000–10,000 ha**; **major** above **10,000 ha** (~38% via major + medium together).
 34. **2018–19** source share (provisional): **tubewells ~48.5%**; wells + tubewells together **~64%**; **canals ~23%**; **tanks ~2.3%**. Wells/tubewells — not canals alone — are the dominant national source.
 35. **Garland Canal** idea = Captain **Dinshaw J. Dastur**. First river-linking thought often tagged to **M. Visvesvaraya**; later **K.L. Rao. Hariyali** (27 Jan **2003**) = watershed / rural water storage. **PMKSY** approved **1 July 2015**.
 36. Peninsula irrigation classic = **tanks / ponds** (hard rock + seasonal rivers). Groundwater resources available for irrigation among major states: **UP** leads (about **40.7 bcm**), then Punjab, Madhya Pradesh, Maharashtra.
 37. **DVC (1948)** is the first multipurpose valley project of independent India; Damodar (दामोदर) = **Sorrow of Bengal**. Tilaiya / Maithon on **Barakar**; Panchet on Damodar.
 38. Oldest hydro = **Sidrapong (Darjeeling, 1897)**; second = **Shivasamudram (Cauvery, 1902)**. Cauvery dispute parties = **TN–KA–KL–Puducherry**.
 39. Chambal chain = **Gandhi Sagar (MP) + Rana Pratap (प्रताप) Sagar + Jawahar Sagar (RJ)**. Telugu Ganga = Krishna water to **Chennai**. Nehru (नेहरू) = **“temples of modern India.”**
 40. **Narmada Bachao (नर्मदा बचाओ) Andolan (नर्मदा बचाओ) / Medha Patkar (मेधा पाटकर)** opposed SSP height. Narmada plan scale ≈ **30 major / 135 medium / 3000 small**.

Must-score facts — lakes, lagoons, Ramsar traps

Lake ↔ type / state

Lake	Lock
Chilika	Largest east-coast lagoon (Odisha)
Sambhar	Largest inland saline (Rajasthan)
Wular	Largest freshwater volume (J&K; Jhelum; tectonic)

Lake	Lock
Loktak	Phumdis; Keibul Lamjao; Sangai (Manipur)
Pulicat	2nd major brackish lagoon (AP–TN)
Kolleru	Mainly freshwater (AP)
Vembanad	Kerala's largest; India's longest ; west-coast kayal
Kabartal	Bihar oxbow Ramsar
Phulhar	Pilibhit; Gomti source link
Kodaikanal	Artificial

Not in UP (common traps)

Rudrasagar (Tripura) · Sultanpur (Haryana) · Surinsar–Mansar (J&K)

World lake traps

Lake	Lock
Onega / Ladoga	Russia (≠ Canada)
Michigan	Wholly USA
Superior	USA + Canada

Topic 5 — Soils

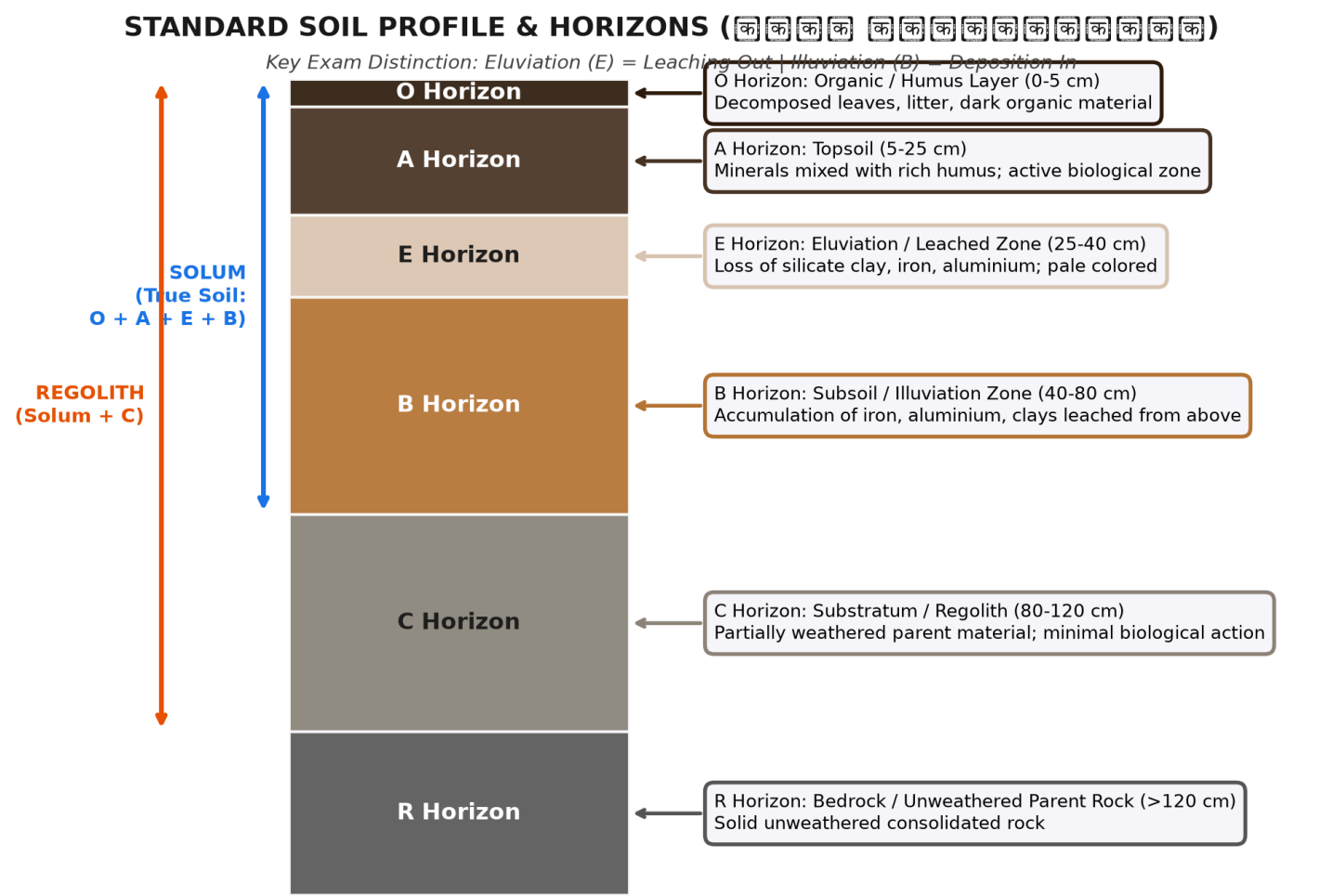
Consolidated — 28 Must-Score Facts

1. Ancient Indian names: **Urvara** = fertile and **Usara** = sterile. Jenny's five factors are Climate, Living organisms, Relief, Parent rock, and Time (**CLORPT**).
2. **Edaphic** means a **soil** factor — not climate, biotic, or topography. Residual / sedentary soils form on bedrock (regur, red). Transported / azonal soils are the alluvium of the plains. 2a. World zonal pairs: **Podzol** = cold temperate woods; **Chernozem** = temperate steppe (स्टेपी); **Laterite** (लेटराइट) = hot humid; **Terra Rossa** = limestone/dolomite weathering; **Sierozem** = mid-latitude desert. 2b. Capillary action strongest in **clay** (Clay > silt/loam > sand). Halophytes = saline soils. Contour bunding = **hill slopes**, not desert margins or flood plains.
3. Profile downward is **O → A (topsoil) → E → B → C → R**. **Eluviation** washes material **out** of A/E; **illuviation** deposits it **into B**. Young khadar (खादर) has weak horizons; mature residual soils show clear horizons.
4. The usual eight-type teaching set is Alluvial, Black / Regur, Red & Yellow, Laterite, Arid, Saline / Usara, Peaty, and Forest / Mountain.
5. Parent material pairs: crystalline rock → red; Cuddapah–Vindhyan → calcareous / clay; Gondwana → immature poor; Deccan (दक्कन) Trap (दक्कन ट्रैप) → regur; extra-peninsular sediment → alluvium. Climate can override rock — granite becomes laterite if wet, or sandy if arid.
6. **Pedocal** soils accumulate lime in arid tracts such as Rajasthan. Himalayan soils develop better on gentler **north-facing** slopes; many south faces are too steep.
7. **Alluvial** (जलोढ़) is India's **largest** soil group (about **40%** in the Class 11 frame). It is rich in potash, **poor in phosphorus**, low in nitrogen–humus, and supports over **40%** of the people. Uttar Pradesh (उत्तर प्रदेश) is about **90%** alluvial. Do not crown red as the largest group.
8. **Khadar** (खादर) is new, fine, and fertile. **Bangar** is older, coarser, and has **more kankar**. **Bhur** is sandy ridges of the western Gangetic belt. **Rarh** is older West Bengal / Jharkhand fringe upland — not active khadar.
9. **Bhabar** (भाबर) (about 8–16 km) is pebble fans where streams **disappear**. **Tarai** (about 15–30 km) lies south, where streams **re-emerge**; it is moist and nitrogen-rich but phosphorus-poor.
10. **Black / regur** soils sit mainly on Deccan and Rajmahal Trap (fissure volcanic); Tamil Nadu (नाडु) black may sit on gneiss / schist. Clay is often **≥62%**, montmorillonite is the key clay, and cotton is the crop. Aliases include Tropical Black Earth / Tropical Chernozem. Black soil (काली मिट्टी) is **not** Himalayan.
11. Keep black soil area about **16.6% / ~5.46 lakh km²** — not a "46 lakh km²" slip. Valley black is more fertile than upland black. Sticky when wet and cracked when dry. Jharkhand black points to **Rajmahal**.
12. **Red / yellow** soils form on crystalline granite–gneiss–quartzite. Colour comes from **ferric oxides** / iron diffusion. They are **not** laterite and are **not** India's largest soil. Upland gravelly red is poor; lower-slope deep red is fertile.
13. **Laterite** was named by Buchanan in 1807 from *later* (brick). It needs high temperature, heavy rain, and a wet–dry rhythm. It is **rich in iron and aluminium**. Strongest on the Western Ghats (पश्चिमी घाट) / Malabar (Kerala), then Maharashtra. Crops: cashew, **tapioca**, tea, coffee, rubber — not Rajasthan or Uttar Pradesh plains.
14. **Desert / arid** soils are about **90–95%** aeolian sand, often **calcareous** in western Rajasthan, with phosphate roughly like alluvium and nitrogen mainly as nitrates. Sand is blown from the Indus (सिंधु) basin and coasts by south-west winds.
15. **Saline / usara** soils carry sodium–potassium–magnesium salts and **reh**. The **Rann** and Gulf of Khambhat mouths are classic crust belts. Gujarat usually leads salt-affected area. Remedy for alkali / sodic soil is **gypsum**, not lime. Acid soils take **lime**.
16. **Peaty / kari** soils of Kerala (Kottayam–Alappuzha / Kuttanad), plus Sundarbans (सुंदरबन) and north Bihar pockets, grow paddy after rains. Their black colour is **not** Deccan cotton regur.
17. **Forest / mountain** soils cover about **8.67%**. Himalaya (हिमालय) soils on the **north face** and Ghats plantation soils often hold rich humus. Mountain soils are thin on steep slopes and loamy in valleys.

18. **Karewa** (करेवा) soils are Kashmir **lacustrine** terraces famous for saffron, almond, and apple — not ordinary Gangetic alluvium.
 19. Texture: sand drains, clay holds, loam is best for crops. Clay particles are smaller than **0.002 mm**. Among copper, iron, manganese, and zinc, **zinc** is the most often deficient micronutrient.
 20. Water erosion ladder: **Splash** → **Sheet** → **Rill** → **Gully** → **Ravine / badland**. Classic badlands sit on the Chambal (चंबल) and the Agra–Etawah (इटवा)–Jalaun belt. Wind / deflation rules the Thar (थार); coastal wave erosion hits Kerala–Tamil Nadu–Odisha–West Bengal–Gujarat coasts.
 21. **Jhum** (shifting cultivation) drives erosion in the north-east and parts of the Ghats. Conservation: contour plough / bund / trench, terrace on steep slopes, strip cropping, **shelter belts** against wind, **check dams** for gullies, mulch / rotation, and **zero tillage plus crop residue**.
 22. **Mitti Bachao** is linked to **Hoshangabad / Narmadapuram, Madhya Pradesh**. World Soil Day is **5 December**. The Central Soil Conservation Board prepares conservation plans.
 23. Uttar Pradesh local names: **Mar / Kabar** = Bundelkhand **black**; **Parua / Rakar (pathari)** = **red**; **Bhonta / Monta** = Vindhyan; **Usar / Reh / Kallar / Thur** = western–central canal tracts.
 24. Tea prefers **slightly acidic**, well-drained soils — not black cotton as a default. Canal irrigation without drainage can create **usar** even on the Indira Gandhi (गांधी) Canal (इंदिरा गांधी नहर) belt.
 25. **Terra Roxa** is Brazil's basaltic coffee soil and **Selva** (सेल्वा) is the Brazilian rainforest — not Indian regur labels.
 26. Earthworms are friends of farmers and raise fertility. The point is that they do **not** have two pairs of hearts.
 27. Climate can make the same granite lateritic in wet belts or sandy in arid Rajasthan. Tamil Nadu can hold black soil even on gneiss under hot–dry conditions.
 28. Saline and sodic are not the same: saline means soluble salts; sodic means exchangeable sodium with high pH. Use gypsum for sodic / **usar** and lime only for acid soils.
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Must-score facts — profile, types, doab (दोआब) belts

Profile / terms



Item	Lock
Profile (down)	O → A → E → B → C → R
Eluviation	Washes out of A/E
Illuviation	Deposits into B
Edaphic	Soil factor
Urvara / Usara	Fertile / sterile (ancient names)
Pedocal	Lime accumulation in arid tracts

Soil type tags

Soil	Lock
Alluvial	Largest (~ 40%); potash-rich; poor in P & N
Khadar	New, fine, fertile
Bangar	Older; more kankar
Bhur	Sandy doab ridges
Bhabar	Pebble fans; streams disappear (~8–16 km)
Tarai	South of Bhabar; streams reappear (~15–30 km)
Black / regur	Deccan Trap; cotton soil

Soil	Lock
Laterite	High rain leaching; brick-like
Red & yellow	Crystalline parent

Topic 6 — Agriculture

Consolidated — 34 Must-Score Facts

1. **Net sown area (NSA)** is land sown at least once a year. **Gross cropped area (GCA)** counts every sowing. **Cropping intensity** = $GCA / NSA \times 100$ (about **111%** in 1950–51 to about **156%** now). About **86%** of holdings are small or marginal, and about half of NSA is still rainfed.
2. Groundwater covers about **64%** of irrigated area (**2018–19**: tubewell ~**48.5%**; canals ~**23%**; tanks ~**2.3%**). Canals suit clayey plains; tube wells dominate north-west and western Uttar Pradesh (उत्तर प्रदेश); tanks suit the peninsula; drip suits horticulture. Minor projects ($CCA \leq 2,000$ ha) create ~**62%** of irrigation potential.
3. **Kharif** (खरीफ) (June–October) includes rice, maize, jowar (ज्वार), bajra (बाजरा), ragi, cotton, jute, groundnut, soy, and tur. **Rabi** (रबी) (October–March) includes wheat, barley, gram, mustard, peas, and linseed, helped by Western Disturbances. **Zaid** (जायद) (March–June) covers melons, cucumber, fodder, and vegetables. Cane is long-duration; tea, coffee, and rubber are perennial.
4. Rice needs hot-wet conditions (often above **20–27°C** and above **100 cm** rain) on clayey alluvium. Methods include transplant, broadcast, drill, **DSR**, and **AWD**. Aus / Aman / Boro are eastern season names. **Azolla** is a biofertiliser. Golden rice carries **Vitamin A**.
5. West Bengal often leads rice **volume**; Punjab leads **yield**. India is usually the world's **second** rice producer after **China**. In **2024–25**, the Union Agriculture Minister claimed India had become **first** — that is a separate recent claim, not the usual textbook rank.
6. Wheat needs cool growing weather and bright ripening, about **50–75 cm** rain, and loam soils. It is a **rabi** crop with two belts: Ganga (गंगा)–Satluj plains and Deccan (दक्कन) black soil (काली मिट्टी). Assam–Wheat is a wrong pair. India is usually world **number two**.
7. Millet leaders: **Jowar = Maharashtra, Bajra = Rajasthan, Ragi = Karnataka**. Maize is food plus feed and mostly kharif. International Year of Millets 2023 marketed millets as **Shree Anna**.
8. Pulses fix nitrogen through **Rhizobium**. Gram is rabi; tur, moong, and urad are kharif. India is the world's top pulse producer.
9. Oilseed leaders: **Groundnut = Gujarat, Mustard = Rajasthan** (rabi), **Soybean = Madhya Pradesh** (kharif). The **Yellow Revolution** (पीली क्रांति) is oilseeds — not the Golden Revolution.
10. Cotton (**white gold**) needs above about **21°C**, **50–100 cm** rain, about **210 frost-free days**, and preferably black soil. It is **kharif**, lasts about **6–8 months**, is about two-thirds (ते-भागा) rainfed, hates waterlogging, and has three zones (north alluvial (जलोढ़), central black, south mixed). India grows all four (चातुर्याम) species (*arboreum*, *herbaceum*, *hirsutum*, *barbadense*); bulk is *hirsutum* / Bt; crop is mostly **medium** staple, not Egyptian ELS.
11. Ahmedabad (अहमदाबाद) lies in the cotton belt, but India's traditional largest textile mill centre is **Mumbai**, not Ahmedabad.
12. Jute (**golden fibre**) needs about **25–35°C**, above **150 cm** rain, alluvial soil, and standing water for **retting**. India grows **white** (*C. capsularis*) and **tossa** (*C. olitorius*); **mesta** is the drier-area allied fibre. The state leader is **West Bengal**. Uttar Pradesh–Jute is wrong. India is world number one in jute.
13. Sugarcane needs about **21–27°C** and **75–150 cm** rain or irrigation. **Uttar Pradesh leads quantity; Maharashtra leads productivity** and cooperatives. Frost and Loo (लू) hurt the north; the south is frost-free. Heavy rain lowers sugar; dry stress makes fibre. Price policy is **FRP** (plus SAP), not cereal MSP. Ratoon means a crop from stubble.
14. Tea is a **plantation** crop needing about **20–30°C**, **150–300 cm** rain, and slopes — Assam, West Bengal, and Nilgiri (नीलगिरि). Coffee needs shade and is strongest in **Karnataka**, then Kerala and Tamil Nadu (नाडु).
15. Rubber needs about **25–35°C** and above **200 cm** rain; the correct state pair is **Kerala**. Pepper (काली मिर्च) and cardamom also concentrate in **Kerala**. World citrus is a Mediterranean specialty.
16. Potato leadership is with **Uttar Pradesh** (उत्तर प्रदेश). CIP-SARC is at **Agra (Singna)**, not Aligarh (अलीगढ़). **Sultana, Gulabi and Kali (काली) Champa (चम्पा)** are **guava** varieties — not grapes.

17. Planning Commission (योजना आयोग) agro-climatic zones = **15**; NARP zones about **127**; agro-ecological regions about **20**. The Trans-Gangetic belt is the classic Green Revolution wheat–rice zone.
18. **CACP recommends** MSP; the **Cabinet decides**. Mandated MSP crops are **22**. Cane uses FRP, not the cereal MSP schedule.
19. Green Revolution = HYV seed + water + fertiliser (Lerma Rojo, Sonora 64, IR-8) in Punjab–Haryana–western Uttar Pradesh. Costs include groundwater crash, monoculture, millet/pulse neglect, and the stubble window. **BGREI** targets eastern rice systems. Norman Borlaug’s Nobel is for **Peace**.
20. **M.S. Swaminathan** (एम.एस. स्वामीनाथन) is the Green / Evergreen face. **Verghese Kurien** is White Revolution (श्वेत क्रांति) / Operation Flood (ऑपरेशन फ्लड) / NDDB / Amul — not milk for Swaminathan.
21. Revolution colours: **Golden** = horticulture + honey; **Grey** = fertiliser; **Yellow** = oilseed; **Black** = petroleum; **Silver** (रूपक) = egg; **Pink** = onion / meat / prawn; **Rainbow** = integrated; **Evergreen** = sustainable agriculture; **Blue** = fish (**Hiralal Chaudhuri**).
22. Institutions: **ICAR** Delhi; **IARI** Pusa; seed chain **Breeder** → **Foundation** → **Certified**; **KVK** at district level. First agri university = **Pantnagar (1960)**. **Agmark** = Act **1937**. FAO grain moisture ≤ **14%**.
23. **Jhum** is north-east shifting cultivation (also podu / bewar / dahiya / kumari / waltre). Organic farming bans synthetics; **Sikkim** is the first fully organic state. Precision farming is GIS / GPS site-specific management.
24. India usually ranks **first** in milk, pulses, and jute, and often **second** in rice, wheat, and cane.
25. Red rot of cane is a **fungus**; citrus canker is a **bacterium**. Command Area Development improves an existing command — it is not a new dam (बांध).
26. Green Revolution costs include monoculture and groundwater stress in Punjab–Haryana. Bio-decomposer sprays are the free fungal stubble-management tool in recent papers.
27. Amartya Sen’s food thesis is **entitlements**, not Swaminathan’s breeding story.
28. Lift irrigation matters where southern canal layouts are irregular. Drip and sprinkler raise water-use efficiency in horticulture and dry tracts.
29. Cotton is **kharif**; wheat is **rabi** — never swap those seasons. Cane is long-duration and straddles seasons.
30. World coffee order often taught for 2016 is Brazil > Vietnam > Colombia > Indonesia. Do not put Gujarat–Tea or Assam–Wheat in a correct-pair list.
31. Mixed farming = crops + livestock. Double cropping = two crops in one year. Parallel cropping classic = **wheat + mustard**. Contract farming pioneer = **Punjab**.
32. Green manure N% fact: **cowpea highest** among common options; sunhemp often max kg N/ha. Fertigation avoids rock / super phosphate. Conservation agri = min till + residue + rotation.
33. Cane points: Sugar Bowl **UP**; breeding **Coimbatore**; first mill **Pratappur 1903**; SSI = WWF (डब्ल्यूडब्ल्यूएफ)–ICRISAT **2009**. Rice Bowl of India = **Krishna (कृष्णा)–Godavari (गोदावरी) delta (डेल्टा)**.
34. Board HQ: Coffee **Bengaluru**, Tea **Kolkata**, Rubber **Kottayam**, Tobacco **Guntur**. History (इतिहास) of Indian Agriculture = **M.S. Randhawa**.

Must-score facts — seasons, crop–state, irrigation

Land-use terms

Term	Lock
NSA	Land sown ≥ once a year
GCA	Counts every sowing
Cropping intensity	(GCA/NSA) × 100
Irrigation share	Groundwater ~ 64% (tubewell heavy)

Season crops

Season	Lock
Kharif (Jun–Oct)	Rice, maize, millets, cotton, jute, groundnut, soy, tur
Rabi (Oct–Mar)	Wheat, barley, gram, mustard, potato

Crop ↔ leader / need

Crop	Lock
Rice	WB volume often; Punjab yield; hot-wet; clayey alluvium
Wheat	Cool grow / bright ripen; ~50–75 cm; rabi
Jowar / Bajra / Ragi	MH / RJ / KA
Groundnut / Mustard / Soy	GJ / RJ (rabi) / MP (kharif)
Cotton	~21°C+; 50–100 cm; ~210 frost-free days; black soil preferred
Pulses	Rhizobium N-fix; India top producer

Topic 7 — Natural Vegetation & Biodiversity Geography

Consolidated — 36 Must-Score Facts

1. "Vegetation is the true index of climate" is **Thornthwaite** (थॉर्नथ्वेट)'s line, not Köppen (कोपेन)'s letter codes. Controls are climate, soil, relief, biotic factors, and time.
2. India is a **mega-biodiversity** country. The four (चातुर्यम) hotspots touching India are **Himalaya** (हिमालय), **Western Ghats** (पश्चिमी घाट)—**Sri Lanka**, **Indo-Burma**, and **Sundaland** (**Nicobar** (निकोबार), not Andaman (अंडमान)). Do not split Himalaya into east and west as if those were the four.
3. Rain ladder: tropical evergreen **>200 cm**; moist deciduous **100–200 cm** (largest area, about **37%**); dry deciduous **70–100 cm** (about **28%**); thorn below about **70–75 cm**. Semi-evergreen is the mix zone. Some notes put wet evergreen above **250 cm** — both figures appear in standard texts.
4. Champion–Seth teaching heads group into five major classes and about **16** sub-types (moist tropical, dry tropical, subtropical montane, temperate montane, alpine).
5. **Dry evergreen** of the Tamil Nadu (नाडु) Coromandel grows on about **100 cm** of **north-east monsoon** with jamun–neem–tamarind. It is not Western Ghat wet rainforest.
6. **Chir pine** belts the western Himalaya at about **1000–2000 m**. Alpine moist belts carry rhododendron–birch; dry alpine juniper sits above about **3500 m**. **Shola** forests are South Indian hill patches.
7. Species diversity (प्रजाति विविधता) **decreases from equator to poles**. Richness is not the same as evenness. Alpha is within one ecosystem (पारिस्थितिकी तंत्र), beta between ecosystems, gamma across a region.
8. India has two realms (Palearctic Himalaya + Indomalayan rest), five biomes, **ten** biogeographic zones, and about **25** provinces.
9. Evergreen forests of the Western Ghats, north-east, and Andamans are multilayered with rosewood, mahogany, and ebony and do **not** shed all leaves in one dry season. Moist deciduous forests (teak, sal, shisham, sandalwood) shed in the dry season and cover the **largest** area.
10. Thorn / scrub of Rajasthan, Gujarat, and rain-shadow tracts holds acacia, babool, ber, and khair (xerophytes). Littoral / mangrove forests of deltas hold **sundari** and **pneumatophores** (वात-मूल) and buffer storms.
11. World vegetation match: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile. Epiphytes peak in equatorial forest; acacia in Sahara; baobab in savanna (सवाना); cedars in Mediterranean.
12. Latest Forest Cover (ISFR 2023 / benchmark for 2025–2026 exams) stands at **21.76%** of India's TGA (Total Forest + Tree Cover = **25.17%**). Top state by forest area is **Madhya Pradesh**; top state by combined green cover gain is **Chhattisgarh**, followed by **Uttar Pradesh** (**#2 gainer, +559 sq km**). **Forest cover is not the same as Recorded Forest Area**. National Forest Policy 1988 aims at about **33%**.
13. Uttar Pradesh (उत्तर प्रदेश) latest forest cover is **6.24%** of geographical area (Total Forest + Tree Cover = **9.96%**). In UP, districts above 20% forest cover include **Chandauli (~21.6%)** and **Shravasti (~20.9%)**, not Bahraich (~10.1%). Jaunpur (जौनपुर)–Ghazipur–Ballia have near-nil forest land, yet Ballia holds **Surha Tal** Ramsar. Uttarakhand (उत्तराखंड) forest cover stands at **45.44%** of geographical area (legal RFA is ~**71%**); top forest area is **Pauri Garhwal**, top % is **Nainital**, and lowest is **Udham Singh Nagar**.
14. National Parks are stricter than Wildlife Sanctuaries. Biosphere Reserves have **core + buffer + transition**. India has **18** biosphere reserves; **Nilgiri (1986)** was the first.
15. **In-situ** (स्थान पर) conservation = National Park, Sanctuary, Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र), sacred grove. **Ex-situ** (स्थानांतरित) = zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक). A botanical garden is **not** in-situ.
16. Chronology fact: Wildlife Protection (वन्यजीव सुरक्षा) Act **1972** → Project Tiger **1973** → Project Elephant (प्रोजेक्ट एलीफेंट) **1992** → Biological Diversity Act **2002**. Project Tiger launched on **1 April 1973**.
17. Park–state pairs: **Simlipal = Odisha** (not Madhya Pradesh); **Nokrek = Meghalaya** (मेघालय) (not Manipur); **Nanda Devi (नंदा देवी) = Uttarakhand** (उत्तराखंड) (not Uttar Pradesh); **Agasthyamalai = Kerala and Tamil Nadu**; Manas = Assam; Keoladeo (केवलादेव) = Rajasthan; Kanha (कान्हा) = Madhya Pradesh.
18. **Dachigam** protects the **Hangul** (Kashmir stag). **Keibul Lamjao** sits on **Loktak** (Manipur) for the Sangai (सांगाई).

19. Uttar Pradesh's **only National Park** is **Dudhwa** (दुधवा) in **Lakhimpur Kheri**. Tiger Reserves are Dudhwa, Pilibhit, Amargarh, and Ranipur — do not invent extra National Parks.
20. Dudhwa Tiger Reserve = Dudhwa National Park (दुधवा राष्ट्रीय उद्यान) + Kishanpur + **Katarniaghat** (Katarniaghat established **1975**). **Chandraprabha** Wildlife Sanctuary is in **Chandauli**.
21. Ramsar traps **not** in Uttar Pradesh: Rudrasagar (Tripura), Sultanpur (Haryana), Surinsar–Mansar (Jammu & Kashmir). Surha Tal / JP Narayan Bird Sanctuary, Ballia is India's **100th Ramsar** and Uttar Pradesh's 13th.
22. **MISHTI** restores mangroves and belongs to Budget **2023–24**, not 2019. Mangrove breathing roots are **pneumatophores**. Sundarbans (सुंदरबन) are the sundari mangrove fact.
23. Coral reefs fact to Andaman & Nicobar, Lakshadweep (लक्षद्वीप), Gulf of Mannar, and Gulf of Kachchh. Coral bleaching (प्रवाल विरंजन) is linked to about **2°C** warming of sea water.
24. **Operation Olivia** is the Coast Guard Olive Ridley programme in Odisha. Nesting season is about **November–May**, not August.
25. Montane vegetation is an altitude elevator: tropical foothills → oak / deodar / chir → alpine → bugyal → snow. Plains rainfall ladders do not replace this altitude logic.
26. Keystone species (कीस्टोन प्रजाति) collapse an ecosystem if lost; flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are public mascots — do not treat them as the same idea.
27. India has about **50,000** rice strains and about **1,000** mango varieties in the usual diversity teaching line.
28. Tea and rubber estates are **plantations**, not climax evergreen forest. Moist and dry deciduous belong to one monsoon-forest family split by rainfall.
29. Tropical evergreen is also called **Selvas**. Typical figures: rain **>200 cm**, mean annual temperature above about **22°C**, humidity near **70%** (many notes write humidity **>77%** and wet evergreen **>250 cm**).
30. **Butea monosperma** is the **flame of the forest**, also called **Dhak** or **Palash**, and is the **state flower of Uttar Pradesh**.
31. **Katha** is obtained from **khair** wood. **Khejri** is the multipurpose “king of the desert” tree used in social forestry (सामाजिक वानिकी).
32. Tree-line is **lower** in the **Western** Himalaya (about **3300–3600 m**) than in the Eastern / Central Himalaya (about **3600–3800 m**). Alpine belts also sit higher in the east because that side is nearer the equator and the sea.
33. **Southern** Himalayan slopes hold **denser** vegetation than northern slopes because they get **more rainfall and sunlight**. Assertion that northern slopes are denser is **false**.
34. **Deodar** is abundant in the **western** Himalayan temperate belt at about **1500–2500 m**. Among common options (deodar, pine/chir, sal, teak), deodar grows at the **highest** elevation.
35. Classic mangrove match: **Achra–Ratnagiri = Maharashtra**; **Coondapur = Karnataka**; **Pichavaram = Tamil Nadu**; **Vembanad = Kerala** (not Karnataka). **Silent Valley (साइलेंट वैली) (Kerala)** = tropical evergreen.
36. **Mahogany** is **not** a Himalayan species (tropical American origin / Indian evergreen belts). Himalayan set = juniper, silver (रूपक) fir, spruce, deodar, birch.

Must-score facts — rain ladder, hotspots, zones

Rain ↔ forest

Forest	Rain / tag
Tropical evergreen	>200 cm
Moist deciduous	100–200 cm (largest area ~ 37%)
Dry deciduous	Lower rain; teak–sal belts
Dry evergreen (TN Coromandel)	~100 cm NE monsoon (उत्तर-पूर्व मानसून)
Thorn / scrub	Rajasthan, Gujarat, rain-shadow; acacia–babool

Biodiversity tags

Item	Lock
"Vegetation is true index of climate"	Thornthwaite
Hotspots touching India	Himalaya · WG–Sri Lanka · Indo-Burma · Sundaland
Chir pine belt	W. Himalaya ~1000–2000 m
Realms	Palearctic (Himalaya) + Indomalayan
Biogeographic zones	About 10 zones

Topic 8 — Minerals, Energy & Industry

Consolidated — 28 Must-Score Facts

1. Most of India's mineral wealth lies **east of the Mangalore–Kanpur (कानपुर) line**. The northern plains are almost empty of major ores. **Chota Nagpur** is the richest mineral belt and is often called India's Ruhr. **Dharwar** rocks are the metallic storehouse; **Gondwana** holds nearly all coal.
2. India is poor in many non-ferrous metals but rich in **bauxite**. Mica's classic dielectric belt is **Koderma–Gaya (गया)–Hazaribagh** in Jharkhand.
3. As per the Ministry of Mines Annual Report **2021–22**, the leading iron-ore states are **Odisha and Chhattisgarh**. Jharkhand and Karnataka do not always lead.
4. Iron-ore mine–state pairs are **Gurumahisani (Odisha)**, **Bailadila (Chhattisgarh)**, **Noamundi (Jharkhand)**, and **Kudremukh (Karnataka)**. Mount Newman in Australia is famous for **iron**, not manganese.
5. The four (चातुर्यम) main iron belts are the Odisha–Jharkhand belt, the Durg–Bastar–Chandrapur belt, the Ballari–Chitradurga belt, and the Maharashtra–Goa (गोवा) belt.
6. Manganese leaders include **Odisha, Maharashtra and Madhya Pradesh**. Postmasburg in South Africa is a manganese centre. **Bhilwara** is mica and zinc (Rampura-Agucha), not manganese.
7. Bauxite feeds aluminium and often sits with **laterite (लेटराइट)**. Odisha leads (Panchpatmali). **MALCO** is at **Mettur, Tamil Nadu (नाडु)**, not Chennai. **HINDALCO** is at **Renukoot, Uttar Pradesh (उत्तर प्रदेश)**. **BALCO** is at Korba; **NALCO** at Damanjodi/Angul.
8. Copper centres are **Khetri (Rajasthan)**, Singhbhum (Jharkhand) and Malanjkhand (Madhya Pradesh). **Panna (Madhya Pradesh)** is diamond. **Sukinda (Odisha)** is chromite. **Zawar** and **Rampura-Agucha** (Bhilwara) are zinc–lead.
9. Uranium's classic mine is **Jaduguda, Jharkhand**. Thorium and monazite sands line the **Kerala–Tamil Nadu** beaches. Lithium finds at **Reasi, Jammu & Kashmir** matter, but India still imports much Li, Ni and Co.
10. About **97%** of Indian coal is **Gondwana** coal in the Damodar (दामोदर)–Son–Mahanadi (महानदी)–Godavari (गोदावरी) valleys. Tertiary coal is mainly in the North-East. Rank rises **peat (<40% C) → lignite (40–55%) → bituminous (60–80%) → anthracite (80–95%)**. Anthracite is scarce (J&K pockets); lignite = **Neyveli**; ~80% of deposits are bituminous **non-coking**.
11. Coalfield centres: **Raniganj (West Bengal)**, **Jharia (Jharkhand, coking)**, Ramgarh (Jharkhand), **Talcher (Odisha)**. **Maolong** coal is in **Meghalaya (मेघालय)**, not Arunachal. **Neyveli** is lignite in Tamil Nadu.
12. Valley–field pairs: Talcher–Mahanadi, Karanpura–Damodar, Singrauli–Son, Singareni–Godavari. **Singrauli** is the oldest large NTPC hub and sits on the **Madhya Pradesh–Uttar Pradesh** border, not northern Chhattisgarh.
13. World coal traps: Ruhr is **Germany**; Donetsk is **Ukraine**; Appalachian is USA; Kuzbass is Russia. **Pegu Yoma (Myanmar)** is mineral oil, not tin. **Kashagan** oil is in **Kazakhstan**, not Kuwait.
14. **Methane** is the main gas in natural gas and **CNG**. **LPG** is propane–butane and is not the same as CNG. The **Black Revolution** means petroleum, not oilseeds.
15. Oil–gas hubs include Digboi/Naharkatiya (Assam), Mumbai High, Ankleshwar (Gujarat), Barmer (Rajasthan) and the Krishna (कृष्णा)–Godavari / Cauvery basins. Digboi is among India's oldest refineries.
16. Refinery–state pairs: Nayara (Gujarat), Manali (Tamil Nadu), Numaligarh (Assam), Tatipaka (Andhra Pradesh), **Mathura (मथुरा) (Uttar Pradesh)**, Barauni (Bihar), Panipat (पानीपत) (Haryana), Jamnagar (Gujarat), Paradip (Odisha), Bina (Madhya Pradesh), Haldia (West Bengal).
17. Conventional energy (पारंपरिक ऊर्जा स्रोत) means coal, oil, gas and nuclear. Non-conventional means solar, wind, tidal, biogas (बायोगैस) and geothermal. Nuclear and geothermal energy (भू-तापीय ऊर्जा) are **not** stored solar energy.
18. Nuclear plant centres: **Tarapur (Maharashtra)**, **Kakrapar (Gujarat)**, **Kudankulam (Tamil Nadu)**, **Kaiga (Karnataka)**, **Rawatbhata (Rajasthan)**, **Kalpakkam (Tamil Nadu)**, **Narora (Uttar Pradesh)**. **Shivasamudra** is hydel, not nuclear.

19. Solar points: electricity from **PV cells**; **Diu** was the first Union Territory to go fully solar-powered; **Cochin** was the first solar airport. The **International Solar Alliance** headquarters is at **Gurugram**, not New Delhi or Paris.
20. Wind leaders include Tamil Nadu (Muppandal), Gujarat (Kutch) and Rajasthan (Jaisalmer). Tidal potential is higher in the **Gulf of Khambhat** than in Kutch.
21. The first cotton-mill cluster grew in **Maharashtra–Gujarat** on cotton, humidity and Mumbai port (बंदरगाह). **Mumbai** was traditionally the largest mill centre; Ahmedabad (अहमदाबाद) is in the belt but was not traditionally the largest. Coimbatore is a Tamil Nadu spinning hub; Kanpur is an Uttar Pradesh mill city.
22. Steel plant centres: **Bhilai (Chhattisgarh)**, **Rourkela (Odisha)**, **Bokaro (Jharkhand)**, **Durgapur (West Bengal)**, Vizag (Andhra Pradesh), Salem (Tamil Nadu), Jamshedpur (Jharkhand). Chotanagpur (छोटानागपुर) is the mineral-heavy industrial heartland.
23. Agency headquarters: ONGC Delhi, MECL Nagpur, NMDC Hyderabad (हैदराबाद), NALCO Bhubaneswar.
24. The **Delhi–Mumbai Industrial Corridor (DMIC)** and the **Eastern and Western Dedicated Freight Corridors** are the main corridor schemes. SEZs are export-oriented zones.
25. In Uttar Pradesh, **Sonbhadra** (सोनभद्र) is the energy capital. **Mathura** (मथुरा) has the major refinery. **Narora** is the nuclear plant. Obra, Anpara and Rihand (रिहंद) are thermal hubs.
26. Shankargarh silica feeds the **Firozabad** glass industry. The **Chunar (चुनार)–Churk–Dalla** belt is cement. **HINDALCO Renukoot** is aluminium. Sonrai (Lalitpur) has a copper occurrence.
27. Uttar Pradesh city industries: Kanpur textile and leather, Moradabad brass, Aligarh (अलीगढ़) padlocks and hardware, Firozabad glass, Meerut (मेरठ) sports goods, Noida–Ghaziabad NCR engineering.
28. Uttar Pradesh coal is mostly **non-coking** Singrauli–Sonbhadra coal. It is not Jharia-type coking coal.

Must-score facts — ore belts, mine–state, energy

Frame

Item	Lock
Mineral wealth map	Mostly east of Mangalore–Kanpur line
Non-ferrous	Poor many metals; rich in bauxite
Mica belt	Koderma–Gaya–Hazaribagh

Iron / Mn / bauxite

Item	Lock
Iron leaders (2021–22 teaching)	Odisha (ओडिशा), Chhattisgarh
Gurumahisani	Odisha
Bailadila	Chhattisgarh
Noamundi	Jharkhand
Manganese leaders	Odisha, MH, MP
Bauxite	Feeds aluminium; laterite link

Energy / industry tags

Item	Lock
Coal belts	Damodar, Mahanadi, Godavari, Son
Petroleum	Mumbai High; Assam; Gujarat
Atomic minerals	Beach sands (monazite) South India

Topic 9 — Transport & Communication

Consolidated — 27 Must-Score Facts

1. Transport moves people and goods; communication moves information. About **95%** of India's trade by **volume** moves by sea.
2. India's first passenger train ran **Mumbai–Thane in 1853**. Indian Railways is organised into about **18 zones** (older books still say 16).
3. Zone–HQ pairs: **Northern Railway — New Delhi**; **North Eastern Railway — Gorakhpur (गोरखपुर)**; **South Eastern Railway — Garden Reach, Kolkata** (not Cuttack); **East Central Railway — Hajipur**; **North Central Railway — Prayagraj (प्रयागराज)**; **South Western Railway — Hubballi**; **East Coast Railway — Bhubaneswar**.
4. The **Western Dedicated Freight Corridor** runs **Dadri–JNPT**. The **Eastern Dedicated Freight Corridor** runs **Ludhiana–Dankuni**.
5. **Amrit Bharat** is a **station modernisation** programme, not a new-lines scheme. About **1,275** stations were identified all-India; in **February 2023**, **149 Uttar Pradesh (उत्तर प्रदेश)** stations were listed.
6. The new **Pamban** bridge is an **RVNL** project and India's first **vertical-lift** railway sea bridge. Hubballi holds both the **South Western Railway HQ** and the **August 2020** railway museum.
7. National Highways are only about **2%** of road length but carry about **40%** of traffic. Rural roads are most of the length and are linked to **PMGSY**.
8. The **Golden Quadrilateral** joins Delhi–Mumbai–Chennai–Kolkata (about **5,846 km**). North–South is Srinagar–Kanyakumari (कन्याकुमारी); East–West is Porbandar–**Silchar**. They meet at **Jhansi (झांसी)**.
9. **Bharatmala** is a **highway** programme. **Sagarmala** is **port-led**. Do not swap the two. **PMGSY (2000)** = rural all-weather roads; **PMBJP** = highway BOT corridors.
10. The **Atal Tunnel** runs under **Rohtang (रोहतांग)** in the **Pir (पीर) Panjal (पीर पंजाल)** (Himachal (हिमाचल) / BRO). A bare claim that it is the “world's longest highway tunnel” is not the safe line.
11. The built **Purvanchal (पूर्वांचल) Expressway** alignment is Lucknow (लखनऊ)–Azamgarh–Mau–Ghazipur–Ballia. It **passes Mau**; Basti is off the main spine.
12. India has **12 major ports** plus many non-major ports. **IWAI** dates from **1986**. The 2016 Act notified **111 National Waterways**.
13. National Waterway pairs: **NW-1** = Ganga (गंगा), Haldia–Prayagraj; **NW-2** = Brahmaputra (ब्रह्मपुत्र), Dhubri–Sadiya; **NW-3** = Kerala West Coast Canal; **NW-4** = Godavari (गोदावरी)–Krishna (कृष्णा) / Kakinada–Puducherry; **NW-5** = Brahmani–Mahanadi (महानदी)–East Coast Canal.
14. A July 2020 containership link from Kolkata toward Agartala used the **Chattogram** route.
15. Port (बंदरगाह)–state pairs: **Paradip (Odisha)**, **Tuticorin / VOC (Tamil Nadu (तामिल)), Kakinada (Andhra Pradesh)**, **Alappuzha (Kerala)**. **JNPT** is the container flagship; **Kolkata–Haldia** is riverine.
16. A **harbour** is sheltered water; a **port** is harbour plus cargo facilities. **Igarka** is in **Russia**, not China.
17. **Kushinagar (कुशीनगर)** became an international airport on **24 June 2020**. **Cochin** was India's first solar-powered airport. **Jewar** is Noida International Airport.
18. FLY91's inaugural flight (18 March 2024) was from **Manohar International Airport** (Goa Mopa).
19. The Motihari–**Amlekhganj** (Nepal) petroleum pipeline was inaugurated in **September 2019**.
20. **PM Gati Shakti** was launched on **13 October 2021** with the “seven engines” model. The **National Logistics Policy** (September 2022) complements it. The **Blue Economy** means ocean-based growth.
21. Optical-fibre communication uses **total internal reflection (TIR)**, not simple refraction.
22. Uttar Pradesh has **no major seaport**. Its rail HQ is **NER Gorakhpur**; its waterway fact is **NW-1** on the Ganga.
23. NHA was set up in **1995**. Expressways are access-controlled highways, distinct from ordinary National Highways.
24. West-coast major ports include Kandla (Deendayal), Mumbai, JNPT, Mormugao, New Mangalore and Cochin. East-coast majors include Kolkata–Haldia, Paradip, Vizag, Chennai, Ennore (Kamarajar) and Tuticorin.

25. **Incredible India** is the MoT tourism brand; **ITDC** dates from **1966**. Match hill stations carefully: Chakrata–UK, Haflong–Assam, Kalimpong–WB, Kufri–HP.
26. **City of Lakes = Udaipur (उदयपुर)** (not Jaisalmer). **Pink City = Jaipur**. **Sabarimala = Kerala**. UNESCO tourist set includes Champaner–Pavagarh, CST Mumbai, Mamallapuram and Konark (कोणार्क).
27. **Srisailem = Nallamala**; Omkareshwar (ओंकारेश्वर) and Pushkar are **not** on Satmala / Mahadeo in the IAS 2015 trap. **Biligiriranga** holds the standing Ranganatha note in Karnataka.

Must-score facts — rail, road, ports, waterways

Rail / freight

Item	Lock
First passenger train	Mumbai–Thane 1853
Zones	~ 18 (older books: 16)
Northern / NER / SER HQ	New Delhi / Gorakhpur (गोरखपुर) / Garden Reach, Kolkata
Western DFC	Dadri–JNPT
Eastern DFC	Ludhiana–Dankuni
Amrit Bharat	Station modernisation (≠ new lines); ~ 149 UP stations
New Pamban	RVNL; first vertical-lift railway sea bridge

Road / schemes

Item	Lock
Golden Quadrilateral	Delhi–Mumbai–Chennai–Kolkata (~5846 km)
NS / EW	Srinagar–Kanyakumari / Porbandar–Silchar
Bharatmala vs Sagarmala	Highways vs port-led
PMGSY	Rural all-weather (2000)
Atal Tunnel	Under Rohtang (Pir Panjal; BRO)
Gati Shakti	13 Oct 2021 ; NLP Sep 2022

Ports / waterways / air

Item	Lock
Sea trade volume	~ 95% of India's trade volume
Major ports	12 + many non-major
NW-1 / NW-2 / NW-3	Ganga Haldia–Prayagraj / Brahmaputra / Kerala W. Coast Canal
Paradip / Tuticorin / Kakinada	Odisha / TN / AP
Harbour vs port	Sheltered water vs harbour + cargo facilities
Kushinagar intl	24 Jun 2020 ; Jewar = Noida greenfield
UP	No major seaport ; rail HQ NER Gorakhpur; NW-1

Topic 10 — Tribes & Institutions

Consolidated — 24 Must-Score Facts

1. Scheduled Tribe (अनुसूचित जनजाति) lists are **state-wise** under **Article 342**. Parliament can amend the lists by law. A community listed in one State need not be listed in another.
2. STs are about **8.6%** of India's population (जनसंख्या) (Census 2011). Among the largest groups are **Bhil, Gond (गोंड) and Santhal (संथाल)**. Madhya Pradesh often leads in absolute (निरपेक्ष) ST population.
3. Lokur Committee criteria for ST listing include primitive traits, distinctive culture, geographical isolation, shyness of contact, and economic backwardness.
4. There are **75 Particularly Vulnerable Tribal Groups (PVTGs)** (Dhebar PTG renamed PVTG in 2006). **Odisha (ओडिशा)** has the most PVTG communities. **PM-JANMAN** is the main outreach mission.
5. PVTG criteria stress pre-agricultural technology, stagnant or declining population, extremely low literacy, and a subsistence economy.
6. The **Fifth Schedule** (5वीं अनुसूची) covers Scheduled Areas in many States. The **Sixth Schedule** (6वीं अनुसूची) covers autonomous councils in **Assam, Meghalaya (मेघालय), Tripura and Mizoram** only.
7. State-tribe pairs: **Angami / Rengma = Nagaland**; **Toda / Paliyan = Tamil Nadu (नाडु)**; **Birhor = Jharkhand**; **Khasi (खासी) = Meghalaya** (not Arunachal); **Yanadi = Andhra Pradesh** (not Rajasthan); **Chenchu = Andhra / Telangana**.
8. **Kuki** is a North-East tribe and is **not** a Uttar Pradesh Scheduled Tribe. **Keria / Kharia** fact to Odisha (also Jharkhand for Kharia). **Bodo = Assam**; **Lepcha = Sikkim**; **Meena = Rajasthan**; **Warli (वारली) = Maharashtra**; **Apatani = Arunachal**.
9. **Khasi and Garo (गारो)** of Meghalaya are classic **matrilineal** societies.
10. Uttar Pradesh has **15** notified ST entries. The classic five are **Tharu, Bhotia, Buksa, Jaunsari and Raji**. UP PVTGs are **Buksa and Raji**.
11. In Uttar Pradesh, **Tharu** live in the **Terai (तराई)**; **Buksa** on the Bijnor fringe; **Saharya** in **Lalitpur**; **Agariya** are linked to **iron smelting** in the Sonbhadra (सोनभद्र)–Mirzapur (मिर्जापुर) belt.
12. After the 2003 Act, many UP ST notifications are **district-limited**. Two clusters matter: Terai north versus Vindhyan / Sonbhadra south.
13. Central Indian heartland tribes include **Gond (गोंड)** (with Maria/Muria), **Bhil**, Baiga, Kol, Korku, Sahariya, Halba, Warli and Meena.
14. Jharkhand core tribes include **Santhal, Munda, Oraon**, Ho, Kharia, Birhor, Bhumij, Asur and **Birjia**. Birjia is **not** an Andaman tribe. Santhali uses the **Oi Chiki** script (लिपि); **Birsa Munda** and the **Hul (हूल) of 1855** are pairs.
15. **Janjatiya Gaurav Diwas** is observed on **15 November** (Birsa Munda's birth anniversary).
16. Andaman **Negrito** natives are Great Andamanese, **Onge, Jarawa** and **Sentinelese** (North Sentinel Island). Nicobar (निकोबार) **Mongoloid** groups are Nicobarese and **Shompen** (Great Nicobar PVTG). **Rengma** is not an island tribe.
17. World habitat pairs: **Bushman = Kalahari**; **Eskimo = northern Canada**; **Pygmy = Congo**; **Masai = East Africa** (not West Africa); **Ainu = Japan**; **Punan = Borneo**; Lapps/Sami = Sweden–Finland. 17a. Also: Semang = **Malaysia** (not Indonesia); Bedouin = Arabia; Kirghiz = Central Asia (not Sudan); Maori = New Zealand; Zulu = South Africa; Bantu ≠ Sahara. **Transhumance** = seasonal valley–mountain herding. 17b. Gypsies' original home = **India**. Africa = cradle of mankind. Cro-Magnon is the most recent among classic fossil-human options.
18. **Khirghiz** are Central Asian, not Sudanese. **Koryak** live in north-east Siberia, not Alaska.
19. **IMD** headquarters is New Delhi (**1875**). **ICAR** headquarters is New Delhi (**1929**). **IARI** is at **Pusa, New Delhi** — it is not the apex ICAR council itself.
20. **FRI** and **ICFRE** are at **Dehradun (देहरादून)**. **WII** is also Dehradun but is not the FRI answer. **NDRI** is at **Karnal** (dairy lab). **IVRI** is at **Izatnagar, Bareilly (UP)**. **NDDB** is at **Anand** and ran **Operation Flood** (ऑपरेशन फ्लड) — it is not NDRI.

21. **IIHR** is at Bengaluru. **CISH** is at Lucknow (लखनऊ). **CSAUAT** is at Kanpur (कानपुर). **Pantnagar** is remembered as the first State Agricultural University tradition.
22. The **Forest Rights Act** is **2006**. **Adi Karmayogi** (June 2025) is a **Ministry of Tribal Affairs** programme.
23. Main tribal belts are the North-Eastern hills, the central Indian plateau, western Bhil country, and the Andaman–Nicobar (अंडमान) isolates.
24. Krishi Vigyan Kendras (KVKs) are the frontline **extension** arms under the ICAR system.

Must-score facts — tribe↔state, PVTG, institutes

Frame

Item	Lock
ST lists	State-wise Art. 342
ST share (2011)	~ 8.6% ; largest groups Bhil, Gond, Santhal
PVTGs	75 ; most in Odisha ; PM-JANMAN outreach
5th vs 6th Schedule (6वीं अनुसूची)	Many States' Scheduled Areas vs Assam–Meghalaya–Tripura–Mizoram councils

Tribe ↔ state

Tribe	Lock
Angami / Rengma	Nagaland
Toda / Paliyan	Tamil Nadu
Birhor	Jharkhand
Khasi / Garo	Meghalaya (मेघालय) (matrilineal)
Kuki	NE (not UP ST)
UP classic five	Tharu, Bhotia, Buksa, Jaunsari, Raji (15 notified)
UP PVTGs	Buksa + Raji
Tharu / Buksa / Saharya	Terai / Bijnor fringe / Lalitpur
Jharkhand core	Santhal, Munda, Oraon, Ho...
Andaman Negrito	Great Andamanese, Onge, Jarawa, Sentinelese

World habitats / institutes

Pair	Lock
Bushman / Eskimo / Pygmy / Masai	Kalahari / N. Canada / Congo / East Africa
IMD / ICAR HQ	New Delhi (1875 / 1929)
IARI	Pusa, New Delhi
FRI / ICFRE / WII	Dehradun (WII ≠ FRI answer)
NDRI / IVRI	Karnal / Izatnagar teaching
CISH	Lucknow (लखनऊ); CSAUAT Kanpur (कानपुर)
FRA	2006
Janjatiya Gaurav Diwas	15 Nov (Birsa birth)

Topic 11 — Population Geography

Consolidated — 27 Must-Score Facts

1. India's first non-synchronous census was in **1872** (some papers key **1871**). The first **synchronous** all-India census was in **1881**.
2. **1921** is the **Great Divide** year of Indian census history (इतिहास). **2011** was the **15th** census and the **7th** after Independence.
3. Census 2011 India population was about **121.09 crore**. Decadal growth **2001–11** was **17.64%**. India holds roughly one-sixth of world population in that count.
4. Arithmetic density is **population / total area**. India's 2011 arithmetic density is **382** persons per km².
5. Physiological density is **population / net sown area**. Agricultural density is **agricultural population / net sown area**. Do not swap these three density types.
6. Among States, **Bihar** has the highest density (**1106**). West Bengal and Kerala follow high; Uttar Pradesh (उत्तर प्रदेश) is **829**. **Arunachal Pradesh** has the lowest State density (**17**). Delhi UT is very dense but is not a "lowest density State (स्यूनतम जनघनत्व)" answer.
7. Core 2011 figures: sex ratio **943**, child sex ratio **919**, literacy **74.04%** (age **7+**), urban (नगरीय) share **31.16%**. Scheduled Castes about **16.6%**; Scheduled Tribes about **8.6%**.
8. **Uttar Pradesh** (उत्तर प्रदेश) is the most populous State and has the largest rural population. **Sikkim** is the least populous State. **Nagaland** showed negative growth in 2001–11.
9. Highest State sex ratio is **Kerala**. Among States in the usual 2011 set, lowest sex ratio is **Haryana**. Lowest child sex ratio (rural + urban) is also **Haryana**.
10. Literacy is highest in **Kerala** and lowest among States in **Bihar**. In Uttar Pradesh, **Shrawasti** has the lowest female literacy among districts.
11. Absolute (निरपेक्ष) growth is $P_2 - P_1$. Growth rate is the percentage change. **Natural growth = CBR - CDR**. Induced change comes from **migration**.
12. Replacement-level **TFR is 2.1** children per woman, not "per thousand" (that wording belongs to birth/death rates). NFHS-4 (2015–16) reported **2.2**. NFHS-5 reports about **2.0**.
13. The National Population Policy **2000** aimed at population stability by **2045**. World Population Day is **11 July**. The World Population Report is associated with **UNFPA**.
14. The demographic dividend window is the large share of working ages **15–59**, not 60+ or 0–6. Dependency compares young plus aged with workers.
15. Urbanisation acceleration in the classic curve is linked to the **second stage** of demographic transition. Among religions, **Jains** are the most urbanised.
16. Demographic Transition Theory is linked to **Thompson** (with **Notestein**). Optimum population is linked to **Edwin Cannan**. Social mal-adjustment is linked to **Henry George**. Malthus argued population grows **geometrically** while food grows **arithmetically**. The Malthusian Theory is a theory of **population**.
17. The largest internal migration stream is **rural → rural**. Female migration is often for marriage; male migration is often for work. **Immigration** means in-movement; **emigration** means out-movement.
18. Push factors drive people from the origin; pull factors attract them to the destination. Out-migration sources are often Uttar Pradesh–Bihar; destinations are often Maharashtra–Delhi–Gujarat.
19. A **census town** needs population $\geq 5,000$, density $\geq 400/\text{km}^2$, and $\geq 75\%$ of **male main workers** in non-agriculture.
20. Census 2011 listed **53** million-plus urban agglomerations. Kanpur (कानपुर) crossed the million mark in **1971**; Lucknow (लखनऊ) in **1981**. 2011 UP UA order is **Kanpur > Lucknow > Ghaziabad > Agra**.
21. Uttar Pradesh district facts (2011): **Prayagraj** (प्रयागराज) most populous; **Ghaziabad** densest; **Jaunpur** (जौनपुर) among the higher sex-ratio districts; **Shrawasti** lowest female literacy.

22. Uttar Pradesh holds about **16.5%** of India's population. Rural share is about **77.7%** against India's **68.84%**. Urban share is about **22.3%** against India's **31.16%**. State sex ratio is **912** against India's **943**. Literacy is about **67.7%** against India's **74.04%**.
23. Crowded belts are the Ganga (गंगा) plain and coasts. Sparse belts are the Himalaya (हिमालय), North-East hills, Thar (थार) and dry interior pockets. Distribution factors include terrain, climate, soil, water, minerals, industry, transport and history.
24. Keep **Census 2011** numbers for Prelims until **Census 2027** results replace them. UN estimates (प्राकलन समिति) that India became the most populous country around **2023** do not rewrite the 2011 tables.
25. A broad-base population pyramid signals high fertility. Ageing shows earlier in southern States than in the high-fertility northern belt.
26. Occupational structure is taught as primary / secondary / tertiary. Primary still takes a large share of India's workforce, with a slow shift toward secondary and tertiary as development proceeds.
27. Malthus's **positive** checks raise the death rate (famine, disease, war). **Preventive** checks lower the birth rate (delayed marriage, moral restraint). **Karl Marx** treated surplus population as a product of capitalism, not a natural food-arithmetic law. **Ester Boserup** argued that population growth can push agricultural intensification.

Must-score facts — density, growth, Census tags

Terms

Term	Lock
Density	Population ÷ area
Physiological density	Population ÷ arable land
Agricultural density	Farmers ÷ arable land
Sex ratio	Females / 1000 males
Literacy	7+ who can read & write with understanding

India Census 2011 spine (teaching)

Item	Lock
Highest density state (सर्वाधिक घनत्व) (usual)	Bihar
Lowest density (large state)	Arunachal teaching
Highest literacy (usual)	Kerala
Phase model	High stationary → early expanding → late expanding → low stationary

Topic 12 — Human Geography (Settlements)

Consolidated — 26 Must-Score Facts

- Human geography studies the **man–environment (पर्यावरण) relationship** and the spatial patterns of human life, settlements and economy.
- Environmental determinism** (Ratzel / Semple) says nature controls culture. **Possibilism** (Vidal de la Blache) says humans choose among nature's options. **Neo-determinism** (Griffith Taylor) is stop-and-go determinism.
- Site** is the local ground of a settlement. **Situation** is its wider regional location relative (सापेक्ष) to routes, resources and other places.
- A **census town** needs population $\geq 5,000$, density $\geq 400/\text{km}^2$, and $\geq 75\%$ of **male main workers** in non-agriculture. It need not be a municipality.
- Rural settlements lean on primary activities and lower density. Urban places are statutory towns or census towns.
- Village types are **clustered / nucleated, semi-clustered, hamleted** and **dispersed**. Clustered villages appear on plains, around Rajasthan water points, and for defence in Bundelkhand / Nagaland — do **not** call the Thar (थार) "dispersed by default."
- Dispersed villages typify **Meghalaya (मेघालय), Uttarakhand (उत्तराखंड), Himachal (हिमाचल) Pradesh, Kerala** and many North-East forest–hill tracts. Hamlet local names include **panna, para, palli, nagla and dhani**.
- Semi-clustered villages often show a dominant-caste centre with lower strata on the flanks (Gujarat / Rajasthan pair). Linear villages follow a road, river, canal or coast.
- Town functions are classed by the **dominant** job: administrative, industrial, transport, commercial, mining, garrison, educational, religious or tourist.
- Class I towns have ≥ 1 **lakh** people (468 towns in 2011 held about **60%** of urban population). Classes II–VI step down from 50–99 thousand to under 5 thousand.
- In the standard Indian size ladder, metropolitan cities are **10 lakh–50 lakh** and mega cities are **above 50 lakh — six** in 2011. A UN megacity is ≥ 1 **crore**. Do not mix these three thresholds.
- 2011 million-plus UA rank starts **Mumbai > Delhi > Kolkata > Chennai > Bengaluru > Hyderabad (हैदराबाद)**. There were **53** million-plus UAs; the smallest in that set is often **Kota**. The **2001** order Mumbai > Kolkata > Delhi is a trap for 2011 stems.
- An urban agglomeration can be a town with outgrowths, two contiguous towns, or a city with adjoining towns and outgrowths. Outgrowths include railway colonies, campuses, ports and cantonments.
- Settlement evolution pairs: ancient **Varanasi (वाराणसी) / Prayag (प्रयाग) / Madurai (मदुरै)**; medieval **Delhi / Agra / Jaipur / Lucknow (लखनऊ)**; modern planned **Chandigarh** (Le Corbusier). Satellite towns include Ghaziabad.
- The **Smart Cities Mission** launched on **25 June 2015** for **100** cities under **MoHUA**, implemented through an SPV. ABD means area-based development (retrofit / redevelopment / greenfield) plus pan-city ICT.
- Uttar Pradesh (उत्तर प्रदेश)'s **Central / Mission Smart City** list is **10**: Lucknow, Kanpur (कानपुर), Prayagraj (प्रयागराज), Agra, Varanasi, Aligarh (अलीगढ़), Bareilly, Jhansi (झांसी), Moradabad and Saharanpur. **Ghaziabad is not** in that Central-10.
- Uttar Pradesh's **State Smart Cities (2019)** are seven Nagar Nigams (including Ghaziabad, Meerut (मेरठ), Gorakhpur (गोरखपुर), Mathura (मथुरा), Ayodhya (अयोध्या), Firozabad, Shahjahanpur (शाहजहाँपुर)). State list (राज्य सूची) does not rewrite Mission-100 keys.
- ISAC-2020 theme pairs: Culture **Indore**, Governance **Vadodara**, Social **Tirupati**, Urban environment **Bhopal**. Best State award went to **Uttar Pradesh** (उत्तर प्रदेश); best cities included Indore and Surat.
- HRIDAY** covers **12** heritage cities. In Uttar Pradesh the pair is **Varanasi and Mathura** — not Prayagraj or Ayodhya.
- SPMRM (Rurban)** was **launched on 21 February 2016** (Cabinet approval 2015). Matching the scheme to **2015** is the classic trap. It is under **MoRD**, not MoHUA.

21. **Rurbanization** is linked to sociologist **G.S. Ghurye**. McLuhan's **Global Village** rests on **transport plus communication**.
22. Scheme chronology: **JNNURM 2005** → Urban Housing Policy **2007** → **AMRUT June 2015** → **Jal Jeevan Mission 2019**. **AMRUT 2.0** is **1 October 2021**. **SAGY** is **2014**.
23. Smart Cities = **MoHUA**; smart village / Rurban = **MoRD**. Keep the ministries separate.
24. **Bhopal** is the classic "not on a major river bank" city trap against Agra, Patna (पटना) or Kolkata.
25. NCR slice of Uttar Pradesh includes Ghaziabad, Noida, Greater Noida and Meerut in the urban-industrial belt.
26. Hamleted villages break the main settlement into secondary units (**panna / para / palli / nagla / dhani**); they are a rural pattern, not a separate census-town category.

Must-score facts — approaches, settlements, activities

Nature & scope

Item	Lock
Human geography	Study of human–environment relations / spatial organisation
Approaches	Welfare, behavioural, radical, humanistic (coaching set)
Primary / secondary / tertiary	Extraction · manufacturing · services
Quaternary / quinary	Knowledge / high decision-making

Settlements

Item	Lock
Rural patterns	Compact, dispersed, linear, circular
Urban functions	Administrative, industrial, commercial, transport, garrison
Primate city / rank-size	Coaching definitions

Topic 13 — Disaster Geography

Consolidated — 26 Must-Score Facts

1. Coaching risk fact: **Risk $\approx$ Hazard $\times$ Vulnerability / Capacity**. A disaster happens when a hazard hits exposed people or assets and capacity is too weak.
2. The disaster-management cycle runs **mitigation $\rightarrow$ preparedness $\rightarrow$ response $\rightarrow$ recovery**. Mitigation is before impact; response is during and just after.
3. Earthquake **focus** is the point **inside** the Earth; **epicentre** is the point on the **surface** above it. **Richter** measures magnitude; **Mercalli** measures intensity / damage.
4. Seismic waves split into **body** (through the Earth) and **surface** (along the ground). Body waves arrive **P then S**; surface waves come last and do most damage. **P-wave** shadow (छाया) is about **103–142°**; **S-wave** shadow lies beyond about **103°** (liquid outer core). The Himalaya (हिमालय) is India's highest seismic belt because of **collision and thrust zones**.
5. India uses seismic **Zones II–V** only (Zone I was dropped in 2002). **Zone V is highest**. A draft Zone VI was **withdrawn in March 2026** — maps still follow the 2016 II–V zones.
6. Zone V covers the North-East, Himalayan pockets, **Kutch** and the **Andaman (अंडमान) & Nicobar (निकोबार)** belt. Much of Rajasthan and the Deccan (देक्कन) sits in Zones II–III.
7. About **59%** of India's landmass is earthquake-prone in NDMA note. Famous shocks: Kangra (कांगड़ा) 1905, Bihar–Nepal 1934, **Koyna 1967 (reservoir-induced)**, **Latur 1993 (peninsular / Killari, Zone III)**, Bhuj 2001, and the **26 December 2004** tsunami.
8. A **tsunami** (Japanese "harbour wave") is caused by **seafloor displacement** (quake, landslide or volcano), not by ordinary wind waves or tides. It races fast in deep water and piles up near the coast. India's warning hub is **INCOIS, Hyderabad (हैदराबाद)**. Hall event: **26 Dec 2004** Sumatra megathrust.
9. **Kilimanjaro** sits on the East African Rift and is **not** part of the Pacific Ring of Fire.
10. Landslide belts are the Himalaya, Western Ghats (पश्चिमी घाट) and North-East. pairs include Kedarnath (केदारनाथ) **2013**, Joshimath subsidence **2023**, and Sikkim GLOF **2023**.
11. Flood types include riverine (Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)), flash floods in hills, urban (नगरीय) floods and coastal storm surge (तूफानी महोर्मि). Drought is classed as meteorological, hydrological and agricultural. IMD drought criteria often use rainfall deficiency **>25%** of normal.
12. IMD **cloudburst** fact is rainfall **≥ 100 mm in one hour**, typically in Uttarakhand (उत्तराखंड)–Himachal (हिमाचल)–J&K–North-East hill belts around **1000–2500 m**.
13. IMD plains heat-wave gate: when **Tmax $\geq 40^\circ\text{C}$** , a departure of **4.5–6.4°C** is a heat wave and **>6.4°C** is severe; or actual **$\geq 45^\circ\text{C}$ / $\geq 47^\circ\text{C}$** . Criteria also use two stations for two days. Cold-wave plains gate is **Tmin $\leq 10^\circ\text{C}$** with a sharp drop.
14. The **Bay of Bengal** produces **more** cyclones than the Arabian Sea. Peak seasons are roughly **May–June** and **October–December**. Arabian Sea storms can still hit Gujarat–Maharashtra (Tauktae / Kandla note).
15. Cyclone-name pairs: **Baguio = Philippines**; Hurricane = USA; Typhoon / Taifu = China–Japan; **Willy-willy (विली-विली) = Australia**; cyclone = North Indian Ocean; Cordonazo = Mexico.
16. 2019 storm–place pairs: Dorian–Bahamas, Hagibis–Japan, Lekima–China, Mitag–Korea, with Indian Ocean names like Amphan, Fani and Tauktae also in the news map.
17. **NDMA** is chaired by the **Prime Minister**. **SDMA** is chaired by the **Chief Minister**. **DDMA** is chaired by the District Magistrate / Collector.
18. Framework years: **DM Act 2005**, **NPDM 2009**, **NDRF 2006** (MHA), **NDMP 2016**, **Sendai Framework 2015–2030** (after Hyogo 2005–15), **CDRI 2019**. International Day for Disaster Risk Reduction is **13 October**.
19. NEC pair = Union Home Secretary. **NIDM** handles training from Delhi. India is **not** a disaster-free country.
20. India's multi-hazard note: **~59%** quake-prone, **~12%** flood-prone, **~5,700 km** cyclone/tsunami coast, **~68%** of cultivable area drought-vulnerable.

21. **Mangroves** cut cyclone and surge impact. **Bhopal 1984** is the classic man-made industrial / chemical disaster.
22. **DPAP** began in **1973–74**. The earlier Community Development Programme pair is **1952**.
23. Uttar Pradesh (उत्तर प्रदेश) sits mainly in seismic **Zones III–IV** (not Zone V as a whole-State label). Eastern districts face river floods; Bundelkhand faces drought. Plains heat waves peak in May–June (**Loo** (लू)).
24. In the 2018 UP paper, the **Gomati** (गोमती) carried the “biological disaster” pollution label among the given rivers.
25. Hazard is potential danger; disaster is when capacity fails. Do not treat the two words as identical.
26. Cold-wave and fog risk in western Uttar Pradesh and the Terai (तराई) peaks in **December–January**; plains heat waves (**Loo**) peak in **May–June**.

Must-score facts — hazard types, DM cycle, India belts

Terms

Term	Lock
Hazard	Potential threat
Vulnerability	Susceptibility
Risk	Hazard × vulnerability
DM cycle	Mitigation → Preparedness → Response → Recovery

India belts

Hazard	Lock
Earthquake	Himalayan / Indo-Gangetic / peninsular zones
Cyclone	Bay of Bengal > Arabian Sea frequency
Flood	Ganga–Brahmaputra plains; flash floods hills
Drought	Rain-shadow / arid west; monsoon failure
Landslide	Himalaya / WG steep slopes

Topic 14 — Earth & Universe

Consolidated — 36 Must-Score Facts

1. Big Bang is linked to **Lemaître** / **Gamow** (~**13.7–13.8 bn yr**) and **Hubble** expansion. Steady State is **Hoyle**. Nebular is **Kant–Laplace**. The Milky Way is a **barred spiral** galaxy.
2. **Chandrasekhar limit** $\approx$ **1.44 M $\odot$** sets the white-dwarf ceiling; beyond that path $\rightarrow$ neutron star / black hole. A galactic year for the Sun is ~**22.5–25 crore** years. IAU lists **88** constellations; **Saptarishi** points to the Pole Star.
3. Solar-system order mnemonic is **MVEMJSUN**. **Venus** is hottest (greenhouse). **Earth** has the highest mean density. **Jupiter** is largest. **Saturn** has the clearest rings. **Mercury** has the fastest orbit and (with Venus) **no moon**.
4. The asteroid belt lies between **Mars and Jupiter**. The largest moon is **Ganymede** (Jupiter — not Saturn). Titan belongs to Saturn; Phobos and Deimos belong to Mars. **Copernicus** is the classic heliocentric name.
5. In the 2022 Prelims set, Saturn's atmosphere is linked with **methane**. Mars is the classic life-candidate planet (including the 2018 subsurface-lake news). **Mangalyaan** was India's first Mars orbiter.
6. A **solar eclipse** occurs at **New Moon**. A **lunar eclipse** occurs at **Full Moon**. Sunlight takes about **8 minutes 20 seconds** to reach Earth. Sun's energy is **nuclear fusion**. One **AU** $\approx$ **149.6 million km**.
7. Earth **rotates** in about **24 hours** (day–night) and **revolves** in about **365¼ days** (year / leap year). **Perihelion** is about **3–4 January**; **aphelion** about **4 July**.
8. Seasons are caused by the **23½° axial tilt**, not by perihelion distance. Equinoxes are about **21 March / 23 September**; solstices about **21 June / 22 December**.
9. One degree of latitude is about **111 km**. Tropics sit at **23½°**; polar circles at **66½°**. Time zones use roughly **15° = 1 hour**. **0° latitude and 0° longitude** meet in the **Atlantic** (Gulf of Guinea).
10. India's Standard Time is based on **82°30' E** near **Mirzapur** (मिर्ज़ापुर) and is **GMT + 5 hours 30 minutes** for the whole country. The International Date Line is near **180°** (zigzag); the west side is a day ahead.
11. Atmosphere layers: **troposphere** (क्षोभमंडल) holds weather; **stratosphere** (समतापमंडल) holds the **ozone** UV shield and is nearly weather-free (**ideal for jet flight**); mesosphere (मध्यमंडल) is coldest; **ionosphere** reflects radio waves and hosts auroras; **exosphere** hosts communication satellites.
12. Earth's atmosphere is heated mainly by **long-wave terrestrial radiation** (भौमिक विकिरण), not by direct short-wave alone. Normal lapse rate is about **6.5°C/km**. Permanent gases = **N₂, O₂, Ar**; CO₂ / O₃ / vapour vary. Albedo = **reflecting** power (fresh snow highest). Max heating belt often ~**20°N**. Same-latitude Agra vs Darjeeling contrast is an **altitude** fact. Cyclone names: Hurricane–USA, Typhoon–China Sea, Willy–Willy (विली-विली)–Australia, Baguio–Philippines. Tornado scale = **Fujita**; hurricane = **Saffir–Simpson**.
13. Diffusion / scattering of light is linked to **dust** in one Prelims set. **Weather** is short-term; **climate** is the long-term average.
14. Greenhouse gases include water vapour, CO₂, CH₄, N₂O, O₃ and CFCs. **Argon is not** a greenhouse gas. **Kyoto** (क्योटो) deals with climate / carbon; **Montreal** deals with ozone.
15. **El Niño** is a warm eastern Pacific / Peru current phase with **less upwelling**, so **plankton and fish fall**. It can disturb the Indian monsoon.
16. Coriolis deflects to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere**. It is zero at the equator (Ferrel's law).
17. Jet streams are mainly high-altitude **westerlies** (polar jet and subtropical jet). The **Tropical Easterly Jet (TEJ)** (उष्णकटिबंधीय पूर्वी जेट) is the summer easterly exception. Speeds can reach **300–500 km/h**. "All jets are easterly" is false.
18. Pressure belts: ITCZ / equatorial **low** (doldrums (डोलड्रम)); subtropical **high** near **30°** (horse latitudes (हॉर्स अक्षांश)); subpolar low near **60°**; polar high. Planetary winds are trades, westerlies and polar easterlies.
19. **Roaring Forties** are strong westerlies near **40°S** (with Furious Fifties / Shrieking Sixties farther south). The ITCZ is a **low**; belts shift with the solstice seasons.

20. Interior shells: **Sial** (continental crust), **Sima** (oceanic crust), **Nife** (core). Discontinuities: crust–**Moho**–mantle–**Gutenberg** (~2900 km, S-wave stop)–**Lehmann**–inner core.
21. P-wave shadow (छाया) is about **103–142°**; S-wave shadow lies beyond about **103°** because the outer core is liquid.
22. **Wegener (1912)** proposed continental drift with **Pangaea** and **Panthalassa**. Plate margins are divergent, convergent or transform. New ocean crust forms by **sea-floor spreading**.
23. **Fold mountains** include the Himalaya (हिमालय), Alps, Andes and Rockies. **Fault / block mountains** include the Vosges, Black Forest, Sierra Nevada and Rhine graben.
24. Volcano types: shield (**Mauna Loa**), composite (**Fuji / Etna / Vesuvius**), cinder (**Paricutin**), fissure (**Deccan** (देक्कन)), caldera (**Krakatoa**). **Stromboli** is the “lighthouse of the Mediterranean.” **Barren Island** is India’s only active volcano (Andaman (अंडमान)).
25. Match pairs: **Rainier = USA**, **Etna = Italy**, **Paricutin = Mexico**, **Apo = Philippines**. **Kilimanjaro** is East African Rift, **not** Ring of Fire.
26. Earthquake–volcano belts include the Pacific **Ring of Fire**, Mid-Atlantic Ridge, Alpine–Himalayan belt and East African Rift. India’s seismic map uses Zones II–V.
27. Earth’s spheres — lithosphere, atmosphere, hydrosphere and biosphere — interact. Rocks cycle among igneous, sedimentary and metamorphic states.
28. Heat budget: about **35/100** of insolation (सूर्यताप) is reflected (albedo) and about **51/100** reaches the surface; surface heating then returns as **long-wave** radiation.
29. Nuclear and geothermal energy (भू-तापीय ऊर्जा) are **not** forms of stored solar energy. India’s Tropic of Cancer (कर्क रेखा) runs through the **middle** of the country — India is not wholly tropical and is not split into equal tropical halves by that line alone.
30. Pluto was reclassified as a **dwarf planet** by the IAU in **2006**. Neptune is the farthest true planet and the coldest among the eight.
31. The Moon always shows roughly the same face because its rotation period $\approx$ its revolution (~**27.3 days**). Weight on the Moon is about **1/6** of weight on Earth. Apollo 11 landed in the **Sea of Tranquility (1969)**.
32. Light-year measures **distance**. Star colour tracks **temperature**. Brittle star is **not** an astronomical object. Earth Day (पृथ्वी दिवस) is **22 April**.
33. **Igneous** = cooled magma/lava (no fossils); **sedimentary** = layered + fossils; **metamorphic** = heat/pressure remake. Granite → **gneiss**; limestone → **marble**; sandstone → **quartzite**; shale → **slate**.
34. Most abundant volcanic gas is **water vapour**. Magma is underground; lava is surface melt. **Stromboli** = Mediterranean lighthouse; **Barren Island** = India’s only confirmed active volcano; Circum-Pacific / **Ring of Fire** has the densest volcanic belt.
35. Earthquake **focus/hypocentre** is the rupture point; **epicentre** is the surface point above it. **P** waves are fastest (all states of matter); **S** waves solids only; surface waves are slowest and most damaging. **Richter** = magnitude (~10 $\times$ amplitude / ~32 $\times$ energy per step); **Mercalli** = intensity.
36. **Tsunami** is Japanese for harbour wave. Ring of Fire $\approx$ **90%** of world quakes. NW India shakes mainly from **Indian–Eurasian** plate convergence. 2004 Indian Ocean and 2011 **Fukushima** are the classic tsunami case studies.

Must-score facts — motions, structure, latitudes

Earth motions / shape

Item	Lock
Rotation	Day–night; west→east
Revolution	Seasons; elliptical orbit

Item	Lock
Shape	Oblate spheroid
Axial tilt	~ 23½°

Structure / circles

Item	Lock
Crust / mantle / core	Silicates · denser rocks · iron–nickel
Tropic of Cancer	23½°N (cuts India)
Equator	0°
International Date Line	~180° (zigzag)
Atmosphere layers	Troposphere → Stratosphere → Mesosphere → Thermosphere

Topic 15 — Geomorphology & Landform Processes

Consolidated — 28 Must-Score Facts

1. **Endogenic** processes (folding, faulting, volcanism, uplift) build relief from inside the Earth. **Exogenic** processes wear and deposit at the surface. **Gradation** = degradation + aggradation.
2. Davis's cycle of erosion is framed as **structure–process–time** (youth–mature–old). Penck and King modify timing, but the landform vocabulary stays -useful.
3. **Igneous** rocks are primary: **granite** is intrusive (acidic); **basalt** is extrusive (basic). Fossils belong to **sedimentary** rocks.
4. Igneous bodies: batholith, laccolith, lopolith, phacolith; a **sill** is parallel to beds; a **dyke** cuts across beds.
5. Metamorphic pairs: **limestone** → **marble**, **sandstone** → **quartzite**, **granite** → **gneiss**, **shale** → **slate / schist**. Foliated rocks include slate–schist–gneiss; marble and quartzite are non-foliated.
6. **Weathering** breaks rock **in situ**. **Erosion** picks up and carries material. **Denudation** is weathering plus erosion. Chemical routes include carbonation, oxidation, hydrolysis and hydration.
7. River load moves by traction, saltation, suspension and solution. Agents of erosion are river, wind, glacier, sea, groundwater (karst) and gravity (mass wasting).
8. Fluvial stages: youth = **V-valley / waterfall / pothole**; mature = **meander / floodplain / levee**; old = **ox-bow / delta (डेल्टा) / peneplain**. Rejuvenation adds knickpoints, terraces and incised meanders.
9. On a meander, the **concave** bank erodes and the **convex** bank builds a **point bar**. A **yazoo** stream is deferred behind a levee. Mountain-foot dumps are **alluvial (जलोढ़) fans**; coalescing fans form a **bajada**.
10. Stream genetic types include consequent (अनुवर्ती), subsequent (परवर्ती), obsequent and resequent. **Antecedent** (पूर्ववर्ती) streams cut rising land; **superimposed** (अध्यारोपित) streams inherit a course from a cover; subsequent streams follow structure.
11. **Bird's-foot** (पक्षी-पाद) delta = **Mississippi**. **Arcuate** (चापाकार) deltas = **Nile** and **Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)**. **Cuspate** pair = Tiber. **Moribund** has two frames: **Bengal subdivision** (UPPCS 2019) vs **Cauvery lobe** in landform match-lists — read the stem. Eastern India mouths favour deltas; western mouths often form **estuaries** (Narmada (नर्मदा) / Tapi (तापी)).
12. Drainage patterns: **folded** beds → **trellis**; homogeneous rock → **dendritic** (वृक्षाकार); dome → **radial**; joints / faults → **rectangular**.
13. The classic fault-valley river tag is the **Damodar** (दामोदर), not the Chambal (चंबल). **Narmada–Tapi** are rift / fault-line pairs as well.
14. The **Imphal** basin is **lacustrine** (lake-filled), not loess or glacial. Kashmir Vale also has lacustrine / structural pairs. **Loess** is wind-laid silt (China belt), not Imphal.
15. Structure landforms include cuesta, hogback, mesa and butte. A **mesa** is broader than a **butte**. Davis's humid old-age plain is a **peneplain** with leftover **monadnocks**; King's arid form is a **pediplain**.
16. Glacial erosional forms: cirque, arête, horn, **U-valley**, hanging valley, fjord, tarn. Depositional forms: moraine, drumlin, esker, kame, outwash, erratic.
17. A **fjord** is a drowned glacial trough. A **ria** is a drowned river valley. Do not treat them as the same estuary (ज्वारनदमुख) type.
18. Aeolian points: **barchan** horns point **downwind**; parabolic dunes often show horns upwind. Yardang, zeugen and mushroom rocks are wind-eroded forms.
19. Coastal deposition builds **spit, bar, tombolo and lagoon** with longshore drift. Coral reefs include fringing, barrier and atoll types. An **atoll's shape** is **horseshoe / ring** around a central lagoon (not triangular or rectangular).
20. Karst needs **limestone plus carbonation**. **Stalactite** hangs from the **ceiling**; **stalagmite** grows from the **floor**. Forms include doline, cave, uvala and polje.
21. River capture vocabulary includes pirate stream, beheaded stream, wind gap and elbow of capture.

22. East Indian rivers typically build **deltas**; west-flowing Narmada and Tapi typically form **estuaries** into the Arabian Sea.
23. Conglomerate has rounded pebbles; breccia has angular fragments. Organic sedimentary rock pair includes coal.
24. Intertrappean beds between Deccan (देक्कन) lava flows hold **land and freshwater** fossils, not marine plant–animal assemblages.
25. Himadri (हिमाद्रि) is fossil-poor crystalline rock; Lesser Himalaya carries marine fossils; Shiwalik (शिवालिक) holds human remains — keep the three belts distinct when landform stems mix fossils.
26. A gorge is a steep mountain cut; a canyon is often the arid stair-step equivalent. U-valley = glacier; V-valley = youthful river.
27. Mekong source is Tibet; its delta is in **Vietnam**, not Cambodia.
28. The rock cycle links magma $\rightleftharpoons$ igneous $\rightarrow$ sedimentary $\rightarrow$ metamorphic $\rightarrow$ melt, so the same material can reappear as a different rock type after burial, melting or recrystallisation.

Must-score facts — agents, landforms, rocks

Endogenic / exogenic

Item	Lock
Endogenic	Diastrophism, volcanism, earthquakes
Exogenic	Weathering, erosion, deposition
Weathering types	Physical, chemical, biological

Agent $\leftrightarrow$ landform

Agent	Landforms
River	V-valley, meander, oxbow, floodplain, delta
Glacier	U-valley, cirque, moraine, fjord
Wind	Yardang, dune, loess
Wave	Cliff, beach, spit, stack
Karst	Sinkhole, cave, stalactite

Rock cycle

Igneous $\rightarrow$ **Sedimentary** $\rightarrow$ **Metamorphic** (with melting / uplift loops)

Topic 16 — Oceans

Consolidated — 34 Must-Score Facts

1. By area the oceans rank **Pacific > Atlantic > Indian > Southern > Arctic**. The Indian Ocean was the Greek **Erythraean Sea** and straddles **both sides of the Equator**.
2. Ocean-floor order from the shore is **shelf → slope → rise (or trench) → abyssal plain**. A **guyot** is a flat-topped seamount. The continental shelf is shallow (about **200 m**), rich in fish and oil, and covers roughly **7.5%** of the ocean floor.
3. India's **west shelf is wider than the east**. The Gujarat–Mumbai belt is widest and holds **Bombay High**. The east shelf is narrower but carries large delta (डेल्टा) fans.
4. The **Telegraphic Plateau** is part of the **North Atlantic Ridge**, not a separate land plateau.
5. Trench–ocean pairs: **Mariana–Pacific, Puerto Rico–Atlantic, Sunda/Java–Indian, and Molloy–Arctic**.
6. **Agulhas** and **Brazil** are **warm** currents. **Humboldt (Peru)** and **California** are **cold**. Do not mark Agulhas or Brazil as cold.
7. The **Benguela** is an **Atlantic** cold current off south-west Africa. It is **not** a Pacific current.
8. Among classic MCQ options, the current linked to the **Indian Ocean** is the **Agulhas**.
9. **El Niño** (अल नीनो) sends warm water off Peru, weakens upwelling, and cuts plankton and fish catch.
10. Open-ocean **salinity** is about **35‰**. The maximum sits near the **Tropics of Cancer and Capricorn**, not on the Equator. The **Red Sea** is high; the **Baltic** is low.
11. Seawater density rises when water is **colder** and **saltier**. Cold salty water sinks and drives the **thermohaline** circulation. **NaCl** is about **77%** of dissolved salts.
12. Surface temperature peaks near the equator. The **thermocline** lies roughly **300–1000 m**. Deep water is cold in every ocean.
13. A **spring tide** forms at **syzygy** (new/full moon) and is **large**. A **neap tide** forms at **quadrature** and is **small**. The tidal period is about **12 h 25 min**, not 12 h 30.
14. A **tidal bore** is famous on the **Hooghly** (हुगली) and the **Amazon**. The **Bay of Fundy** has the world's greatest tidal range. India's tidal-energy coast fact is the **Gulf of Khambhat**.
15. **Upwelling** is strongest on **west coasts**: Peru, California, Canary, Benguela, and Somalia.
16. **Constructive** waves build beaches; **destructive** waves erode them. A **tsunami** (सुनामी) is not a tide.
17. Ocean currents shape climate (mild or foggy coasts, west-coast deserts), concentrate fish where warm and cold meet or upwelling occurs, and affect navigation routes.
18. Deepest deposit is **red clay**. **Globigerina** and **pteropod** oozes are **calcareous**. **Diatom** and **radiolarian** oozes are **siliceous**. **Manganese nodules** sit on abyssal plains, especially in the Pacific and the Central Indian Ocean Basin.
19. The **Suez Canal** (Egypt (मिस्र), **1869**) joins the **Mediterranean** and the **Red Sea**. It is a **sea-level** cut — ships stay at one water height and do **not** climb stepped chambers. Lakes north→south: **Manzala → Timsah → Great Bitter → Little Bitter**. Ends: **Port (बंदरगाह) Said** (N) and **Suez** (S). India–Europe sea route is shorter by about **7000 km**. It does **not** itself touch the Atlantic or the open Indian Ocean.
20. The **Panama Canal** (**1914**) joins the **Atlantic/Caribbean** and the **Pacific**. Ships **climb stepped chambers** and use **Gatun Lake** to cross the isthmus. The **Kiel Canal** (Germany) joins the **North Sea** and the **Baltic**.
21. UNCLOS belts: **territorial sea 12 nm, contiguous zone 24 nm, EEZ 200 nm**. The continental shelf starts at **200 nm** and may extend to **350 nm**.
22. Classic fishery centres: **Grand Banks** (Labrador meets Gulf Stream), **Dogger Bank**, Peru upwelling, and the **Kuroshio–Oyashio** mix.
23. India's **operational** Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri** (गंगोत्री) is not the operational pair. **Himadri** (हिमाद्रि) is India's **Arctic** station, not Antarctic.

24. The **Antarctic Treaty** dates to **1959/61**. India joined in **1983**. India is an **observer** in the Arctic Council.
25. Key straits: **Hormuz** (Persian Gulf ↔ Gulf of Oman; **Iran north / Oman south**; ~25% seaborne oil), **Malacca**, **Gibraltar**, **Bering**, **Bab-el-Mandeb**, **Palk**, and the **10° Channel** (दस डिग्री). Persian Gulf coasts = Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — **not Oman**.
26. The **Carlsberg Ridge** lies in the north-west Indian Ocean. The **Ninetyeast Ridge** is not the Mid-Atlantic Ridge.
27. The International Seabed Authority manages the seabed **Area** beyond national zones (HQ Jamaica). India's inland waterway **NW-1** is the **Ganga** (गंगा) from Haldia to Prayagraj (प्रयागराज).
28. **Drake Passage** is not the Strait of Magellan. Panama opened in **1914** with stepped chambers; Suez opened in **1869** as a sea-level cut.
29. Hydrosphere ≈ **71%** of Earth. Usable fresh ≈ **<1%** of all water. Of fresh water, ice >> groundwater >> rivers/lakes.
30. **Datum line** = mean-sea-level height/depth zero. Abyssal plains ≈ most extensive ocean-floor province (~**76%**).
31. **Sargasso** = North Atlantic, **no coast**. **Red Sea** = axial trough. **NAD** = Blanket of Europe. **OMT** (26°C isotherm) helps monsoon rainfall forecast.
32. Lake salinity pairs: Assal / Van / Dead Sea / Great Salt (लवणाध्यक्ष) Lake. **Halocline** = salinity gradient with depth. **Diamantina** = Indian Ocean.
33. Türkiye clock: **N = Black**, **S = Mediterranean**, **W = Aegean**, **NW = Marmara**. West→east ladder: **Mediterranean** → **Black** → **Caspian** → **Aral** ("My Black Cat Ate").
34. **Gaza** faces the **Mediterranean**. Baltic coasts = Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland, Sweden — **not Norway**. **Jordan** has no Mediterranean coast (only **Aqaba** / Red Sea).

Must-score facts — relief, currents, El Niño

Ocean relief

Feature	Lock
Continental shelf	Shallow margin
Slope / rise	Steep then gentle to floor
Abyssal plain	Deep flat floor
Trench	Deepest (e.g. Mariana)
Mid-ocean ridge	Divergent volcanism

Currents / climate

Item	Lock
Open-ocean salinity	~ 35‰ ; max near Tropics (not Equator)
Warm / cold currents	Affect coasts & fog / desert margins
Agulhas	Into Indian Ocean (S. Africa teaching)
El Niño	Warm water off Peru; weak upwelling; fish crash
Tides	Spring / neap; lunar–solar alignment

Topic 17 — World Rivers & Lakes

Consolidated — 28 Must-Score Facts

1. **New Orleans** stands on the **Mississippi**, not the Missouri. The Missouri joins the Mississippi at **St Louis**.
2. **Budapest** is on the **Danube**. **Cologne** is on the **Rhine**. India's **Hyderabad** (हैदराबाद) is on the **Musi**, not the Godavari (गोदावरी) or Paleru.
3. The **Mekong** rises in Tibet, flows **south / south-east**, and builds its delta (डेल्टा) in **southern Vietnam**. It does **not** flow south-west and the delta is **not** in Cambodia.
4. Direction pairs: **Amur** north-east; **Syr Darya** north-west into the Aral; **Angara** north out of Baikal; **Volga** from the Valdai Hills into the **Caspian**.
5. The main USA–Mexico border river is the **Rio Grande**. The **Colorado** is the usual trap option.
6. **Lake Onega** is in **Russia**, not Canada. **Lake Michigan** lies wholly in the **USA**. **Maracaibo** is in Venezuela. **Baikal** is in Russia.
7. Superlatives: the **Nile** is longest for Prelims; the **Amazon** has the largest discharge; the **Congo** is deepest and crosses the Equator **twice**; the **Yangtze** is Asia's longest; the **Volga** is Europe's longest.
8. Delta shapes: **Mississippi** = bird's-foot (पक्षी-पाद); **Nile**, **Hwang Ho**, and **Niger** = arcuate (चापाकार) (bow).
9. The **Danube** crosses the most countries. The **Rhine** is Europe's busiest navigation artery.
10. **Khartoum** is where the White and Blue Nile meet. **Aswan** created **Lake Nasser**. **GERD** sits on the **Blue Nile** in Ethiopia.
11. Dam pairs: **Three Gorges–Yangtze**, **Aswan–Nile**, **Itaipu–Paraná**, **Hoover–Colorado**, **Kariba–Zambezi**.
12. The **Caspian** is the largest lake. **Superior** is the largest freshwater lake by area. **Baikal** is deepest, holds the most fresh volume, and is among the oldest.
13. **Tanganyika** is second deepest and the longest freshwater lake. **Titicaca** is the highest navigable lake. The **Dead Sea** is the lowest land surface and is hypersaline (about **34%** salt (लवणाध्यक्ष)).
14. Great Lakes west to east: **Superior** → **Michigan** → **Huron** → **Erie** → **Ontario** (mnemonic: Super Man Helps Every One).
15. Falls: **Angel** (Venezuela) is highest; **Victoria** is on the Zambezi; **Niagara** is US–Canada; **Iguazu** is Brazil–Argentina.
16. The **St Lawrence** is the Great Lakes seaway. The **Rhine–Main–Danube** canal links the Black Sea system to the Rhine.
17. The **Aral Sea** shrinks mainly because the **Amu Darya** and **Syr Darya** were diverted. **Lake Chad** is also shrinking.
18. The **Niger** is the paradox river that first flows inland. The Congo is also known as the **Zaire**.
19. The **Darling Range** is a highland of south-west Australia. It is **not** the same as the **Murray–Darling** river system.
20. **Endorheic** lakes have no ocean outlet — Caspian, Aral, Dead Sea, Chad, and Eyre.
21. More city–river pairs: **Baghdad–Tigris**, **Paris–Seine**, **London–Thames**, **Cairo–Nile**.
22. Lake types include tectonic/rift, glacial, crater, lagoon, and oxbow. Do not treat every famous lake as simply a "fresh lake".
23. Continent spine: Nile (Africa), Amazon (South America), Yangtze (Asia), Mississippi–Missouri (North America), Volga (Europe), Murray–Darling (Australia).
24. Navigation classics include the Rhine, Danube, Volga, St Lawrence, Yangtze, and Mississippi.
25. **Limpopo** crosses the Tropic of Capricorn **twice**. **Congo/Zaire** crosses the Equator **twice**.
26. **Mahaweli** = longest Sri Lanka river; Sri Lanka drainage is **radial** from central highlands.
27. City–river desk: Phnom Penh–Mekong; Hanoi–Red; Yangon–Irrawaddy; Madrid–Manzanares; Bangkok–Chao Phraya; Basra–Shatt al-Arab.
28. **Grand Canyon** = Colorado. **Chisapani Gorge** = Nepal (Karnali). Largest delta = **Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)**. Inselberg ≠ glacier.

Must-score facts — city–river, deltas, lakes

City ↔ river

City	River
New Orleans	Mississippi (≠ Missouri)
Budapest	Danube
Cologne	Rhine
Hyderabad (India)	Musi (≠ Godavari)
Khartoum	White + Blue Nile meet

Direction / border

River	Lock
Mekong	Tibet → SE; delta S. Vietnam
Amur	North-east
Syr Darya	North-west into Aral
Rio Grande	Main USA–Mexico border (≠ Colorado trap)

Superlatives / deltas

Item	Lock
Longest (Prelims)	Nile
Largest discharge	Amazon
Bird's-foot delta	Mississippi
Arcuate	Nile, Hwang Ho, Niger
Onega	Russia (≠ Canada)
Michigan	Wholly USA

Topic 18 — World Landforms

Consolidated — 28 Must-Score Facts

1. Area order (largest → smallest): **Asia** → **Africa** → **North America** → **South America** → **Antarctica** → **Europe** → **Australia**. Australia is the **smallest** continent; Africa has the **most countries** (~54).
2. Mean elevation leader is **Antarctica** (~2300 m). Europe has the **highest share of plains** in its area. **Guyana** is in **South America**, not Africa.
3. The **Andes** are the world's **longest** fold chain. The **Himalaya** (हिमालय) are the **highest**. **Aconcagua** stands in the Andes of Argentina. Length ladder pair: Andes > Rockies > Great Dividing Range > Himalaya.
4. **Pyrenees** = Spain–France; **Alps** = Switzerland / central Europe (**not England**); **Apennine** = Italy; **Balkan** = Bulgaria; **Urals** mark Europe–Asia. **British Columbia** = "Sea of Mountains."
5. **Toubkal** (Atlas) is in **Morocco**. **Hoggar** is in **Algeria**. **Stanley / Rwenzori** is in Uganda. **Kilimanjaro** is in **Tanzania**. **Darling Range** = SW Australia.
6. **Chimborazo** is in **Ecuador**. The Atlas system is North African — do not place it in France or Spain. **Sierra Nevada** = block mountain (not young fold like Rockies/Alps/Himalaya).
7. **Kilimanjaro** sits on the **East African Rift** and is **not** in the Pacific Ring of Fire. **Etna** is in Italy / the Mediterranean. Tertiary young folds: Alps, Andes, Himalaya, Rockies, Atlas; Appalachians = **Caledonian / old**.
8. **Tibet** is the highest large plateau (~4500 m). The **Pamir** is the "**Roof of the world**". The **Altiplano** is Bolivia–Peru. The **Meseta / Madrid** is Spain. **Patagonia Plateau** = mineral storehouse pair.
9. The **Colorado Plateau** holds the Grand Canyon. The **Columbia Plateau** is a lava plateau. Do not swap the two. **Telegraphic Plateau** sits on the **North Atlantic Ridge** (not Indian Ocean).
10. Grassland names: **Pampas** = Argentina; **Campos** = Brazil; **Llanos** = Venezuela–Colombia; **Puszta** (पुस्ज़्टा) = Hungary; **Prairie** (प्रेयरी) = North America; **Steppe** (स्टेपी) = Eurasia; **Veld** = South Africa; **Downs** = Australia.
11. **Borneo** is shared by three countries and is **not** a volcanic dump. **Java** and **Sumatra** sit on a volcanic arc.
12. Island rank by size: **Greenland** > **New Guinea** > **Borneo** > **Madagascar** > **Baffin** > **Sumatra** > **Honshu** > **Victoria** > **Great Britain** > **Ellesmere**. Australia is a continent, not ranked as an island in that list. **Honshu** = Japan's largest; **Faroe** = Sheep Islands (Denmark).
13. The **Gobi** lies in **Mongolia and China** only. It is a cold desert, not Russia or Kazakhstan.
14. Hot-desert size order: **Great Sandy** < **Gobi** < **Arabian** < **Sahara**. The **Sahara** is the largest hot desert. **Thar** (थार) = densest populated desert; **Atacama** = driest; **Europe** has the least desertification (मरुस्थलीकरण) problem among continents.
15. **Gibson** = Australia (not Brazil). **Sonoran** = USA. **Taklamakan** = China. **Karakum** = Turkmenistan (also Kazakhstan fringe note).
16. The **Atacama** (N Chile / S Peru — not southern Chile alone) is the driest. The **Namib** is a fog coast. **Patagonia** is Argentina's rain-shadow / temperate desert.
17. Hot deserts favour **west coasts** because of the Horse Latitudes (हॉर्स अक्षांश) plus cold ocean currents (~15–30°).
18. "Land of Big Games" points to the tropical **savanna** (सवाना).
19. Sclerophyll scrub: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile.
20. **Epiphytes** mark equatorial forest; **baobab** marks savanna; **cedars** mark the Mediterranean; **acacia** marks the Sahara fringe.
21. In Brazil, **Selva** (सेल्वा) is the rainforest and **Terra Roxa** is the famous coffee soil — both are Brazil facts.
22. The **Mediterranean** climate has **winter rain** on five **west coasts** near **30–45°**. Summer is dry under the subtropical high.
23. **Taiga** is boreal conifer forest. **Tundra** (टुंड्रा) is treeless. Climate letters: Savanna **Aw**, Steppe **BS**, Tundra **ET**.
24. Volcano pairs: **Rainier** USA; **Etna** Italy; **Paricutin** Mexico; **Apo** Philippines; also Fuji (Japan), Pinatubo (Philippines), St Helens (USA).

25. Peak continents: **Elbrus** = Europe (Caucasus / Russia); **Mont Blanc** = Alps; **Denali / McKinley** = North America; **Kosciuszko** = Australia mainland; **Cook / Aoraki** = New Zealand Southern Alps.
26. Temperate grasslands (Prairie, Steppe, Pampas, Veld, Downs) sit on **chernozem**-type wheat soils. Residual and dome mountains are worn or laccolith leftovers, not young fold belts.
27. **Death Valley** (California) = rift valley, extreme heat, Devil's Golf Course salt (लवणाध्यक्ष) pan. **Silicon Valley** = California chip belt. **Great Artesian Basin** = Australia. Blind valley / sinkhole = **karst**.
28. **Arakan Yoma** = Myanmar (India–Myanmar hills: Naga (नागा), Chin, Patkai (पटकाई) / Lushai — not Khasi (खासी) as the boundary trio). **Golan Heights** = Middle East (SW Syria / occupied note). **Black Forest** = Germany (east of Rhine; Vosges west).

Must-score facts — continents, mountains, grasslands

Continent area (large → small)

Asia → Africa → N. America → S. America → Antarctica → Europe → Australia

Mountain / plateau tags

Item	Lock
Longest fold chain	Andes
Highest chain	Himalaya
Pyrenees	Spain–France
Alps	Central Europe (≠ England)
Kilimanjaro	East African Rift (≠ Ring of Fire)
Roof of the world	Pamir
Highest large plateau	Tibet
Colorado vs Columbia Plateau	Grand Canyon vs lava plateau

Grasslands

Name	Region
Pampas	Argentina
Campos	Brazil
Llanos	Venezuela–Colombia
Puszta	Hungary
Prairie / Steppe / Veld	N. America / Eurasia / S. Africa

Topic 19 — World Regional Geography

Consolidated — 28 Must-Score Facts

1. Continent area order: **Asia > Africa > North America > South America > Antarctica > Europe > Australia/Oceania.**
 2. The **Nobi** and **Kanto** plains are in **Japan**, not Korea.
 3. Iraq's **Sunni Triangle** is **Baghdad, Tikrit, and Ramadi**. **Basra** (Shia south) is the trap.
 4. West Asia mountains west to east: **Pontic → Zagros → Hindu Kush → Karakoram (काराकोरम).**
 5. Central Asia capitals: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**, Kazakhstan **Astana**.
 6. The **Kara Kum** desert is in **Turkmenistan**.
 7. **Borneo** is shared by Indonesia, Malaysia, and Brunei and is **not** a volcanic island dump.
 8. Philippines cane and coconut history (इतिहास) fact: **Spanish and Americans.**
 9. Korea: **Seoul** south, **Pyongyang** north, roughly the **38th parallel**. Nobi/Kanto are not Korean plains.
 10. Palestine map: **Gaza** on the Egypt (मिस्र) side, **West Bank** on the Jordan side, with the **Jordan River / Dead Sea** belt. Neighbours include Israel, Jordan, Egypt, Lebanon, and Syria.
 11. Western Europe has **westerlies** and rain in **all months**.
 12. The **Suez Canal** shortened the India–Europe sea route by about **7000 km**.
 13. **Cape Verde**'s capital is **Praia**. **Bamako** is Mali's capital — do not swap them.
 14. Maghreb capitals: Morocco **Rabat**, Algeria **Algiers**, Tunisia **Tunis**.
 15. Brazil points: **Selva (सेल्वा)** rainforest and **Terra Roxa** coffee soil are both true.
 16. Australia's **interior is desert**. The **north is tropical**, not temperate. The **Darling Range** lies in **south-west** Australia.
 17. India's operational Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri (गंगोत्री)** is not the operational answer.
 18. **Madeira** is Atlantic Portugal and is **not** Caribbean.
 19. Indonesia west to east: **Sumatra → Java → Bali (बलि) → Lombok**.
 20. The **Mekong** delta (डेल्टा) is in **southern Vietnam**, not Cambodia.
 21. South America's landlocked pair is **Bolivia** and **Paraguay**. Uruguay, Peru, and Suriname have coasts.
 22. **Igarka** is in **Russia**, not China.
 23. New Zealand's capital is **Wellington**. NZ has North and South Islands with **Cook Strait** and is **not** an Australian state.
 24. **Ethiopia** is landlocked after Eritrea's secession; **Eritrea** holds the Red Sea coast. The Horn is Ethiopia–Somalia–Eritrea–Djibouti.
 25. Only **double-landlocked** states in the usual set are **Uzbekistan** and **Liechtenstein**.
 26. Capitals that are not the country's most-famous tourist city: Turkey **Ankara**, Australia **Canberra**, Brazil **Brasília**, UAE **Abu Dhabi**, NZ **Wellington**.
 27. Boundary parallels: **38th** ≈ Korea; **49th** ≈ Canada–USA.
 28. South Asia map: Nepal and Bhutan are landlocked; Maldives and Sri Lanka are islands; **Kabul** is Afghanistan's capital. The Caucasus trio is Georgia–Armenia–Azerbaijan; Istanbul is not Turkey's capital.
 29. **Esperanto** is an artificial world auxiliary language. **Tamil** is an official language of **Singapore**. Spanish is official in Chile/Colombia/Cuba — **not** Congo. **Bahasa** = Indonesia (not Thailand). Mandarin leads classic L-1 speaker counts; English often leads total L-1+L-2.
-

Must-score facts — capitals, deserts, regional traps

Central Asia capitals

Country	Capital
Uzbekistan	Tashkent
Tajikistan	Dushanbe
Kyrgyzstan	Bishkek
Turkmenistan	Ashgabat

Regional locks

Item	Lock
Nobi / Kanto plains	Japan (≠ Korea)
Sunni Triangle	Baghdad–Tikrit– Ramadi (≠ Basra)
W Asia mountains W→E	Pontic → Zagros → Hindu Kush → Karakoram
Kara Kum	Turkmenistan
Borneo	Indonesia + Malaysia + Brunei
Korea	Seoul south / Pyongyang north (~38th parallel)
Gaza / West Bank	Egypt side / Jordan side

Topic 20 — World Agriculture

Consolidated — 20 Must-Score Facts

1. In plantation MCQs the classic plantation crop is **tea**, not wheat, rice, or maize. Plantation means estate, capital, monoculture, hired labour, and export.
2. Coffee producer order often taught for **2016** is **Brazil > Vietnam > Colombia > Indonesia**. Brazil is mainly **arabica** on Terra Roxa; Vietnam is bulk **robusta**.
3. Major cocoa producers are **Côte d'Ivoire, Ghana (घन), and Cameroon**. **Latvia** is not a cocoa producer. Cocoa originated in the Amazon.
4. India's rubber fact is **Kerala**. Rubber (*Hevea*) originated in the Amazon and moved to South-East Asia. Malaysia's **Kinta Valley** is **tin**, not rubber.
5. Shifting-cultivation names: **Jhum** (NE India), **Podu** (Andhra/Odisha), **Ladang** (Malaysia), **Milpa** (Mexico), **Roca** (Brazil), **Chena** (Sri Lanka), **Caingin** (Philippines).
6. The world **citrus** belt is **Mediterranean**, not equatorial.
7. Philippines cane and coconut history (इतिहास) fact: **Spanish and Americans**.
8. Wrong Indian crop–state dumps include Gujarat–tea, UP–jute, and Assam–wheat. The right fact is **Kerala–rubber** (and Assam for tea volume).
9. **Oil palm** majors are **Indonesia and Malaysia**. **Olive** majors are Spain, Italy, and Greece.
10. World **sugarcane** leader is **Brazil**. **Sugar beet** belongs to temperate **Europe**.
11. For rice and wheat, **China and India** lead in **volume**, but classic **exporters** are often other countries (Thailand/Vietnam for rice; Russia/USA/Canada and others for wheat).
12. **Intensive subsistence** = monsoon wet rice. **Extensive commercial** = Prairie (प्रेयरी)/Pampas wheat. **Mixed farming** = crops plus livestock (West Europe / US Midwest).
13. **Black tea** is fermented; **green tea** is not. The pluck fact is **two leaves and a bud**. Tea likes **20–30°C, 150–300 cm** rain, slope, and acid soil.
14. **IRRI** is in the Philippines. Borlaug's Nobel was for **Peace**. **FAO** HQ is **Rome**.
15. Banana: India produces heavily; **Ecuador** is a classic shipper. Silk and wool volume often centres on **China**; Australia is famous for **Merino** wool.
16. Soy export triangle = **USA–Brazil–Argentina**. India leads world **milk volume**; New Zealand and the Netherlands dominate the dairy-export story.
17. **Von Thünen** is market rings around a city. **Whittlesey** is the thirteen world agricultural types — do not merge the two models.
18. Five Mediterranean **west coasts** carry the citrus / vine / olive world belt.
19. Maize fact: **US Corn Belt**; USA > China > Brazil in volume. Long-staple cotton = Egypt (मिस्र)/Sudan; short staple = India–Pakistan belt.
20. Ranching belts include Pampas, Prairie, Veld, and Downs. Shifting cultivation is humid-tropics slash-and-burn with short crop years and long fallow.
21. **Golden Crescent** = Afghanistan–Iran–Pakistan opium. **Shahtoosh** = Chiru. **Fazenda** = Brazil plantation (not jhum). **Taungya** = Myanmar. **Mocha** = Yemen; **Kona** = Hawaii.
22. Apiculture = bees; viticulture = grapes; olericulture = vegetables. India often leads milk + banana + castor; China tea/tobacco/silk; Brazil cane + coffee #1; Thailand rubber.

Must-score facts — plantation, shifting, leaders

Crop / system tags

Item	Lock
Classic plantation crop (MCQ)	Tea (estate/capital/labour)
Coffee order (2016 teaching)	Brazil > Vietnam > Colombia > Indonesia
Cocoa majors	Côte d'Ivoire, Ghana, Cameroon
India rubber	Kerala (<i>Hevea</i> Amazon origin)
Oil palm majors	Indonesia, Malaysia
Olive majors	Spain, Italy, Greece
Sugarcane leader	Brazil ; sugar beet = temperate Europe
Citrus belt	Mediterranean

Shifting cultivation names

Name	Region
Jhum	NE India
Podu	Andhra / Odisha
Ladang	Malaysia
Milpa	Mexico

Topic 21 — World Minerals & Energy

Consolidated — 20 Must-Score Facts

1. Coalfield–country pairs: **Appalachian–USA**, **Lancashire–England**, **Ruhr–Germany**, **Kuzbass–Russia**. **Donetsk** is Ukraine’s classic coal answer among common options.
2. **Mount Newman / Pilbara / Hamersley** = **iron** in Australia. **Krivoy Rog** = Ukraine iron. **Lorraine** = France. **Kiruna** = Sweden. **Mesabi** = USA.
3. Germany–Normandy iron is **wrong**. **Normandy** is in **France**.
4. **Chile** leads copper with northern Andes **porphyry** deposits (Chuquibambilla, El Teniente). Also fact Katanga (DRC), Copperbelt (Zambia), and Bingham (USA).
5. Malaysia’s **Kinta Valley** = **tin** (cassiterite). Myanmar’s **Pegu Yoma** = **mineral oil**, not tin.
6. **Kashagan** oil is in **Kazakhstan**, not Kuwait. **Burgan** is Kuwait. Also fact Ghawar/Dhahran (Saudi), Kirkuk/Zubair (Iraq), Haft Kel (Iran), Baku (Azerbaijan).
7. Natural gas’s main constituent is **methane**. **LPG** is not the same as **CNG**.
8. **Postmasburg** (South Africa) = **manganese**. **Iodine** fact = **Chile**. Steel volume top = **China**.
9. Nuclear and geothermal energy (भू-तापीय ऊर्जा) are **not** “stored solar”. Wind, biomass, and hydro are solar-linked renewables. Coal, oil, and gas are conventional non-renewables.
10. **Hematite** is the red bulk ore; **magnetite** is black and highest grade. **Weipa** = bauxite; **Newman** = iron; **Witwatersrand** = gold; **Morocco** = phosphate.
11. Qatar’s **North Field** and Iran’s **South Pars** are one continuous Gulf gas giant.
12. **Itaipu** = Brazil–Paraguay hydel. **Three Gorges** = China on the Yangtze. **OPEC** HQ = **Vienna**. Original OPEC five (1960): Iran, Iraq, Kuwait, Saudi Arabia, Venezuela.
13. **Lithium Triangle** = Chile–Argentina–Bolivia. Cobalt volume = **DRC**. REE processing fact = **China**. Uranium volume often facts on **Kazakhstan**.
14. Iron exporters = Australia (**Pilbara**) and Brazil (**Carajás**). China mines iron and still imports heavily.
15. Ancient **shields** host many metals. Sedimentary **basins** host coal, oil, and gas.
16. Karaganda (Kazakhstan) is a classic coal city. Tin also centres on Bangka–Belitung (Indonesia) and the Andes of Bolivia.
17. Energy classification: coal/oil/gas = conventional non-renewable; wind/biomass/hydro = renewable solar-linked; nuclear and geothermal stand apart from “stored solar” wording.
18. Pegu Yoma must not be dumped into the SE-Asia tin belt — it is an **oil** fact.
19. Chile copper Assertion–Reason pattern: both copper leadership and porphyry geology can be true, and geology can explain leadership.
20. North Sea oil/gas = UK–Norway belt; West Siberia = Russia’s giant hydrocarbon province.
21. **German silver (रूपक)** contains **no silver**. **Peace Pipeline** = Iran–Pakistan. Brent = North Sea light crude. Uranium City = **Canada**. Rare earth processing = **China**.

Must-score facts — coalfields, ores, energy types

Coalfield ↔ country

Field	Country
Appalachian	USA
Lancashire	England
Ruhr	Germany

Field	Country
Kuzbass	Russia

Ore / oil tags

Item	Lock
Pilbara / Newman / Hamersley	Australia iron
Krivoy Rog	Ukraine iron
Chile copper	Andes porphyry (Chuquicamata)
Kinta Valley	Malaysia tin
Pegu Yoma	Myanmar oil (≠ tin)
Kashagan	Kazakhstan oil (≠ Kuwait)
Burgan	Kuwait
Postmasburg	South Africa manganese
Hematite / magnetite	Red bulk / black high-grade

Energy

Item	Lock
Natural gas main	Methane
LPG ≠ CNG	Different fuel forms
Nuclear / geothermal	Not "stored solar"

Topic 22 — World Industries

Consolidated — 20 Must-Score Facts

1. City–industry pairs: **Osaka–cotton**, **Detroit–auto**, **Cuba–cigar**, **St Petersburg–shipbuilding**.
2. **Igarka** is in **Russia** (Yenisei timber), not China. **Rotterdam** is in the **Netherlands**. **Montevideo** is Uruguay. **Jakarta** is Indonesia.
3. The **Suez Canal** joins the **Mediterranean** and **Red Sea** and cut India–Europe distance by about **7000 km**. Lakes north to south: **Manzala** → **Timsah** → **Great Bitter** → **Little Bitter**.
4. **Panama** joins Atlantic/Caribbean and Pacific **with facts**. **Kiel** joins the North Sea and Baltic. Suez is sea-level with **no facts**.
5. **Chinook** (चिनुक) is a warm dry wind of the Rockies. **Foehn** is the Alps equivalent. Both can be true together.
6. **Mistral** is southern France, **not** Australia. **Shamal** is Arabia, **not** Austria. **Brickfielder** (ब्रिकफील्डर) = Australia.
7. **Ruhr** = Germany heavy industry. **Lancashire** = cotton. **Yorkshire** = wool. **Pittsburgh–Great Lakes** = steel. **Detroit** = autos.
8. Japan belts: **Keihin** = Tokyo–Yokohama; **Hanshin** = Osaka–Kobe; **Chukyo** = Nagoya autos. Nicknames: Osaka = Manchester of Japan; Nagoya = Detroit of Japan; Kawasaki = Pittsburgh of Japan; Ivanovo = Russian Manchester.
9. **Footloose** industry example = electronics. **Aluminium** seeks cheap hydel. **Steel** seeks ore and coking coal (weight-losing near raw (रॉ) material).
10. **Entrepôt** classics = Singapore, Rotterdam, Hong Kong. **Duisburg** is an inland Rhine port (बंदरगाह), not a Dutch sea mouth.
11. More wind pairs: **Santa Ana** = California; **Haboob** = Sudan; **Yamo** = Japan; **Leveche** = Spain; **Bora** (बोरा) = Adriatic cold; **Harmattan** (हरमट्टन) = West Africa; **Sirocco** = Sahara→Med; **Khamsin** (खामसिन) = Egypt (मिस्र).
12. A **willy-willy** (विली-विली) is a cyclone name, not a local wind. **Akron** = tyres. **Toulouse** = Airbus.
13. **Silicon Valley** (California) = electronics / IT — not Detroit.
14. Shipbuilding volume leaders are **China–South Korea–Japan**. St Petersburg remains the classic European city.
15. Weight-losing industries sit near raw material; bottling and market-oriented industries sit near consumers. Oil refining often seeks the port or pipeline end.
16. Italy's industrial triangle is the **Po Basin / Milan–Turin–Genoa** belt.
17. Port Said is at the **north** end of Suez; Suez town is at the **south**. Panama uses facts and **Gatun Lake**.
18. Break-of-bulk points concentrate at ports. Textiles often seek labour and market; steel is the classic weight-losing case.
19. Pearl River Delta (डेल्टा) is China's classic coastal manufacturing belt in this chapter's map diet.
20. Shanghai leads container volume among common port MCQs; Rotterdam sits at the Rhine mouth.

Must-score facts — city–industry, canals, winds

City ↔ industry

City	Industry
Osaka	Cotton
Detroit	Auto
Cuba	Cigar
St Petersburg	Shipbuilding
Ruhr	Heavy industry (Germany)

City	Industry
Lancashire / Yorkshire	Cotton / wool
Pittsburgh–Great Lakes	Steel

Japan belts

Belt	Lock
Keihin	Tokyo–Yokohama
Hanshin	Osaka–Kobe
Chukyo	Nagoya autos

Canals / winds / industry type

Item	Lock
Suez	Med ↔ Red Sea; ~7000 km shorter India–Europe
Panama	Atlantic/Caribbean ↔ Pacific
Kiel	North Sea ↔ Baltic
Chinook / Foehn	Rockies / Alps warm dry winds
Mistral	S. France (≠ Australia)
Footloose	Electronics; aluminium seeks hydel; steel seeks ore+coal
Entrepôt	Singapore, Rotterdam, Hong Kong

Topic 23 — Political & Map-Based Geography

Consolidated — 30 Must-Score Facts

1. UNCLOS: **territorial sea 12 nm, contiguous zone 24 nm, EEZ 200 nm**. The continental shelf may extend to **350 nm**, but that does **not** push EEZ water beyond 200.
2. **Innocent passage** applies in the territorial sea. **Transit passage** applies in international straits.
3. **McMahon Line** = India–China (1914). **Durand Line** = Pakistan–Afghanistan (1893). **Radcliffe Line** = 1947 India–Pakistan/Bangladesh.
4. India has **seven** land neighbours (Pakistan, Afghanistan via Wakhan/PoK, China, Nepal, Bhutan, Myanmar, Bangladesh). Maritime neighbours are **Sri Lanka** and the **Maldives**.
5. Longest Indian **state** coastline = **Gujarat**. Longest land border = **Bangladesh**. Shortest land border = **Afghanistan**. Mainland plus islands coastline $\approx$ **7516 km**.
6. The **Suez Canal** joins Med and Red Sea, shortens India–Europe by about **7000 km**, and lakes run Manzala → Timsah → Great Bitter → Little Bitter.
7. **Panama** = Atlantic–Pacific canal. **Kiel** = North Sea–Baltic.
8. Straits: **Hormuz** = Gulf oil; **Malacca** = Indian Ocean–South China Sea; **Gibraltar** = Med–Atlantic; **Bosporus** = Black Sea–Marmara; **Bering** = Russia–USA.
9. Area ladder: Russia > Canada > USA > China > Brazil > Australia > **India (7th)**. Longest world coastline = **Canada** (then Indonesia, Russia, Philippines, Japan, Australia in the usual set).
10. Central Asia capital set: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**.
11. **Bolivia** is landlocked among common South America traps. **Laos** = only SE Asia landlocked. **Nobi/Kanto** = Japan. **Igarka** = Russia.
12. **Thorntwaite** (थॉर्न्थ्वेट) is the true **vegetation** climate index. **Köppen** gives letter-code climate classes. Mediterranean = **winter rain**. Western Europe = rain **all months** plus westerlies (**Cfb**).
13. India is the **seventh**-largest country, about **2.4%** of world land, with the Tropic of Cancer (कर्क रेखा) through the middle — so India is **not** wholly tropical. The Tropic does **not** cross **Uttar Pradesh** (उत्तर प्रदेश).
14. **Cape Verde** capital = **Praia**. **Bamako** = Mali. Only common double-landlocked states: **Uzbekistan** and **Liechtenstein**.
15. Uttar Pradesh's only foreign neighbour is **Nepal**.
16. Largest landlocked country by area = **Kazakhstan**. Most populous landlocked = **Ethiopia**. **Lesotho** is an enclave inside South Africa.
17. **49th Parallel** $\approx$ USA–Canada. **38th Parallel** $\approx$ Koreas. Maginot $\approx$ France–Germany; Rio Grande $\approx$ USA–Mexico.
18. Equator traps: Egypt (मिस्र) and Mexico are **not** equatorial countries in the usual MCQ sense.
19. India's nine coastal states include Gujarat as the longest; Telangana is not coastal.
20. Oder–Neisse is the historic (ऐतिहासिक) Germany–Poland line; do not confuse it with McMahon or Durand.
21. Shelf rights beyond 200 nm (up to 350) are seabed rights — they do not create a wider EEZ water column.
22. Horn of Africa = Djibouti, Eritrea, Ethiopia, Somalia (**not Sudan**). Balkans = Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Greece, Kosovo, Montenegro, North Macedonia, Romania, Serbia, Slovenia, European Türkiye — **Austria not** Balkan. Oceania excludes **Indonesia**. Caspian five exclude Armenia/Iraq.
23. Capitals $\neq$ famous cities: Australia **Canberra**; Japan **Tokyo**; Nigeria **Abuja**; Myanmar **Naypyidaw**; Slovenia **Ljubljana** (Bratislava = Slovakia).
24. Old names: Siam→Thailand; Formosa→Taiwan; Gold Coast→Ghana (घना); Dutch Guiana→**Suriname**; Southern Rhodesia→Zimbabwe.
25. Greenland = Denmark politically / N America geographically. Gaza borders **Egypt + Israel**. Afghanistan does **not** border Russia.
26. Most megacities are coastal as **ocean gateways**. Alaska = USA; Malta = Mediterranean; Baikonur = **Kazakhstan**.

27. **Norway** = Land of the Midnight Sun (Arctic Circle). **Japan** = Land of the Rising Sun. **Finland** = Thousand Lakes. **(South) Korea** = Morning Calm. **Thailand** = White Elephants.
28. **South America** = Bird Continent. **Sri Lanka** = Mistress of the Eastern Sea / Pearl of the Indian Ocean. **Singapore** = Gateway to Asia. **Istanbul** = Gateway to the West.
29. City tags: **Venice** = canals; **Osaka** = Manchester of the East; **San Francisco** = Golden Gate; **Chicago** = City of Smoke; **Buenos Aires** = Paris of South America; **St. Petersburg** = Venice of the North.
30. **Pamir** = Roof of the World. **Baikal** = Pearl of Siberia. **Bahrain** = Island of Pearls. **Aberdeen** = Oil Capital of Europe. **Ninety East Ridge** = Indian Ocean.

Must-score facts — UNCLOS, borders, straits

UNCLOS distances

Zone	Distance
Territorial sea	12 nm
Contiguous zone	24 nm
EEZ	200 nm
Continental shelf max	Up to 350 nm (conditions)
Innocent passage	Territorial sea
Transit passage	International straits

Borders / lines

Line	Lock
McMahon	India–China (1914)
Durand	Pakistan–Afghanistan (1893)
Radcliffe	1947 India–Pakistan
Longest Indian state coastline	Gujarat
Longest land border	Bangladesh
Shortest land border	Afghanistan (teaching)
Area ladder	Russia > Canada > USA > China > Brazil > Australia > India (7th)

Straits

Strait	Lock
Hormuz	Gulf oil
Malacca	Indian Ocean–South China Sea
Gibraltar	Med–Atlantic
Bosporus	Black Sea–Marmara

Uttarakhand — Topic 1 — Location, Relief & Structure

Must-Score Facts

1. Lat-long $\sim 28^{\circ}43' - 31^{\circ}27' \text{ N}$ · $77^{\circ}34' - 81^{\circ}02' \text{ E}$; HP west, Nepal/Tibet E–N, UP south.
 2. South→north full stack: Tarai → Bhabar → Shiwalik → Dun → Lesser → Great Himalaya.
 3. Bhabar = streams sink; Tarai = streams reappear, marshy, fertile.
 4. Between MBT and MCT = **Lesser Himalaya**; Shiwalik = south of MBT.
 5. Dehradun = classic dun; Nanda Devi $\sim 7816 - 7817 \text{ m}$ (Chamoli); Kamet $\sim 7756 \text{ m}$.
 6. Garhwal = west; Kumaon = east.
-

Uttarakhand — Topic 2 — Climate & Drainage

Must-Score Facts (High-Yield UKPCS)

1. **West to East River Alignment (UKPCS 2025 Q109):**

\text{Yamuna (West)} \rightarrow \text{Bhilangana (Tehri (टिहरी))} \rightarrow \text{Alaknanda (Central)} \rightarrow \text{Gori Ganga (East Pithoragarh)}

2. **Panch Prayag Confluence Formula:** - **Vishnuprayag:** Alaknanda + Western Dhauliganga (धौलीगंगा) (Chamoli) - **Nandaprayag:** Alaknanda + Nandakini (Chamoli) - **Karnaprayag:** Alaknanda + Pindar (Chamoli) - **Rudraprayag:** Alaknanda + Mandakini (मंदाकिनी) (Rudraprayag) - **Devprayag:** Alaknanda + Bhagirathi (Tehri Garhwal (गढ़वाल)) \rightarrow becomes **Ganga**

3. **Panch Kedar (Body Parts of Shiva Worshipped):** - **Kedarnath (केदारनाथ):** Hump (*Prishtha*) — Rudraprayag - **Madhyamaheshwar:** Navel (*Nabhi*) — Rudraprayag - **Tungnath:** Arms (*Bhuja*) — Rudraprayag (highest Shiva temple: 3,680 m) - **Rudranath:** Face / Mukha (*Mukh*) — Chamoli - **Kalpeshwar:** Hair lock (*Jata*) — Chamoli (only Kedar open all 12 months)

4. **Panch Badri (All in Chamoli District):** - **Vishal Badri (Badrinath (बद्रीनाथ)):** Main Dham consecrated by Adi Shankara (अद्वैतवाद). - **Yogdhyan Badri:** Pandukeshwar (where Pandu meditated). - **Bhavishya Badri:** Subhai / Tapovan (future shrine). - **Vridha Badri:** Animath (ancient seat before Badrinath). - **Adi Badri:** Karnaprayag (cluster of 16 ancient stone shrines).

Uttarakhand — Topic 3 — Vegetation & Wildlife

Must-Score Facts

1. Nanda Devi BR: WHS + snow leopard / musk deer / blue sheep.
 2. Askot–Pithoragarh; Binsar–Almora; Govind–Uttarkashi; Nandhaur–Nainital.
 3. Corbett 1936; Rajaji 1983; Gangotri 1989; Govind **1990**.
 4. Forest richness explained by topography and climate (A/R both true, R explains A).
 5. Swapping Askot/Binsar districts loses the match code.
 6. **Birch** = Himalayas (UKPCS 2006).
 7. **Deodar** abundant in western Himalayan temperate zone **1500–2500 m**; highest elevation among deodar / pine / sal / teak.
 8. **Lisa** comes from **pine** (resin → turpentine).
 9. **Khair** = **sub-tropical** forest type (UKPCS 2021); also the katha tree.
 10. **Bugyal** = alpine meadow above the tree-line.
-

Uttarakhand — Topic 4 — Minerals, Power & Industry

Must-Score Facts

1. Silica sand ≠ Pauri Garhwal — use Dehradun, Nainital, Tehri Garhwal.
 2. Dolomite — Dehradun, Pithoragarh.
 3. Soapstone — Chamoli, Bageshwar.
 4. Graphite — Almora, Nainital.
 5. Hydropower is the core power narrative; Tehri is the flagship on Bhagirathi.
 6. Industry concentrates in Tarai plains and Dehradun belt.
-

Uttarakhand — Topic 5 — Agri, AH & Irrigation

Must-Score Facts

1. Animal husbandry is a **supplementary** occupation in Uttarakhand (उत्तराखंड), not the sole primary livelihood for most households.
 2. On cultivated-area / AH stems, often **only the first statement** holds — read both carefully.
 3. Major sugarcane / irrigated plains district rivalry often pits **Haridwar** against **Udham Singh Nagar**.
 4. Plains lead cultivated area; hills lead terrace / horticulture stories.
 5. Tarai = canal–tubewell; hills = springs / guls.
-

Uttarakhand — Topic 6 — Population, SC/ST & Settlements

Must-Score Facts (High-Yield UKPCS)

1. **Sex Ratio Descending Trap (UKPCS 2025 Q111):**

\text{Almora (1142)} \rightarrow \text{Rudraprayag (1114)} \rightarrow \text{Tehri (टिहरी) Garhwal (1077)} \\ \rightarrow \text{Haridwar (880)}

(7 hill districts have sex ratio > 1000 due to heavy male out-migration; plains districts Haridwar 880 and Dehradun 889 have lowest).

2. **Negative Decadal Growth (Ghost Villages):** Uttarakhand has two hill districts that registered an absolute (निरपेक्ष) population contraction between 2001 and 2011: **Pauri Garhwal (-1.41%)** and **Almora (-1.28%)**.

3. **Density Contrast: Haridwar (801)** has 20 times the population density of **Uttarkashi (41)**.

4. **Literacy Extremes:** - Overall Literacy: **Dehradun** highest (84.25%); **US Nagar** lowest (73.10%). - Male Literacy: **Rudraprayag** highest in India/state (93.90%). - Female Literacy: **Uttarkashi** is the only district with female literacy under 60% (59.74%).

5. **Vanrawat / Raji Lifestyle (UKPCS 2025 Q113):** Forest-dwelling primitive tribal (आदिवासी) group concentrated in **Pithoragarh**; historically sustained by **shifting cultivation** (*Jhuming*) until restricted by state forest laws.

Uttarakhand — Topic 7 — Transport, Tourism & Hazards

Must-Score Facts (High-Yield UKPCS)

- Airport & Airstrip Catalog (UKPCS 2025 Q103):** - *Jolly Grant Airport*: Dehradun (main commercial terminal). - *Pantnagar Airport*: Udham Singh Nagar (उधम सिंह नगर) (Kumaon (कुमाऊँ) gateway). - *Naini Saini Airport*: Pithoragarh. - *Chinyalisaur (Maa Ganga (गंगा)) Airstrip*: Uttarkashi. - *Gauchar Airstrip*: **Chamoli** (not Rudraprayag).
 - Silkyara–Barkot Bi-Directional Tunnel**: Length = **4.531 km**; located on NH-134 under the Char Dham project; collapsed in Nov 2023, resulting in the successful 17-day rescue of 41 workers.
 - High Mountain Passes (La / Dhura / Darra)**: - *Sinla Pass*: Connects Darma and Byans valleys in Pithoragarh. - *Kalindi Pass*: Glaciated alpine pass connecting **Gangotri (Uttarkashi) to Badrinath (बद्रीनाथ) (Chamoli)**. - *Lipulekh Pass*: Pithoragarh to Tibet (tri-junction of India, Nepal, and China). - *Traill's Pass*: Between Pindari (पिंडारी) Glacier (Bageshwar) and Milam (मिलाम) (Pithoragarh).
 - Alpine Pastures (Bugyals)**: - *Bedni Bugyal*: Chamoli — largest bugyal in the state; Brahma Kamal (ब्रह्मकमल) flourishes here. - *Auli (औली) Bugyal*: Chamoli — international winter skiing destination. - *Dayara Bugyal*: Uttarkashi — site of the traditional butter-throwing folk (लोक) festival (*Anduri Utsav*).
 - Seismic Vulnerability**: Uttarakhand lies in **Seismic Zones IV and V** (highest vulnerability to catastrophic earthquakes in the Central Seismic Gap).
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